

Introductory Algebra



Time for 2IA :)

Contents

1	Integers	1
1.1	Induction	1
1.2	Divisibility and Prime Factorization	5
1.2.1	Division Algorithm	6
1.2.2	Greatest common divisor. Primes and coprime integers	7
1.2.3	Euclidean Algorithm	11
1.2.4	Prime Factorization	13
1.3	Integers modulo n	16
1.3.1	Modular arithmetic	20
1.3.2	Solving linear equations and systems in $\mathbb{Z}_n$	22
2	Permutations	31
2.1	What is a permutation?	31
2.1.1	Product of permutations	33
2.1.2	Inverse permutations	37
2.1.3	Disjoint permutations	38
2.2	Cycles	40
2.3	The Alternating Group	47
3	Groups	53
3.1	What is a group?	53
3.1.1	Cayley tables	55
3.1.2	Inverses. Exponent Laws. Cancellation Laws	58
3.1.3	A taste of isomorphic groups	64
3.2	Subgroups	66
3.3	Cyclic groups	72

3.3.1	Some important properties of Cyclic Groups	76
3.3.2	Fundamental Theorem of Finite Cycle Groups	78
3.4	Homomorphisms and isomorphisms	83
4	Lagrange's Theorem	91
4.1	Cosets	91
4.2	Cosets properties	92
4.3	Lagrange's Theorem	95
5	Factor groups	97
5.1	Normal subgroups	97
5.2	Factor groups	100
5.3	The First Isomorphism Theorem	104
A	Some useful stuffs	109
A.1	Sets	109
A.2	Maps	111
A.3	Binary operations	113
A.4	Equivalence relations	116

Chapter 1

Integers

In this chapter, we revisit the so called **principle of mathematical induction** and how to prove a statement by induction. We will also talk about the **Well Ordering Principle**.

1.1 Induction

What do we mean by INDUCTION? Let us consider the following sequence of integers:

$$\begin{aligned} 1 &= 1, \\ 1 + 3 &= 4, \\ 1 + 3 + 5 &= 9, \\ 1 + 3 + 5 + 7 &= 16, \\ &\vdots \end{aligned} \tag{S}$$

Does (S) follow any pattern? Look at carefully its left and right sides:

<u>LEFT SIDE</u>	<u>RIGHT SIDE</u>
Sum of the first n odd integers	$1^2, 2^2, 3^2, \dots, n^2$

Keeping in mind that the n -th odd integer is $2n - 1$, we might wonder whether

$$1 + 3 + 5 + \dots + (2n - 1) \stackrel{?}{=} n^2 \tag{p_n}$$

But **how can we prove the validity of (p_n)** ?

PRINCIPLE OF MATHEMATICAL INDUCTION. Let p_n be a property or statement defined for each $n \in \mathbb{N}$. Suppose that the following two conditions hold.

- (1) **Basis Step:** p_1 is true.
- (2) **Inductive Step:** if p_k is true, then p_{k+1} is true. ($k \geq 1$)

Then the property p_n is true for every $n \in \mathbb{N}$.

Definition 1.1. The proof of a result that applies the Principle Mathematical Induction is called a **proof by induction**. The assumption that p_k is true is called the **induction hypothesis**.

Example 1.2. Prove by induction that

$$1 + 3 + 5 + \dots + (2n - 1) = n^2, \text{ for all } n \geq 1$$

Solution: Let p_n be the property $1 + 3 + 5 + \dots + (2n - 1) = n^2$. It is clear that the base step, (the validity of p_1) holds since $1 = 1^2$. Assume that p_k is true and let us show that p_{k+1} is also true. The induction hypothesis tells us that $1 + 3 + 5 + \dots + (2k - 1) = k^2$. Using this we obtain that

$$1 + 3 + 5 + \dots + (2k - 1) + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2,$$

proving the validity of p_{k+1} and the induction principle applies to get that p_n is true for all $n \geq 1$.

We have started the Principle of Mathematical Induction at 1 but we can indeed replace it by any integer m . More concretely:

GENERAL PRINCIPLE OF MATHEMATICAL INDUCTION. If $m \in \mathbb{N}$ and $p_m, p_{m+1}, \dots$ are statements such that

- (1) **Base Step:** p_m is true.
- (2) **Inductive Step:** if p_k is true, then p_{k+1} is true, where $k \geq m$.

Then p_n is true for each $n \geq m$.

To prove the General Principle of Mathematical Induction, apply the Principle of Mathematical Induction to the family of statements:

$$t_n = p_{m+n-1}, \quad n \geq 1.$$

A variant of the principle of mathematical induction is the following:

STRONG INDUCTION. Let $m \in \mathbb{N}$ and for each $n \geq m$, let p_n be a statement. If the following conditions are satisfied:

- (1) p_m is true.
- (2) If $k \geq m$ and all of $p_m, p_{m+1}, \dots, p_k$ are true, then p_{k+1} is also true.

Then p_n is true for every $n \geq m$.

PROOF. Given $n \geq m$, we define the following statement:

$$t_n : p_m, p_{m+1}, \dots, p_n \text{ are true}$$

Note that t_m is true by (1). Assume that t_k is true for some $k \geq m$, and let us show that t_{k+1} is so. Note that the validity of t_k means that $p_m, p_{m+1}, \dots, p_k$ are true. Then (2) implies that p_{k+1} is true, so t_{k+1} is true, as desired. An application of the General Principle of Mathematical Induction concludes the proof. $\square$

Definition 1.3. We say that $p \in \mathbb{N}$ is **prime number** (or simply **prime**) if $p \neq 1$ and the only natural numbers divisors of p are 1 and p . A natural number ≥ 2 which is not prime is called **composite**.

For instance, the first few primes are 2, 3, 5, 7, 11, 13, ...

Example 1.4. Use strong induction to show that every integer $n \geq 2$ is a product of (one or more) primes.

Solution: For $n \geq 2$, consider the statement

$$p_n : n \text{ is a product of primes.}$$

It is clear that p_2 is true, since 2 is a prime. Take $k \geq 2$ and assume that $p_2, p_3, \dots, p_k$ are true, i.e., 2, 3, ..., k are all products of primes. In order to apply Strong Induction, we need to show that $k+1$ is a product of primes. We distinguish two cases:

- CASE 1: $k+1$ is a prime. In this case we are done! :)

- CASE 2: $k + 1 = ab$, for some $2 \leq a, b \leq k$.

Applying the Strong Induction hypothesis, we get that a and b are product of primes, so is $ab = k + 1$.

At this point, you might be wondering what is the difference between *Induction* and *Strong Induction*, or in other words, when one should use *Strong Induction*. Sometimes, like in the previous example, in order to prove the validity of the statement p_{k+1} it is not enough to assume that the statement p_k is true. We might need to use that some other statement p_i is true, for $i \leq k$. Okay, you might be now thinking “*how do I know that?*” If you are trying to prove that p_{k+1} is true assuming only that p_k holds, and you can not go very far in the proof (your argument does seem natural/ smooth and you are stuck), then most likely, you should give *Strong Induction* a go.

We close this section with a property of the natural numbers that turns out to be equivalent to the principle of induction. It states that every nonempty set of $\mathbb{N}$ has a smallest integer. More precisely:

WELL-ORDERING AXIOM/ LEAST INTEGER PRINCIPLE.

If X is a nonempty subset of $\mathbb{N}$, then there exists $a \in X$ such that $a \leq x$ for all $x \in X$.

THEOREM 1.5. *The following principles are equivalent:*

- (i) *Well-Ordering*
- (ii) *Induction*
- (iii) *Strong Induction*

PROOF. (i) $\Rightarrow$ (ii)

Let $p_1, p_2, \dots$ be statements such that p_1 is true and $p_k \Rightarrow p_{k+1}$ for every $k \geq 1$. We have to show that p_n is true for every $n \geq 1$. To do so we define the set

$$X = \{n \geq 1 \mid p_n \text{ is false}\}.$$

In other words, X consists on the natural numbers n such that p_n is false. Our goal is to show that $X = \emptyset$. Assume on the contrary that $X \neq \emptyset$. Then

by the Well-Ordering Principle X has a smallest element, say m . Notice that $m > 1$ since p_1 is true, so $m - 1 \geq 1$. Notice that $m - 1 \notin X$ because m is the least element of X . Thus p_{m-1} is true and so is p_m (because $p_{m-1} \Rightarrow p_m$). But this is impossible because p_m is false, since $m \in X$. Thus $X = \emptyset$ and hence p_n is true for all $n \geq 1$.

(ii) $\Rightarrow$ (iii)

We have already proved this.

(iii) $\Rightarrow$ (i)

Let X be a nonempty subset of $\mathbb{N}$ and let us show that X contains a least element. For $n \geq 1$, define the statement

$$p_n : \text{ if } n \in X \text{ then } X \text{ has a least element.}$$

Notice that p_1 is true; in fact, if $1 \in X$, then 1 is the least element of X . Take $k \geq 1$ and assume that $p_1, \dots, p_k$ are true and let us show that p_{k+1} is also true. Suppose then that $k + 1 \in X$ and let us prove that X has a least element. We consider two cases:

- (1) $k + 1$ is the least element of X . Then there is nothing to prove.
- (2) There exists $m \in X$ such that $m < k + 1$. In this case, then p_m is true, and since $m \in X$, this implies that X has a least element. In any case, we have proved that if $k + 1 \in X$, then X has a least element, that is, p_{k+1} is true.

Applying Strong Induction we can conclude that p_n is true for all $n \geq 1$. Keeping in mind that X is nonempty, we get that X has a least element. $\square$

1.2 Divisibility and Prime Factorization

In this section, we will cover the following topics:

- Long division of integers. Division Algorithm.
- Gcd, relatively primes, primes, Bézout's Lemma. Euclidean Algorithm.
- Prime Factorization. Lcm.

1.2.1 Division Algorithm

Let me start with an easy question:

What do we mean by writing $\frac{22}{7}$ in the form $3 + \frac{1}{7}$?

Well ... We are just saying that 22 can be written as $22 = 3 \cdot 7 + 1$. In other words, **22 leaves a remainder of 1 when divided by 7**. This is a very important property of the integers, known as the *Division Algorithm*; we will see that is a consequence of the Well-Ordering Axiom.

DIVISION ALGORITHM. Let n and $d \geq 1$ be integers. Then there exist unique integers q and r such that $n = qd + r$ and $0 \leq r < d$.

PROOF. We divide the proof into two parts:

PART 1: Existence of q and r

Consider the set $X = \{n - td \mid t \in \mathbb{Z}, n - td \geq 0\}$. We claim that X is non-empty. In fact:

- If $n \geq 0$, then $n \in X$ since $n = n - 0 \cdot d \geq 0$.
- If $n < 0$, then $n - nd \in X$ since $n - nd = n(1 - d) \geq 0$.

Hence X is non-empty, and the Well-Ordering Axiom applies to get that X has a least element r . Then $r \geq 0$ and $r = n - qd$ for some $q \in \mathbb{Z}$. Thus, $n = qd + r$. It remains to show that $r < d$. Suppose on the contrary that $r \geq d$. Then $r - d \in X$ since $r - d \geq 0$ and

$$r - d = n - qd - d = n - (q + 1)d.$$

This contradicts the minimality of r (because $r - d < r$) and so $r < d$.

PART 2: Uniqueness of q and r

Suppose that we can also write $n = q'd + r'$ for some integers q' and r' such that $0 \leq r' < d$. We can assume (without loss of generality) that $r \geq r'$. Then subtracting both expressions we obtain that

$$0 = (q - q')d + r - r',$$

which implies that $(q' - q)d = r - r'$. On the other hand, we know that $0 \leq r - r' \leq r < d$, and $q' - q$ is an integer. So necessarily $r = r'$ and $q = q'$. $\square$

Definition 1.6. For n and $d \geq 1$, we call the integers q and r in the *Division Algorithm* the **quotient** and **remainder**, respectively.

Example 1.7. If we divide $n = -17$ by $d = 5$, the result is

$$-17 = (-4) \cdot 5 + 3,$$

so the quotient is -4 and the remainder is 3 .

If both n and r are positive, the familiar process of long division is an algorithm for finding the quotient q and the remainder r . If we have a calculator, we can easily find q and r as follows:

1. Find n/d .
2. Let q be the largest integer that is less than or equal to n/d .
3. Take $r = n - qd$.

Example 1.8. Find the quotient and remainder if $n = 4187$ and $d = 129$.

Solution: We have that $n/d = 32.457$ approximately. So $q = 32$ and $r = n - dq = 4187 - 129 \cdot 32 = 59$. Thus, $4187 = 32 \cdot 129 + 59$.

1.2.2 Greatest common divisor. Primes and coprime integers

Let us revisiting a very important concept: divisibility. If n and d are two integers with $d \neq 0$. We say that n is **divisible by** d , or that d is a **divisor of** n , if there exists an integer q such that $n = qd$. We will write $d|n$. We also say that n is a **multiple of** d . If d does not divide n , we write $d \nmid n$.

Example 1.9. Observe that $2|108$, or 108 is divisible by 2 since $108 = 2 \cdot 54$. But $5 \nmid 108$ since there is no integer q for which $108 = 5q$.

★ **NOTE:** $1|n$ and $n|0$, for all integers n .

$$r = 0 \Leftrightarrow d|n$$

THEOREM 1.10. Let m, n and d be integers.

- (1) $n|n$ for all n .

$$(2) \quad d|m, m|n \Rightarrow d|n.$$

$$(3) \quad d|n, n|d \Rightarrow d = \pm n.$$

$$(4) \quad d|n, d|m \Rightarrow d|(xn + ym), \text{ for all integers } x \text{ and } y.$$

PROOF. (1) It is clear that $n|n$ since $n = 1 \cdot n$.

(2) Assume that $d|m$, $m|n$, and let us show that $d|n$. Then there exist two integers q, q' such that $m = qd$ and $n = q'm$. From here we get that

$$n = q'm = q'(qd) = (q'q)d,$$

which proves that $d|n$, as desired.

(3) Assume that $d|n$ and $n|d$. Then we can find integers q and q' satisfying that $n = qd$ and $d = q'n$. From here we get that

$$n = qd = qq'n \Rightarrow n(1 - qq') = 0 \Rightarrow qq' = 1 \Rightarrow q = q' = 1 \text{ or } q = q' = -1.$$

Thus either $d = n$ or $d = -n$.

(4) Assume that $d|n$ and $d|m$, and take two integers x and y . Our assumptions allow us to find two integers q and q' such that $n = qd$ and $m = q'd$. Thus:

$$xn + ym = xqd + yq'd = (xq + yq')d,$$

which implies that $d|(xn + ym)$. □

Expressions of the form $xn + ym$, where x and y are integers, are called **linear combinations** of n and m .

Example 1.11. If $d \geq 1$ is such that $d|(3k + 5)$ and $d|(7k + 2)$ for some k , show that $d = 1$ or that $d = 29$.

Solution: Apply Theorem 1.10 (4) to conclude that d divides the linear combination $7(3k + 5) - 3(7k + 2) = 29$. From here, we can obtain that $d = 1$ or $d = 29$.

Definition 1.12. Let m and n be integers. We say that an integer d is a **common divisor of m and n** if $d|m$ and $d|n$.

Example 1.13. Consider the positive divisors of 36 and 84:

POSITIVE DIVISORS OF 36

1, 2, 3, 4, 6, 9, 12, 18, 36

POSITIVE DIVISORS OF 84

1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84

COMMON DIVISORS OF 36 AND 84

1, 2, 3, 4, 6, 12

★ **NOTE!** The **largest common divisor 12** is a **multiple** of all the other positive common divisors.

Definition 1.14. Let m and n be integers, not both zero. An integer d is called the **greatest common divisor of m and n** , written $d = \gcd(m, n)$ if it satisfies the following:

- (1) $d \geq 1$.
- (2) $d|m$ and $d|n$.
- (3) $k|m$ and $k|n \Rightarrow k|d$.

We say that m and n are **coprime** or **relatively prime** if $\gcd(m, n) = 1$.

For example, $\gcd(36, 84) = 12$ and $\gcd(-9, 15) = 3$. Note that 9 and 16 are coprime since $\gcd(9, 16) = 1$.

The following results will be useful when proving identities involving the gcd.

BÉZOUT'S LEMMA. Let m and n be two integers, not both zero. Then $d = \gcd(m, n)$ exists and $d = xm + yn$ for some integers x and y .

PROOF. Let $X = \{sm + tn \mid s, t \in \mathbb{Z} \text{ and } sm + tn \geq 1\}$. Note that X is non-empty, since $m^2 + n^2 \in X$. By the Well-Ordering Axiom we can find a least element in X , say d . Then $d \geq 1$ and $d = xm + yn$ for integers x and y ; if $k|m$ and $k|n$, then $k|(xm + yn) = d$ by Theorem 1.10 (4). This shows that any common divisor of m and n is a divisor of d . It remains to show that d is a common divisor of m and n . Here, we will show that $d|n$. One can show that $d|m$ similarly. Apply the *Division Algorithm* to find $q, r \in \mathbb{Z}$ such that $n = qd + r$ and $0 \leq r < d$. Then

$$r = n - qd = n - q(xm + yn) = (-qx)m + (1 - qy)n,$$

which says that r is a linear combination of m and n . If $r > 0$ we would have that $r \in X$, which contradicts the minimality of d in X . Thus $r = 0$. $\square$

Direct consequences of Bézout's Lemma are:

COROLLARY 1.15. *Two integers m and n are relatively prime if and only if there exist integers x and y such that $xm + yn = 1$. In particular, any two consecutive integers k and $k + 1$ are relatively prime.*

PROOF. If m and n are relatively prime, then the result follows by Bézout's Lemma. Suppose now that there exist integers x and y such that $xm + yn = 1$, and let $d = \gcd(m, n)$. Then an application of Theorem 1.10 (4) tells us that $d \mid (xm + yn)$. But $xm + yn = 1$, so $d \mid 1$ and since $d \geq 1$ (by the definition of the gcd) we have that $d = 1$, proving that m and n are relatively prime. $\square$

COROLLARY 1.16. *If $d = \gcd(m, n)$, then m/d and n/d are relatively prime.*

Caution! If $d \neq 1$ is such that $d = xm + yn$ for some integers x and y , we can not conclude that $d = \gcd(m, n)$. In other words, **the converse of Bézout's Lemma does not hold** in general. For example, $12 = 1 \cdot 4 + 1 \cdot 8$, and $\gcd(4, 8) = 4 \neq 12$.

THEOREM 1.17. *Let m and n be relatively prime integers. Then for any integer k*

$$(1) \quad m \mid k, \quad n \mid k \Rightarrow mn \mid k.$$

$$(2) \quad m \mid kn \Rightarrow m \mid k.$$

PROOF. By Bézout's Lemma, we have $1 = xm + yn$, for x and y integers.

(1) If $k = qm$ and $k = pn$ for some integers p and q , then

$$k = 1 \cdot k = (xm + yn)k = xmk + ynk = xm(pn) + yn(qm) = (xp + yq)mn.$$

Hence $mn \mid k$, proving (1).

(2) Let $nk = qm$, where q is an integer. Then

$$k = 1 \cdot k = (xm + yn)k = xmk + ynk = xmk + y(qm) = (xk + yq)m,$$

which shows that $m \mid k$, as desired. $\square$

It is well known that if the product mn of two integers m and n is even, then either m or n is even. This fact can be rephrased as “if $2 \mid mn$ then either $2 \mid m$ or $2 \mid n$ ”. As we will prove now, this statement indeed holds for any prime.

EUCLID'S LEMMA. Let p be a prime.

- (1) If $p|mn$ where m and n are integers, then either $p|m$ or $p|n$.
- (2) If $p|m_1m_2\ldots m_r$ where each m_i is an integer, then $p|m_i$ for some i .

PROOF. (1) Let $d = \gcd(m, p)$. Then $d|p$, which implies that either $d = 1$ or $d = p$ (because p is a prime). If $d = p$, then $p|m$; otherwise if $d = 1$, then m and p are coprime, and Theorem 1.17 (2) applies to get that $p|n$.

(2) follows by induction on r ; in fact, the base case $r = 2$ is (1). Suppose that (2) is true for $r - 1$, and let $p|m_1m_2\ldots m_{r-1}m_r$ (for $r > 2$). Then from (1) (or the base case of the induction) we obtain that either $p|m_1m_2\ldots m_{r-1}$ or $p|m_r$. If $p|m_r$, then we are done; otherwise if $p|m_1m_2\ldots m_{r-1}$, then applying the inductive hypothesis we obtain that $p|m_i$ for some $i \in \{1, \ldots, r-1\}$, concluding the proof. $\square$

★ **NOTE!** Euclid's Lemma fails for non primes! For example, 6 is a divisor of $3 \cdot 4$, but 6 does not divide 3 or 4.

1.2.3 Euclidean Algorithm

Euclid provided in his book *Elements* (written 300 BC) an algorithm (known nowadays as the Euclidean algorithm) to compute the gcd of two positive integers m and n . The key is the following result:

PROPOSITION 1.18. If $m = qn + r$ then $\gcd(m, n) = \gcd(n, r)$.

PROOF. Write $d = \gcd(m, n)$ and $k = \gcd(n, r)$. Then $k|(qn + r)$, i. e., $k|m$. Thus k is a common divisor of m and n , so $k|d$ since $d = \gcd(m, n)$. A similar argument (using the linear combination $r = -qn + m$) shows that $d|k$, so $d = \pm k$ by Theorem 1.10 (3). Thus $d = k$ because both d and k are positive. $\square$

Example 1.19. For $m = 140$ and $n = 21$, we have that $140 = 6 \cdot 21 + 14$, that is, $q = 6$ and $r = 14$. Proposition 1.18 tells us that $\gcd(140, 21) = \gcd(21, 14) = 7$.

EUCLIDEAN ALGORITHM. Let m and n be integers, not both zero. We apply the *Division Algorithm* repeatedly:

$$\begin{aligned} m &= q_1n + r_1 & \text{divisor: } n, & \text{remainder: } r_1 \\ n &= q_2r_1 + r_2 & \text{divisor: } r_1, & \text{remainder: } r_2 \\ r_1 &= q_3r_2 + r_3 \\ &\vdots & & \end{aligned}$$

At each stage we divide the divisor at the previous stage by the remainder, so the **remainders form a decreasing sequence of nonnegative integers**:

$$n > r_1 > r_2 > r_3 > \dots \geq 0.$$

It is clear that (in at most n steps) we will reach a remainder of 0. Assume that r_t is the last nonzero remainder, then the last two equations are:

$$r_{t-2} = q_tr_{t-1} + r_t, \quad \text{and} \quad r_{t-1} = q_{t+1}r_t + 0.$$

Now, Proposition 1.18 gives

$$\gcd(m, n) = \gcd(n, r_1) = \gcd(r_1, r_2) = \dots = \gcd(r_{t-1}, r_t) = r_t$$

Thus $\gcd(m, n)$ is the **LAST NONZERO REMAINDER**.

Moreover, we can **express** $\gcd(m, n) = r_t$ as a **linear combination of m and n** (finding the coefficients in *Bézout's Lemma*) by eliminating the remainders $r_{t-1}, r_{t-2}, \dots$, successively from these equations.

Example 1.20. Find $\gcd(268, 112)$ and express it as a linear combination of 268 and 112.

Solution: We will use the Euclidean Algorithm. To do so, we first apply the *Division Algorithm* repeatedly until we reach a zero remainder. In our case, we obtain the following:

$$268 = 2 \cdot 112 + 44$$

$$112 = 2 \cdot 44 + 24$$

$$44 = 1 \cdot 24 + 20$$

$$24 = 1 \cdot 20 + 4$$

$$20 = 5 \cdot 4 + 0.$$

Next we look at the last remainder before the zero remainder. Since it is 4,

the *Euclidean Algorithm* allows us to conclude that

$$\gcd(268, 112) = 4.$$

To write 4 as a linear combination of 112 and 268, we eliminate the remainders as follows:

$$268 = 2 \cdot 112 + 44 \quad \Rightarrow \quad 44 = 268 - 2 \cdot 112 \quad (1.1)$$

$$112 = 2 \cdot 44 + 24 \quad \Rightarrow \quad 24 = 112 - 2 \cdot 44 \quad (1.2)$$

$$44 = 1 \cdot 24 + 20 \quad \Rightarrow \quad 20 = 44 - 1 \cdot 24 \quad (1.3)$$

$$24 = 1 \cdot 20 + 4 \quad \Rightarrow \quad 4 = 24 - 1 \cdot 20 \quad (1.4)$$

$$20 = 5 \cdot 4 + 0.$$

Starting at the bottom, we can write the gcd as a linear combination of two larger and larger numbers:

$$\begin{aligned} 4 &\stackrel{(1.4)}{=} 24 - 20 \stackrel{(1.3)}{=} 24 - (44 - 24) = 2 \cdot 24 - 44 \stackrel{(1.2)}{=} 2(112 - 2 \cdot 44) - 44 = \\ &= 2 \cdot 112 - 5 \cdot 44 \stackrel{(1.1)}{=} 2 \cdot 112 - 5(268 - 2 \cdot 112) = 268 \cdot (-5) + 112 \cdot 12 \end{aligned}$$

1.2.4 Prime Factorization

PRIME FACTORIZATION THEOREM.

- (1) Every integer $n \geq 2$ is a product of (one or more) primes.
- (2) The factorization is unique up to the order of the factors.
That is, if $n = p_1 p_2 \dots p_r = q_1 q_2 \dots q_s$, where all the p_i and q_j are primes, then $r = s$ and the q_j can be relabelled so that $p_i = q_i$ for all $i = 1, 2, \dots, r$.

PROOF. (1) We have already proved it. (See Example 1.4.)

(2) Assume on the contrary that (2) fails. It means that we can find an integer ≥ 2 admitting two different factorizations into primes. Then by the Well-Ordering Axiom we can choose $m \geq 2$ to be the smallest integer admitting two different factorizations into primes:

$$m = p_1 p_2 \dots p_r = q_1 q_2 \dots q_s,$$

Then m is not a prime (otherwise, it would only admit a prime factorization; the trivial one) and so $r \geq 2$, $s \geq 2$. We have $p_1 | q_1 q_2 \dots q_s$, so $p_1 | q_j$ (for some j) by *Euclid's Lemma*. By relabelling the q_j , we may assume that $p_1 | q_1$, which yields that $p_1 = q_1$ because both are primes, so

$$\frac{m}{p_1} = p_2 \dots p_r = q_2 \dots q_s,$$

is an integer (smaller than m) that admits two distinct factorizations into primes. It contradicts the choice of m , and therefore proves (2). $\square$

For any integer $n \geq 2$, the *Prime Factorization Theorem* asserts that n can be written uniquely in the form:

$$n = p_1^{n_1} p_2^{n_2} \dots p_r^{n_r},$$

where the p_i are the distinct prime divisors of n , and $n_i \geq 1$ for each i . For example, $60 = 2^2 \cdot 3 \cdot 5$. If n has only one prime divisor, we call it a **prime power**. For instance, $9 = 3^2$, $8 = 2^3$. We say that n is **square free** if all its exponents are equal to 1, that is $n_i = 1$ for all i . Hence, any prime is square free, as are $6 = 2 \cdot 3$ and $70 = 2 \cdot 5 \cdot 7$.

Example 1.21. $504 = 2^3 \cdot 3^2 \cdot 7$, and this is the **only** way to factor 504 into primes.

The Prime Factorization Theorem provides a list of all positive divisors of an integer n when its prime factorization is known. For example, if $n = 12 = 2^2 \cdot 3$, these divisors are 1, 2, 3, 4, 6, 12, and they can be written as

$$\begin{aligned} 1 &= 2^0 \cdot 3^0, & 3 &= 2^0 \cdot 3^1 \\ 2 &= 2^1 \cdot 3^0, & 6 &= 2^1 \cdot 3^1 \\ 4 &= 2^2 \cdot 3^0, & 12 &= 2^2 \cdot 3^1 \end{aligned}$$

Thus they can all be expressed as $2^r \cdot 3^s$, where $0 \leq r \leq 2$ and $0 \leq s \leq 1$.

THEOREM 1.22. *Let n be an integer with prime factorization*

$$n = p_1^{n_1} p_2^{n_2} \dots p_r^{n_r},$$

where the p_i are all distinct primes and $n_i \geq 1$ for each i . Then the positive divisors of n are precisely the integers d of the following form:

$$d = p_1^{d_1} p_2^{d_2} \dots p_r^{d_r}, \text{ with } 0 \leq d_i \leq n_i \text{ for all } i.$$

PROOF. It follows by Euclid's Lemma and the Prime Factorization Theorem.

Fill in the details yourself. □

Definition 1.23. Let $n_1, n_2, \dots, n_r$ be positive integers. The **least common multiple** $\text{lcm}(n_1, n_2, \dots, n_r)$ of $n_1, \dots, n_r$ is the (positive) common multiple of $n_1, \dots, n_r$ that is a divisor of any other common multiple of $n_1, \dots, n_r$.

We can use the **Prime Factorization Theorem** to compute the gcd and the lcm. Let us see an example:

Example 1.24. Find $d = \text{gcd}(12, 20, 18)$ and $m = \text{lcm}(12, 20, 18)$.

Solution: We first compute the prime factorization of 12, 20 and 18:

$$12 = 2^2 \cdot 3^1 \cdot 5^0$$

$$20 = 2^2 \cdot 3^0 \cdot 5^1$$

$$18 = 2^1 \cdot 3^2 \cdot 5^0$$

To find d we consider the product of common primes (2 in this case) power to the minimum exponents (i.e. 1). Thus:

$$d = \text{gcd}(12, 20, 18) = 2^1 = 2$$

While to compute m we look at the product of all primes power to the maximum exponents, that is:

$$m = \text{lcm}(12, 20, 18) = 2^2 \cdot 3^2 \cdot 5^1 = 180$$

This method is collected in the following result.

THEOREM 1.25. Let $\{a, b, c, \dots\}$ be a finite set of positive integers, and let the factorization into primes

$$a = p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$$

$$b = p_1^{b_1} p_2^{b_2} \dots p_r^{b_r}$$

$$c = p_1^{c_1} p_2^{c_2} \dots p_r^{c_r}$$

$$\vdots = \quad \quad \quad \vdots$$

where an exponent is zero if such a prime does not occur. Then

$$\gcd(a, b, c) = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}, \quad \text{lcm}(a, b, c) = p_1^{m_1} p_2^{m_2} \dots p_r^{m_r},$$

where $k_i = \min(a_i, b_i, c_i, \dots)$, $m_i = \max(a_i, b_i, c_i, \dots)$ for each i .

COROLLARY 1.26. Let a, b be positive integers. Then

$$\text{lcm}(a, b) \cdot \gcd(a, b) = ab.$$

★ In summary to **compute** $\gcd(a, b)$ we have seen **two methods**:

METHOD 1: Using the Factorization into Primes (see Theorem 1.25).

METHOD 2: Applying the Euclidean Algorithm.

If we find $\gcd(a, b)$ by using Method 1, the corollary above can be used to **compute** $\text{lcm}(a, b)$.

At this point it is worth to mention that in general

$$\text{lcm}(a, b, c) \cdot \gcd(a, b, c) \neq abc.$$

We close this section with another application of the Prime Factorization Theorem.

EUCLID'S THEOREM. There are infinitely many primes.

PROOF. Suppose, on the contrary, that there are only n primes: $p_1, p_2, \dots, p_n$. Consider the integer $m = 1 + p_1 p_2 \dots p_n \geq 2$. The first part of the Prime Factorization Theorem allows us to conclude that some prime must divide m , say p_i . From $p_i | m$ and $p_i | p_1 p_2 \dots p_n$, we get that $p_i | (m - p_1 p_2 \dots p_n)$, that is, $p_i | 1$. But this is impossible, and so there are infinitely many primes. $\square$

1.3 Integers modulo n

In this section, we will introduce a very important equivalence relation on $\mathbb{Z}$, called **congruence** modulo (an integer) n , and study its equivalence classes.

Recall that two integers a and b have the **same parity** if both are even or both are odd, that is, $2 | (a - b)$. Congruence modulo n generalizes this situation, letting any integer n play the role of 2.

Definition 1.27. Let $n \geq 2$, a and b be integers. We say that a and b are **congruent modulo n** if $n|(a - b)$. In such a case, we write $a \equiv b \pmod{n}$ and refer to n as the **modulus**.

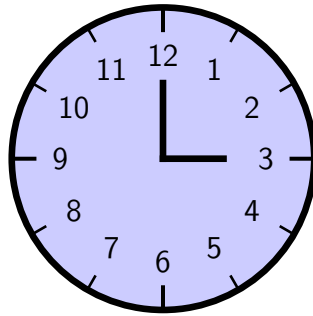
Remark 1.28. We have taken $n \geq 2$, since the cases $n = 0$ and $n = 1$ are not interesting. The congruence $a \equiv b \pmod{1}$ is always true, and

$$a \equiv b \pmod{0} \Leftrightarrow 0|(a - b) \Leftrightarrow a = b$$

Example 1.29. $2 \equiv 5 \pmod{3}$, $21 \equiv 16 \pmod{5}$, $-4 \equiv 2 \pmod{6}$.

Example 1.30. Let a and b be two positive integers, then $a \equiv b \pmod{10}$ if and only if they have the same last digit, since $10|(b - a)$. In general, $a \equiv b \pmod{10^n}$ if and only if they have same last n digits. For instance, $533 \equiv 1433 \pmod{100}$.

Remark 1.31. Since congruences will play an important role in your algebra journey, at this point it is worth stopping for a moment and think what we are indeed doing. So, I want you to imagine a wall clock that has struck three, like in this figure:



What will the time be in four hours? and in seven hours? Obviously, the answers will be $3 + 4 = 7$ and $3 + 7 = 10$, respectively. But be careful here, you all know that once we reach to 12, we will be back at 1. In other words, on a clock when the numbers get to 12, they start again at 0. So, in ten hours, the time will be 1 (which is not $3 + 10 = 13$); the same will happen after 22 hours (it will be 1 instead of 25). Note that $13 \equiv 1 \pmod{12}$ and $25 \equiv 1 \pmod{12}$. What you have been doing is computing the remainder when divided by 12:

$$13 = 1 \cdot 12 + 1, \quad 25 = 2 \cdot 12 + 1.$$

We can view congruence modulo n like a clock with n numbers on it instead of 12. Then two integers will be congruent modulo n if they end up in the same place on the clock.

THEOREM 1.32. *Congruence modulo n is an equivalence relation on $\mathbb{Z}$:*

- (a) $a \equiv a \pmod{n}$ for every integer a .
- (b) $a \equiv b \pmod{n}$ implies $b \equiv a \pmod{n}$.
- (c) $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ implies $a \equiv c \pmod{n}$.

PROOF. (a) is clear since $n|(a - a) = 0$.

(b) If $a \equiv b \pmod{n}$, then $n|(a - b)$. From here we derive that $n|(b - a)$, that is, $b \equiv a \pmod{n}$.

(c) Suppose that $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. This means that $n|(a - b)$ and $n|(b - c)$. Applying Theorem 1.10 (4) we obtain that $n|((a - b) + (-1)(b - c))$, that is, $n|(a - c)$, showing that $a \equiv c \pmod{n}$, as required. $\square$

Definition 1.33. The equivalence class $[a]_n$ or $[a]$ of an integer a with respect to congruence modulo n is called its **residue class modulo n** .

$$[a]_n = \{x \in \mathbb{Z} \mid x \equiv a \pmod{n}\}.$$

We say that a is a **generator of $[a]_n$** . Let us try to rephrase the sentence “being congruent to a modulo n ” in a more simple way:

$$x \in [a]_n \Leftrightarrow x \equiv a \pmod{n} \Leftrightarrow n|(x - a) \Leftrightarrow x - a = qn, \text{ for } q \in \mathbb{Z}.$$

Notice that we have proved that $x \in [a]_n$ if and only if the remainder when dividing x by n is a . In other words, $[a]_n$ consists on the integers whose remainder is a when divided by n .

Two residue classes $[a]_n$ and $[b]_n$ are equal if and only if $a \equiv b \pmod{n}$. In fact, if $a \equiv b \pmod{n}$ and $x \in [a]_n$, then $x \equiv a \pmod{n}$. Using that congruence modulo n is an equivalence relation we obtain that $x \equiv b \pmod{n}$, which means that $x \in [b]_n$. This proves that $[a]_n \subseteq [b]_n$; the other containment can be shown in a similar way. The converse trivially holds.

The set consisting of all the residue classes modulo n is denoted by $\mathbb{Z}_n$, and is called the **set of integers modulo n** .

THEOREM 1.34. *Let $n \geq 2$ be an integer.*

- (1) *If $a \in \mathbb{Z}$, then $[a]_n = [r]_n$ for some $0 \leq r \leq n - 1$.*
- (2) *The residue classes $[0]_n, [1]_n, \dots, [n - 1]_n$ are all distinct.*

PROOF. (1) For $a \in \mathbb{Z}$, apply the *Division Algorithm* to get $a = qn + r$, where $0 \leq r \leq n - 1$. From here we derive $n|(a - r)$, and so $a \equiv r \pmod{n}$, which is equivalent to say that $[a]_n = [r]_n$.

(2) Suppose that $[r]_n = [s]_n$ for some $0 \leq r, s \leq n - 1$. We can assume (without loss of generality) that $r \leq s$. Now $[r]_n = [s]_n$ implies that $n|(s - r)$, or that $s - r$ is a multiple of n . But $0 \leq s - r \leq n - 1$ yields $r = s$. $\square$

COROLLARY 1.35. $\mathbb{Z}_n$ has n elements:

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n - 1]_n\}$$

Example 1.36. $\mathbb{Z}_2$ has two elements, which are the residue classes of the remainders obtained when divided by 2. There only two possible remainders, namely, 0 and 1. So, $\mathbb{Z}_2 = \{[0]_2, [1]_2\}$, where

$$\begin{aligned} [0]_2 &= \{x \in \mathbb{Z} \mid x \equiv 0 \pmod{2}\} = \text{integers with remainder 0 when divided by 2} \\ &= \text{the set of even integers,} \\ [1]_2 &= \{x \in \mathbb{Z} \mid x \equiv 1 \pmod{2}\} = \text{integers with remainder 1 when divided by 2} \\ &= \text{the set of odd integers} \end{aligned}$$

Example 1.37. Find $[48]$ and $[-16]$ in $\mathbb{Z}_7$.

Solution: We know that $\mathbb{Z}_7 = \{[0], [1], \dots, [6]\}$. So, the question here is the following: What of such elements equals $[48]$ (respectively, $[-16]$)? Keeping in mind the proof of Theorem 1.34 (1), all we need to do is to find the remainder when divided 48 (respectively, -16) by 7:

$$\begin{aligned} 48 &= 6 \cdot 7 + 6 \Rightarrow 48 \equiv 6 \pmod{7}, \\ -16 &= (-3)7 + 5 \Rightarrow -16 \equiv 5 \pmod{7}, \end{aligned}$$

Hence: $[48] = [6]$ and $[-16] = [5]$ in $\mathbb{Z}_7$.

Pay special attention to Theorem 1.34 (1). Its proof is what we call in Mathematics constructive, since it provides a method to find elements in $\mathbb{Z}_n$, as we have seen in Example 1.37; which, by the way, it is important.

1.3.1 Modular arithmetic

Let $n \geq 2$ be an integer. We can define an addition (respectively, a multiplication) on the set of integers $\mathbb{Z}_n$ modulo n by adding (respectively, multiplying) the generators of the corresponding residue classes. More precisely, for $[a], [b] \in \mathbb{Z}_n$ we define:

$$[a] + [b] = [a + b] \quad \text{and} \quad [a] \cdot [b] = [ab].$$

For instance, $[3]_6 + [5]_6 = [3 + 5]_6 = [8]_6 = [2]_6$, since $8 \equiv 2 \pmod{6}$. Before studying the properties of these two new operations, we need to check that they do not depend on the generator of the residue classes. For example, $[9]_6 = [3]_6$, so $[9]_6 + [5]_6$ should produce the same result that $[3]_6 + [5]_6$, that is, $[2]_6$. Let us check this out:

$$[9]_6 + [5]_6 = [9 + 5]_6 = [14]_6 = [2]_6, \text{ since } 14 \equiv 2 \pmod{6}.$$

We prove this fact, in a more general way, in the next result.

THEOREM 1.38. *Let $n \geq 2$ be an integer, and $a_i, b_i \in \mathbb{Z}$ (for $i = 1, 2$) such that $a_1 \equiv a_2 \pmod{n}$ and $b_1 \equiv b_2 \pmod{n}$. Then:*

- (1) $a_1 + b_1 \equiv a_2 + b_2 \pmod{n}$,
- (2) $a_1 b_1 \equiv a_2 b_2 \pmod{n}$,
- (3) $a_1^m \equiv a_2^m \pmod{n}$, for all $m \geq 1$.

PROOF. (1) From $a_1 \equiv a_2 \pmod{n}$ and $b_1 \equiv b_2 \pmod{n}$, we get that $n \mid (a_1 - a_2)$ and $n \mid (b_1 - b_2)$. An application of Theorem 1.10 (4), we obtain that

$$n \mid (a_1 - a_2 + (b_1 - b_2)) \Rightarrow n \mid ((a_1 + b_1) - (a_2 + b_2)),$$

which implies $a_1 + b_1 \equiv a_2 + b_2 \pmod{n}$, as required.

(2) Applying Theorem 1.10 (4) we obtain that

$$n \mid ((a_1 - a_2)b_1 + a_2(b_1 - b_2)) = a_1 b_1 - a_2 b_2,$$

which implies that $a_1 b_1 \equiv a_2 b_2 \pmod{n}$, as desired.

Notice that (3) follows from (2). □

Notation 1.39. From now on, to simplify our calculations, we will denote the residue class $[a]_n$ simply by $\bar{a}$.

Example 1.40. Below you could find the addition and multiplication tables of $\mathbb{Z}_6$. We will talk more about these tables (usually called *Cayley tables*) when we introduce groups in Chapter 3.

+	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$
$\bar{5}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$

$\times$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$
$\bar{1}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{2}$	$\bar{0}$	$\bar{2}$	$\bar{4}$	$\bar{0}$	$\bar{2}$	$\bar{4}$
$\bar{3}$	$\bar{0}$	$\bar{3}$	$\bar{0}$	$\bar{3}$	$\bar{0}$	$\bar{3}$
$\bar{4}$	$\bar{0}$	$\bar{4}$	$\bar{2}$	$\bar{0}$	$\bar{4}$	$\bar{2}$
$\bar{5}$	$\bar{0}$	$\bar{5}$	$\bar{4}$	$\bar{3}$	$\bar{2}$	$\bar{1}$

Example 1.41. Compute the remainder when 4^{119} is divided by 7.

Solution: Here, we are going to learn a way to solve this problem, without having to use a calculator. Suppose for a moment we have found $0 \leq r \leq 6$ satisfying that $\overline{4^{119}} = \bar{4}^{119} = \bar{r}$ in $\mathbb{Z}_7$, or $[4^{119}]_7 = [r]_7$ using our “old” notation. Then, in particular, $4^{119} \in [r]_7$, which means that the remainder of 4^{119} when divided by 7 is r . Therefore, our goal is to find such r .

To do so, we first find $n \geq 1$ such that $\bar{4}^n = \bar{1}$ in $\mathbb{Z}_7$. We will justify the existence of such n later on. By now, let us just find it:

$$\bar{4}^2 = \overline{16} = \bar{2}, \text{ since } 16 \equiv 2 \pmod{7},$$

$$\bar{4}^3 = \bar{4}^2 \cdot \bar{4} = \bar{2} \cdot \bar{4} = \bar{8} = \bar{1}, \text{ since } 8 \equiv 1 \pmod{7}.$$

So, $n = 3$ satisfies that $\bar{4}^3 = \bar{1}$ in $\mathbb{Z}_7$. Here is the trick now: we apply the *Division Algorithm* with 119 and 3 to obtain that $119 = 3 \cdot 39 + 2$. Lastly, using how modular operations work we have that

$$\bar{4}^{119} = \bar{4}^{3 \cdot 39 + 2} = (\bar{4}^3)^{39} \cdot \bar{4}^2 = \bar{2}$$

Hence, the remainder when divided 4^{119} by 7 is 2.

Some properties of the modular operations (that is, the addition and multiplication in $\mathbb{Z}_n$) are collected in the following result. Please give its proof a try, and let me know whether you encounter any problems.

THEOREM 1.42. *Let $n \geq 2$ be a fixed modulus and $a, b, c \in \mathbb{Z}$. Then the following hold in $\mathbb{Z}_n$.*

- (1) $\bar{a} + \bar{b} = \overline{b + a}$ and $\bar{a}\bar{b} = \overline{ba}$.
- (2) $\bar{a} + (\bar{b} + \bar{c}) = (\bar{a} + \bar{b}) + \bar{c}$ and $\bar{a}(\bar{b}\bar{c}) = (\bar{a}\bar{b})\bar{c}$.
- (3) $\bar{a} + \bar{0} = \bar{a}$ and $\bar{a}\bar{1} = \bar{a}$. We call $\bar{0}$ the **zero** of $\mathbb{Z}_n$, while $\bar{1}$ is said to be the **unity** of $\mathbb{Z}_n$.
- (4) $\bar{a} + \overline{-a} = \bar{0}$. We usually write $-\bar{a}$ to denote $\overline{-a}$.
- (5) $\bar{a}(\bar{b} + \bar{c}) = \bar{a}\bar{b} + \bar{a}\bar{c}$.

Remark 1.43. In the previous result, we have highlighted some of the properties of the modular operations, which should be familiar to you, since the addition and multiplication in $\mathbb{Z}$ also have them. One could say that this was pretty normal. But, not everything will be “that normal” unfortunately.

There are **several differences** between the **arithmetic** of $\mathbb{Z}_n$ and that of $\mathbb{Z}$.

For example:

- 0 and 1 are the only integers satisfying that $k^2 = k$, while in $\mathbb{Z}_6$ the elements $\bar{3}$ and $\bar{4}$ have that property:

$$(\bar{3})^2 = \bar{9} = \bar{3}, \text{ since } 9 \equiv 3 \pmod{6},$$

$$(\bar{4})^2 = \overline{16} = \bar{4}, \text{ since } 16 \equiv 4 \pmod{6},$$

- **the cancellation law holds in $\mathbb{Z}$:** if $ab = ac$ and $a \neq 0$, then $b = c$. In general, **we do not have cancellation in $\mathbb{Z}_n$** . For instance, in $\mathbb{Z}_6$ we have that $\bar{4} \cdot \bar{2} = \bar{4} \cdot \bar{5}$, where $\bar{4} \neq \bar{0}$ and $\bar{2} \neq \bar{5}$.
- If $ab = 0$ in $\mathbb{Z}$, then either $a = 0$ or $b = 0$. This need not hold in $\mathbb{Z}_n$. For instance, in $\mathbb{Z}_6$ we have that $\bar{2} \cdot \bar{3} = \bar{0}$, and $\bar{2} \neq \bar{0}$, $\bar{3} \neq \bar{0}$.

1.3.2 Solving linear equations and systems in $\mathbb{Z}_n$

Now we will learn how to solve linear equations and systems in $\mathbb{Z}_n$. Let us by noticing the following:

$$\bar{a}x = \bar{c} \text{ in } \mathbb{Z}_n \iff ax \equiv c \pmod{n}$$

$$\left. \begin{array}{l} \bar{a}x + \bar{b}y = \bar{r} \\ \bar{c}x + \bar{d}y = \bar{s} \end{array} \right\} \text{ in } \mathbb{Z}_n \Leftrightarrow \begin{cases} ax + by \equiv r \pmod{n} \\ cx + dy \equiv s \pmod{n} \end{cases}$$

$$\left. \begin{array}{l} \bar{x} = \bar{r} \text{ in } \mathbb{Z}_n \\ \bar{x} = \bar{s} \text{ in } \mathbb{Z}_m \end{array} \right\} \Leftrightarrow \begin{cases} x \equiv r \pmod{n} \\ x \equiv s \pmod{m} \end{cases}$$

We start by introducing a notion that will be very useful for our purposes.

Definition 1.44. Let $n \geq 2$ and a, b be integers. A **residue class** $\bar{b} \in \mathbb{Z}_n$ is called an **inverse** of $\bar{a}$ if $\bar{a}\bar{b} = \bar{b}\bar{a} = \bar{1}$ in $\mathbb{Z}_n$.

The following result provides a necessary and sufficient condition for an element of $\mathbb{Z}_n$ to have an inverse in $\mathbb{Z}_n$. The proof is constructive, which means that it provides a method to find such an inverse.

THEOREM 1.45. Let a and n be integers with $n \geq 2$. Then $\bar{a}$ has an inverse in $\mathbb{Z}_n$ if and only if a and n are relatively prime. Moreover, the inverse of $\bar{a}$ in $\mathbb{Z}_n$ is the residue class of the coefficient of a in Bézout's Lemma.

PROOF. We have that

$$\bar{b}\bar{a} = \bar{1} \text{ in } \mathbb{Z}_n \Leftrightarrow ba \equiv 1 \pmod{n} \Leftrightarrow 1 - ba = qn \Leftrightarrow 1 = ba + qn,$$

for some integer q , which is equivalent to say that $\gcd(a, n) = 1$. $\square$

Example 1.46. Find the inverse of $\overline{16}$ in $\mathbb{Z}_{35}$.

Solution: Let us start by noticing that 16 and 35 are relatively prime, since $\gcd(16, 35) = 1$. So, $\overline{16}$ has an inverse in $\mathbb{Z}_{35}$ by Theorem 1.45. Following the proof of Theorem 1.45, we need to find the coefficient of 16 in Bézout's Lemma. We know (Bézout's Lemma) that there are integers b and c such that $b \cdot 16 + c \cdot 35 = 1$. Keeping in mind that $\overline{35} = \bar{0}$ in $\mathbb{Z}_{35}$ we have that

$$\overline{b \cdot 16 + c \cdot 35} = \bar{b} \cdot \overline{16} + \bar{c} \cdot \overline{35} = \bar{b} \cdot \overline{16} = \bar{1},$$

that is, the inverse of $\overline{16}$ in $\mathbb{Z}_{35}$ is the residue class of b in $\mathbb{Z}_{35}$. In order to find b , we apply the Euclidean Algorithm:

$$35 = 2 \cdot 16 + 3, \quad 16 = 5 \cdot 3 + 1,$$

which implies that

$$1 = 16 - 5 \cdot 3 = 16 - 5 \cdot (35 - 2 \cdot 16) = 11 \cdot 16 + (-5) \cdot 35$$

We can then conclude that $\overline{11}$ is the inverse of $\overline{16}$ in $\mathbb{Z}_{35}$. In fact:

$$\overline{11} \cdot \overline{16} = \overline{11 \cdot 16} = \overline{176} = \overline{1}$$

in $\mathbb{Z}_{35}$, since $176 \equiv 1 \pmod{35}$ because $176 - 1 = 175$ and $35|175$.

Example 1.47. Find the elements in $\mathbb{Z}_9$ that have inverses.

Solution: We know that

$$\mathbb{Z}_9 = \{\overline{0}, \overline{1}, \overline{2}, \overline{3}, \overline{4}, \overline{5}, \overline{6}, \overline{7}, \overline{8}\}$$

Theorem 1.45 tells us that the elements in $\mathbb{Z}_9$ with inverses are the residue classes $\bar{r}$ such that r is coprime to $9 = 3^2$, that is, 3 does not divide r . Thus:

$$\overline{1}, \overline{2}, \overline{4}, \overline{5}, \overline{7}, \overline{8}$$

are the only elements in $\mathbb{Z}_9$ having inverses. As an exercise, check that $\overline{1}$ and $\overline{8}$ are both self-inverse ($\overline{1} \cdot \overline{1} = \overline{1}$ and $\overline{8} \cdot \overline{8} = \overline{1}$), $\overline{2}$ and $\overline{5}$ are inverses of each other ($\overline{2} \cdot \overline{5} = \overline{1}$) as are $\overline{4}$ and $\overline{7}$ ($\overline{4} \cdot \overline{7} = \overline{1}$).

In what follows we find necessary and sufficient condition to be imposed on n to guarantee that every nonzero element in $\mathbb{Z}_n$ has an inverse.

THEOREM 1.48. *The following are equivalent for an integer $n \geq 2$.*

- (1) *Every element $\bar{a} \neq \bar{0}$ in $\mathbb{Z}_n$ has an inverse.*
- (2) *If $\bar{a}\bar{b} = \bar{0}$ in $\mathbb{Z}_n$, then either $\bar{a} = \bar{0}$ or $\bar{b} = \bar{0}$.*
- (3) *n is a prime.*

PROOF. (1) $\Rightarrow$ (2). Assume that $\bar{a}\bar{b} = \bar{0}$ in $\mathbb{Z}_n$. If $\bar{a} = \bar{0}$, then we are done. So, assume that $\bar{a} \neq \bar{0}$ in $\mathbb{Z}_n$. Then by (1) $\bar{a}$ has an inverse, say $\bar{c}$. Thus:

$$\bar{a}\bar{b} = \bar{0} \Rightarrow \bar{c}(\bar{a}\bar{b}) = \bar{0} \Rightarrow (\bar{c}\bar{a})\bar{b} = \overline{1} \cdot \bar{b} = \bar{b} = \bar{0},$$

which implies $\bar{b} = \bar{0}$ and (2) follows.

(2) $\Rightarrow$ (3). Suppose on the contrary that n is not prime. Then $n = ab$ for some $2 \leq a, b < n$. But then

$$\bar{a}\bar{b} = \overline{ab} = \bar{n} = \bar{0},$$

where $\bar{a}, \bar{b} \neq \bar{0}$. This contradicts (2), and so n is a prime.

(3) $\Rightarrow$ (1). Take $\bar{a} \neq \bar{0}$ in $\mathbb{Z}_n$. Then $\gcd(a, n) = 1$ since n is prime and $\bar{a} \neq \bar{0}$. Theorem 1.45 finishes the proof. $\square$

Let us come back now to the equation $\bar{a}x = \bar{c}$ in $\mathbb{Z}_n$. Clearly, if $\bar{a}$ has an inverse in $\mathbb{Z}_n$, then the equation $\bar{a}x = \bar{c}$ has a solution in $\mathbb{Z}_n$. Applying Theorem 1.45, we can then conclude that if $\gcd(a, n) = 1$ a solution exists. Moreover, $x = \bar{b}\bar{c}$ is a solution, where $\bar{b}$ is the inverse of $\bar{a}$ in $\mathbb{Z}_n$. At this point a natural question arises: *does the equation $\bar{a}x = \bar{c}$ have a unique solution?* See the next result for the answer!

THEOREM 1.49. *Let a and n be integers with $n \geq 2$. If a and n are relatively prime, then for every integer c the equation $\bar{a}x = \bar{c}$ can be solved for x (in $\mathbb{Z}_n$). In fact, $x = \bar{b}\bar{c}$, where $\bar{b}\bar{a} = \bar{1}$ in $\mathbb{Z}_n$. Moreover, any two solutions are congruent modulo n .*

PROOF. The first part follows from Theorem 1.45. Rewrite $\bar{a}x = \bar{c}$ as a congruence equation $ax \equiv c \pmod{n}$, and suppose that x_1 and x_2 are two solutions:

$$ax_1 \equiv c \pmod{n}, \quad ax_2 \equiv c \pmod{n}.$$

By the symmetry and transitivity properties of congruences we get that $ax_1 \equiv ax_2 \pmod{n}$, which says that $n|a(x_1 - x_2)$, and so $n|(x_1 - x_2)$, since $\gcd(a, n) = 1$. Thus $x_1 \equiv x_2 \pmod{n}$, as desired. $\square$

COROLLARY 1.50. *If p is prime, then the congruence $ax \equiv c \pmod{p}$ is always solvable provided that a is not divisible by p .*

Example 1.51. Solve the equation $\bar{16}x = \bar{9}$ in $\mathbb{Z}_{35}$.

Solution: From Example 1.46 we know that $\bar{11}$ is the inverse of $\bar{16}$ in $\mathbb{Z}_{35}$. Multiplying both sides of the given equation by $\bar{11}$ we obtain

$$x = \bar{11}\bar{9} = \bar{99} = \bar{29}, \quad \text{in } \mathbb{Z}_{35},$$

since 29 is the remainder obtained when divided 99 by 35: $99 = 2 \cdot 35 + 29$.

Example 1.52. Solve the following system of equations in $\mathbb{Z}_{11}$:

$$\bar{5}x + \bar{8}y = \bar{2}$$

$$\bar{3}x + \bar{2}y = \bar{1}$$

Solution: We proceed like usual, and perform some algebraic manipulations to eliminate with one the variables. We begin by multiplying the second equation by $\bar{4}$:

$$\bar{3}x + \bar{2}y = \bar{1} \xrightarrow{\times \bar{4}} \bar{12}x + \bar{8}y = \bar{4} \Leftrightarrow x + \bar{8}y = \bar{4}.$$

Next, subtract the resulting equation from the first equation to obtain that

$$\bar{4}x = -\bar{2} = \bar{9} \tag{1.5}$$

It remains to compute the inverse of $\bar{4}$ in $\mathbb{Z}_{11}$. Note that such an inverse exists since 4 and 11 are relatively prime. We follow the process explained in the proof of Theorem 1.45, and apply the Euclidean Algorithm.

$$11 = 2 \cdot 4 + 3, \quad 4 = 1 \cdot 3 + 1 \Rightarrow 1 = 4 - 1 \cdot 3 = 4 - 1 \cdot (11 - 2 \cdot 4) = 3 \cdot 4 + (-1) \cdot 11$$

We have obtained that $3 \cdot 4 + (-1) \cdot 11 = 1$. Thus the inverse of $\bar{4}$ in $\mathbb{Z}_{11}$ is $\bar{3}$. Coming back to (1.5) we have that

$$x = \bar{3} \cdot \bar{9} = \bar{27} = \bar{5} \text{ in } \mathbb{Z}_{11}.$$

To find y , we substitute $x = \bar{5}$ in the equation $\bar{3}x + \bar{2}y = \bar{1}$:

$$\bar{2}y = \bar{1} - \bar{3}x = \bar{1} - \bar{3} \cdot \bar{5} = \bar{1} - \bar{15} = -\bar{14} = \bar{8} \tag{1.6}$$

Multiplying both sides of (1.6) by $\bar{6}$ (the inverse of $\bar{2}$ in $\mathbb{Z}_{11}$) we get that

$$y = \bar{6} \cdot \bar{8} = \bar{48} = \bar{4}.$$

The problem of solving our next congruence equations is known as the *Classical Chinese Remainder Theorem*. We first need a preliminary result:

LEMMA 1.53. Let a, b, c be integers such that a, b are relatively prime, and $a|c, b|c$. Then $ab|c$.

PROOF. By Bézout's Lemma we can find $s, t \in \mathbb{Z}$ such that $as + bt = 1$. On the other hand, from $a|c, b|c$ we have $c = ak_1 = bk_2$ for some $k_1, k_2 \in \mathbb{Z}$. The next trick is quite useful, so pay attention. We will use that $c = 1 \cdot c$ and then replace 1 by $as + bt$:

$$\begin{aligned} c &= 1 \cdot c = (as + bt)c = asc + btc = (as)(bk_2) + (bt)(ak_1) = \\ &= (ab)(sk_2 + tk_1), \end{aligned}$$

which implies that $c|ab$, as desired. $\square$

CLASSICAL CHINESE REMAINDER THEOREM. Let m, n, s, t be integers with $m \geq 2$ and $n \geq 2$. Assume that m and n are **relatively prime**. Then the simultaneous congruences:

$$\begin{aligned} x &\equiv s \pmod{m} \\ x &\equiv t \pmod{n} \end{aligned}$$

have a common solution. Moreover, any two solutions are congruent modulo mn .

PROOF. Notice that every solution of the congruence equation $x \equiv s \pmod{m}$ has the form $x = s + km$, for some integer k . Thus, all we need to do is to show that the congruence equation $s + km \equiv t \pmod{n}$, or equivalently, $km \equiv t - s \pmod{n}$ has a solution. But this follows from Theorem 1.49 since $\gcd(m, n) = 1$. We have proved that the congruence equations $x \equiv s \pmod{m}$ and $x \equiv t \pmod{n}$ have a common solution. It remains to show that any two solutions are congruent modulo mn . Suppose that both x and y are common solutions. Then:

$$\begin{aligned} x &\equiv s \pmod{m} \text{ and } y \equiv s \pmod{m} &\Rightarrow x &\equiv y \pmod{m}, \\ x &\equiv t \pmod{n} \text{ and } y \equiv t \pmod{n} &\Rightarrow x &\equiv y \pmod{n}, \end{aligned}$$

which imply that both m and n divide $x - y$. Applying Lemma 1.53 with $a = m, b = n$ and $c = x - y$, we obtain that $mn|(x - y)$, and so $x \equiv y \pmod{mn}$, finishing the proof. $\square$

Example 1.54. Find all the solution to the simultaneous congruences:

$$x \equiv 7 \pmod{8}$$

$$x \equiv 11 \pmod{15}$$

Solution: Let us first notice that $\gcd(8, 15) = 1$. So, we can proceed like in the proof of the Classical Chinese Remainder Theorem. We look at the solutions of the congruence $x \equiv 7 \pmod{8}$. Each of its solutions has the form $x = 7 + 8k$, for some integer k . Substituting this into the equation $x \equiv 11 \pmod{15}$ we get:

$$7 + 8k \equiv 11 \pmod{15} \Rightarrow 8k \equiv 4 \pmod{15}$$

We need to find the inverse of $\bar{8}$ in $\mathbb{Z}_{15}$. To do so, we need to find the coefficient of 8 in Bézout's Lemma. From $2 \cdot 8 + (-1) \cdot 15 = 1$, we get that

$$\bar{1} = \overline{2 \cdot 8 + (-1) \cdot 15} = \bar{2} \cdot \bar{8} + \overline{-115} = \bar{2} \cdot \bar{8}, \text{ in } \mathbb{Z}_{15},$$

which says that the inverse of $\bar{8}$ in $\mathbb{Z}_{15}$ is $\bar{2}$. From here, we obtain that

$$k \equiv 16k \equiv 8 \pmod{15}.$$

Hence $x = 7 + 8 \cdot 8 = 71$ is a solution. The Chinese Remainder Theorem applies to get that every solution has the form $71 + 120n$, for $n \in \mathbb{Z}$.

Example 1.55. Find all the solutions to the simultaneous congruences:

$$x \equiv 2 \pmod{5}$$

$$3x \equiv 5 \pmod{13}$$

Solution: This example might appear a bit different at first because the presence of coefficient different from 1 in the second equation, although we will solve it in the same way. Since $\gcd(5, 13) = 1$, the Chinese Remainder Theorem tells us that there is a common solution. We look at the solutions of the congruence equation $x \equiv 2 \pmod{5}$, which have the form $x = 5k + 2$, for some integer k . Substituting $x = 5k + 2$ into $3x \equiv 5 \pmod{13}$, we obtain the following congruence equation:

$$3(5k + 2) \equiv 5 \pmod{13} \Rightarrow 15k + 6 \equiv 5 \pmod{13} \Rightarrow 2k \equiv -1 \pmod{13}.$$

We need to find the inverse of $\bar{2}$ in $\mathbb{Z}_{13}$, which exists because $\gcd(2, 13) = 1$. Since $1 = 7 \cdot 2 + (-1) \cdot 13$, we have that the inverse of $\bar{2}$ in $\mathbb{Z}_{13}$ is $\bar{7}$. Multiplying the congruence equation $2k \equiv -1 \equiv 12 \pmod{13}$ by 7 we obtain

$$14k \equiv -7 \pmod{13} \xrightarrow{14 \equiv 1 \pmod{13}} k \equiv -7 \pmod{13} \xrightarrow{-7 \equiv 6 \pmod{13}} k \equiv 6 \pmod{13}.$$

Thus, a common solution is $x = 5 \cdot 6 + 2 = 32$. Moreover, any other solution has to be congruent to 32 modulo $5 \cdot 13 = 65$. In other words, all the common solutions are the elements of $\overline{32}$ in $\mathbb{Z}_{65}$, that is:

$$\dots, -98, -33, 32, 97, 162, \dots$$

The following result, sometimes called *Fermat's Little Theorem*, is due to Pierre Fermat. It is very important in Number Theory and in Computer Science, since it provides a useful (an efficient) way to test whether an integer p is not a prime (check whether $10^{p-1} \not\equiv 1$).

FERMAT'S THEOREM. Let p be a prime. Then

$$a^{p-1} \equiv 1 \pmod{p}, \quad \text{for every } a \not\equiv 0 \pmod{p}$$

PROOF. Let a be an integer such that $a \not\equiv 0 \pmod{p}$, or equivalently, $\bar{a} \neq \bar{0}$ in $\mathbb{Z}_p$. By Theorem 1.48 the element $\bar{a}$ has an inverse $\bar{b}$ in $\mathbb{Z}_p$. Multiplying all the nonzero elements in $\mathbb{Z}_p$ by $\bar{a}$ we obtain

$$\bar{a}\bar{1}, \bar{a}\bar{2}, \dots, \overline{ap-1}.$$

All the elements above are nonzero and distinct:

$$\bar{a}\bar{r} = \bar{a}\bar{s} \xrightarrow{\times \bar{b}} \bar{r} = \bar{s}$$

Since $\{\bar{a}\bar{1}, \bar{a}\bar{2}, \dots, \overline{ap-1}\} \subseteq \mathbb{Z}_p \setminus \{\bar{0}\}$ and $|\{\bar{a}\bar{1}, \bar{a}\bar{2}, \dots, \overline{ap-1}\}| = |\mathbb{Z}_p \setminus \{\bar{0}\}|$, we have that

$$\{\bar{a}\bar{1}, \bar{a}\bar{2}, \dots, \overline{ap-1}\} = \mathbb{Z}_p \setminus \{\bar{0}\} = \{\bar{1}, \bar{2}, \dots, \overline{p-1}\}$$

In particular, the product of all the elements should be the same:

$$(\bar{a}\bar{1})(\bar{a}\bar{2}) \dots (\overline{ap-1}) = \bar{a}^{p-1} \cdot \bar{1} \cdot \bar{2} \dots \overline{p-1} = \bar{1} \cdot \bar{2} \dots \overline{p-1} \Rightarrow \bar{a}^{p-1} \overline{(p-1)!} = \overline{(p-1)!}$$

The element $\overline{(p-1)!}$ is a nonzero element of $\mathbb{Z}_p$, so it has an inverse in $\mathbb{Z}_p$. Multiplying both sides of the equation above by its inverse we get $\bar{a}^{p-1} = \bar{1}$, as desired. $\square$

COROLLARY 1.56. *If p is a prime, then*

$$a^{p^k} \equiv a \pmod{p},$$

for all integers a and $k \geq 1$.

PROOF. Let p be a prime and a any integer. If $a \equiv 0 \pmod{p}$, then the result trivially holds. Suppose then that $a \not\equiv 0 \pmod{p}$. In such a case, we can apply Fermat's Theorem to obtain that $a^{p-1} \equiv 1 \pmod{p}$.

We proceed by induction on k . For $k = 1$, we apply Theorem 1.38 (2) with the congruences $a^{p-1} \equiv 1 \pmod{p}$ and $a \equiv a \pmod{p}$ to obtain that

$$a^p = a \cdot a^{p-1} \equiv a \pmod{p},$$

proving the base case $k = 1$. Suppose now that $a^{p^{k-1}} \equiv a \pmod{p}$. Applying now Theorem 1.38 (3) we obtain that

$$(a^{p^{k-1}})^p \equiv a^{p^{k-1} \cdot p} \equiv a \pmod{p},$$

Now since $(a^{p^{k-1}})^p = a^{p^{k-1} \cdot p} = a^{p^k}$, we obtain that $a^{p^k} \equiv a \pmod{p}$, as desired. $\square$

Chapter 2

Permutations

This chapter focusses on the study of *permutations*, which are bijective maps from a set onto itself. Groups consisting of permutations were in fashion in mid-19th century, and the mathematicians back then were only studying such groups.

Permutations of finite sets are used in Geometry, Statistics and in elementary Algebra, among many other branches in Mathematics. They have many applications in Science and Technology.

2.1 What is a permutation?

A **permutation of a set** X is a bijection $\alpha : X \rightarrow X$. For an integer $n \geq 1$, we write $X_n = \{1, 2, \dots, n\}$; the set of all permutations of X_n is denoted by S_n , and is called the **symmetric group of degree n** .

★ DOES IT CORRESPOND WITH OUR IDEA OF PERMUTATIONS?

Let us consider the numbers 1, 2, 3. A permutation of 1, 2, 3 is a rearrangement of them in a definite order. We have the six following possibilities:

1 2 3 1 3 2 2 1 3 2 3 1 3 1 2 3 2 1

Consider now $X_3 = \{1, 2, 3\}$ and let us think about the bijective maps $X_3 \rightarrow X_3$:

$1 \mapsto 1$	$1 \mapsto 1$	$1 \mapsto 2$	$1 \mapsto 2$	$1 \mapsto 3$	$1 \mapsto 3$
$2 \mapsto 2$	$2 \mapsto 3$	$2 \mapsto 1$	$2 \mapsto 3$	$2 \mapsto 1$	$2 \mapsto 2$
$3 \mapsto 3$	$3 \mapsto 2$	$3 \mapsto 3$	$3 \mapsto 1$	$3 \mapsto 2$	$3 \mapsto 1$

Notice that we have got the same results but expressed in a different way!

In what follows, we are going to introduce a more convenient way to write down permutations. This notation is known as the *Matrix-Type Notation*. We are no longer interested in writing permutations as bijective maps but as matrices. Saying this, please do not forget that they are bijections, since we will be using that all the time.

MATRIX-TYPE NOTATION. We write $\sigma \in S_n$ as a matrix as follows:

- in the top row we list the elements $1, 2, \dots, n$ of X_n ;
- in the bottom row we put the numbers after shuffling, that is, $\sigma(1), \sigma(2), \dots, \sigma(n)$.

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix} = \begin{pmatrix} 1 & 2 & \dots & n \\ \downarrow & \downarrow & & \downarrow \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix}$$

In other words, the elements of X_n are listed in the top row, and beneath each element of X_n is its image.

Example 2.1. The permutation $\sigma \in S_4$ defined by $\sigma(1) = 3$, $\sigma(2) = 1$, $\sigma(3) = 4$, $\sigma(4) = 2$ in matrix notation is:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}.$$

The permutation $\beta \in S_6$ given by $\beta(1) = 6$, $\beta(2) = 3$, $\beta(3) = 1$, $\beta(4) = 5$, $\beta(5) = 2$ and $\beta(6) = 4$ is expressed in matrix notation as

$$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 3 & 1 & 5 & 2 & 4 \end{pmatrix}.$$

THEOREM 2.2. The symmetric group S_n of degree n has $|S_n| = n!$ elements.

PROOF. Since permutations in S_n are bijective maps $X_n \rightarrow X_n$, we have that two permutations σ and τ in S_n are equal if and only if $\sigma(k) = \tau(k)$ for every $k \in X_n$. On the other hand, in order to construct a permutation

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix}$$

we have to choose the numbers $\sigma(1), \sigma(2), \dots, \sigma(n)$ from X_n , so that they are all distinct. Hence we have n choices for $\sigma(1)$, then $n - 1$ choices for $\sigma(2)$, then $n - 2$ choices for $\sigma(3)$, and so on. Thus, in total σ can be built in $n(n - 1) \dots 2 \cdot 1 = n!$ different ways, which shows that $|S_n| = n!$ $\square$

2.1.1 Product of permutations

Let γ and σ be permutations in S_n . Recall that both are bijections from X_n to X_n , thus their **composition** $\gamma \circ \sigma$ is bijective, as well. So, $\gamma \circ \sigma$ is again a **permutation** in S_n . To find $\gamma \circ \sigma$, note that we first apply σ and then γ :

$$(\gamma \circ \sigma)(k) = \gamma(\sigma(k)), \quad \text{for all } k \in X_n.$$

We write $\gamma\sigma$ to denote $\gamma \circ \sigma$, and refer to it as the **product of the permutations γ and σ** .

It is much more practical to compute $\gamma\sigma$ using the matrix expressions of γ and σ . For instance, let us compute $\gamma\sigma$ for

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 5 & 1 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 2 & 3 \end{pmatrix}.$$

We follow these steps:

1. look at 1 in the 1st row of the matrix of σ (element in position 11 of σ);
2. look at the element a below 1 in the matrix of σ ; in this case, $a = 2$;
3. locate the element a in the top row of γ , and go from top to bottom row to find $\gamma(a)$; in this example $\gamma(2) = 4$;
4. we have obtained that $(\gamma\sigma)(1) = 4$;
5. repeat steps 1, 2, 3 with 2 (then with 3, 4 and 5) in the first row of the matrix of σ (element in position 12).

Continuing in this way we obtain that

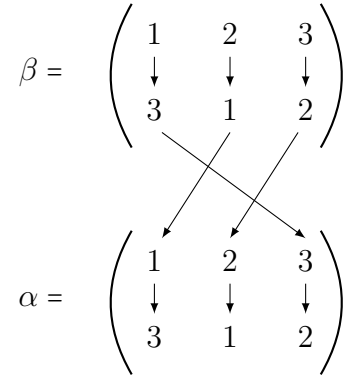
$$\gamma\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 1 & 3 & 5 \end{pmatrix}.$$

In the next example, we will learn a more “visual” and intuitive method to compute the product of two permutations.

Example 2.3. Find $\alpha \circ \beta$, for $\alpha = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$.

Solution: We follow the steps above, we look at 1 in the top row of β , and the element below 1 is $\beta(1) = 3$, then we find 3 in the top row of α , and the element below 3 in α is $\alpha(3) = 2$. This gives us $(\alpha\beta)(1) = 2$; do the same thing with the elements 2 and 3 in the top row of β . At the beginning, it would be a good idea to proceed as follows:

1. stack one permutation on top the other one, putting the permutation on the right-most permutation first; in this case, we begin with β ;
2. follow the direction of the arrows, like in the figure below.

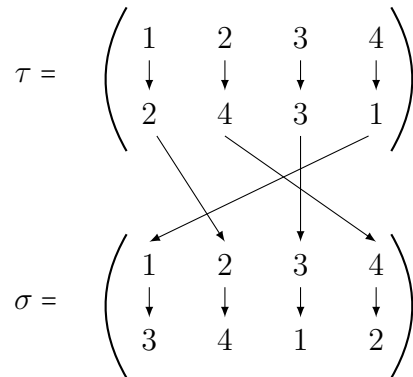


We have obtained that $\alpha\beta = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$.

Example 2.4. Compute $\sigma\tau$ and $\tau\sigma$ for

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}, \quad \tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix}.$$

Solution: To compute $\sigma\tau$ we stack τ on top of σ and follow the arrows as follows:



We have obtained that

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix}.$$

We compute now $\tau\sigma$, stacking σ on top of τ and following the arrows:

$$\begin{array}{c} \sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 4 & 1 & 2 \end{pmatrix} \\ \tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 4 & 3 & 1 \end{pmatrix} \end{array}$$

This produces:

$$\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix}.$$

Notice that $\sigma\tau \neq \tau\sigma$.

We say that **two permutations σ and τ in S_n commute** if $\sigma\tau = \tau\sigma$. As we have already seen two permutations need not commute. On the other hand, if σ , τ and μ are permutations in S_n then we always have that $(\sigma\tau)\mu = \sigma(\tau\mu)$, since the composition of maps is associative.

Definition 2.5. The **identity permutation** ε in S_n is defined as

$$\varepsilon = \begin{pmatrix} 1 & 2 & \dots & n \\ 1 & 2 & \dots & n \end{pmatrix}.$$

In other words, ε is just the identity map: $\varepsilon(k) = k$ for every $k \in X_n$. It is easy to verify that $\varepsilon\sigma = \sigma = \sigma\varepsilon$, for all $\sigma \in S_n$. So ε plays the role in S_n that the number 1 plays for multiplication of numbers.

Example 2.6. Let us find **all the elements of the symmetric group S_3** . By Theorem 2.2 we know that S_3 has six elements. Clearly, one of them is the identity permutation $\varepsilon = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$. Another element of S_3 is $\alpha = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$, and so are $\alpha^2, \alpha^3, \dots$. Let us compute those:

$$\begin{array}{cc}
\alpha = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 2 & 3 & 1 \end{pmatrix} & \alpha^2 = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 3 & 1 & 2 \end{pmatrix} \\
\alpha = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 2 & 3 & 1 \end{pmatrix} & \alpha = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 2 & 3 & 1 \end{pmatrix}
\end{array}$$

We have obtained only one new element, namely $\alpha^2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$, since $\alpha^3 = \varepsilon$.

Notice that $\beta = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$ is also in S_3 , and so are $\beta^2, \beta^3, \dots$

$$\begin{array}{c}
\beta = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 1 & 3 & 2 \end{pmatrix} \\
\beta = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 1 & 3 & 2 \end{pmatrix}
\end{array}$$

Since $\beta^2 = \varepsilon$, we do not need to compute any more powers of β . Let us now look at the products $\alpha\beta$ and $\alpha^2\beta$, which are also elements in S_3 .

$$\begin{array}{cc}
\beta = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 1 & 3 & 2 \end{pmatrix} & \beta = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 1 & 3 & 2 \end{pmatrix} \\
\alpha = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 2 & 3 & 1 \end{pmatrix} & \alpha^2 = \begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 3 & 1 & 2 \end{pmatrix}
\end{array}$$

We have that

$$\alpha\beta = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \quad \alpha^2\beta = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

Let us recap, so far we got

$$\varepsilon, \quad \alpha, \quad \alpha^2, \quad \beta, \quad \alpha\beta, \quad \alpha^2\beta$$

Since all the elements above are different, and $|S_3| = 6$, from here we can conclude that

$$S_3 = \{\varepsilon, \alpha, \alpha^2, \beta, \alpha\beta, \alpha^2\beta\}.$$

2.1.2 Inverse permutations

Every $\sigma \in S_n$ is a bijective map $\sigma : X_n \rightarrow X_n$, and so there exists a unique map $\sigma^{-1} : X_n \rightarrow X_n$ (which is an element of S_n), called the **inverse** of σ , such that $\sigma(\sigma^{-1}(k)) = k$, $\sigma^{-1}(\sigma(k)) = k$. These equations tell us that **each of σ and σ^{-1} reverses the action of the other**. So, we can easily obtain **the matrix expression of σ^{-1}** from the matrix of σ by reading up:

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ \uparrow & \uparrow & & \uparrow \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix}$$

For instance, the inverse permutation of $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 1 & 8 & 3 & 2 & 5 & 6 & 7 \end{pmatrix} \in S_8$ can be found by looking at the second row of σ and reading up to its top row:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ 4 & 1 & 8 & 3 & 2 & 5 & 6 & 7 \end{pmatrix}$$

From here we obtain that

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 5 & 4 & 1 & 6 & 7 & 8 & 3 \end{pmatrix}.$$

We check below that $\sigma^{-1}\sigma = \varepsilon$. Similarly, one can check that $\sigma\sigma^{-1} = \varepsilon$.

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 4 & 1 & 8 & 3 & 2 & 5 & 6 & 7 \end{pmatrix}$$

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 5 & 4 & 1 & 6 & 7 & 8 & 3 \end{pmatrix}$$

2.1.3 Disjoint permutations

We say that an element $k \in X_n$ is **fixed** by a permutation $\sigma \in S_n$ if $\sigma(k) = k$. Otherwise if $\sigma(k) \neq k$, we say that k is **moved** by σ . Two permutations σ and τ are **disjoint** if there is no element of X_n moved by both.

Examples 2.7.

1. The identity permutation $\varepsilon \in S_n$ is the only permutation that fixes all the elements of X_n .

2. $\begin{pmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 2 & 3 & 4 & \dots & n & 1 \end{pmatrix}$ moves every element of X_n .

3. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{pmatrix}$ moves 1, 3, and 5, and fixes 2 and 4.

4. $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix}$ are disjoint permutations, since σ moves 1 and 3, while τ moves 2 and 4.

An important property of disjoint permutations is the following:

THEOREM 2.8. *Disjoint permutations commute.*

Take σ and τ as in Example 2.7. You can easily check that

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} = \tau\sigma$$

In order to prove Theorem 2.8, we first need a preliminary lemma.

LEMMA 2.9. *If k is moved by σ , then $\sigma(k)$ is also moved by σ .*

PROOF. Let $k' = \sigma(k)$ and assume on the contrary that σ fixes k' . Then

$$\sigma(k') = k' = \sigma(k),$$

which implies that $k' = k$, since σ is one-to-one. This shows that $\sigma(k) = k$, which says that k is fixed by σ , a contradiction. $\square$

PROOF OF THEOREM 2.8. Let σ and τ be two disjoint permutations in S_n . To prove that they commute, we must show that $(\sigma\tau)(k) = (\tau\sigma)(k)$, for all $k \in X_n$. Let M_σ and M_τ be the set of elements of X_n that are moved by σ and τ , respectively. Then $M_\sigma \cap M_\tau = \emptyset$, since we are assuming that σ and τ are disjoint. For $k \in X_n$, we distinguish three cases:

- Case 1: $k \in M_\sigma$.

Then $\sigma(k) \in M_\sigma$ by Lemma 2.9, and so $k, \sigma(k) \notin M_\tau$. This means that k and $\sigma(k)$ are fixed by τ , that is, $\tau(k) = k$ and $\tau(\sigma(k)) = \sigma(k)$. Thus:

$$\begin{aligned}(\sigma\tau)(k) &= \sigma(\tau(k)) = \sigma(k), \\ (\tau\sigma)(k) &= \tau(\sigma(k)) = \sigma(k),\end{aligned}$$

and $(\sigma\tau)(k) = (\tau\sigma)(k)$.

- Case 2: $k \in M_\tau$. Proceed like in Case 1.
- Case 3: $k \notin M_\sigma$ and $k \notin M_\tau$.

Then k is fixed by σ and τ , that is, $\sigma(k) = k$ and $\tau(k) = k$. Thus:

$$\begin{aligned}(\sigma\tau)(k) &= \sigma(\tau(k)) = \sigma(k) = k, \\ (\tau\sigma)(k) &= \tau(\sigma(k)) = \tau(k) = k,\end{aligned}$$

and $(\sigma\tau)(k) = (\tau\sigma)(k)$.

In any case, we have showed that $(\sigma\tau)(k) = (\tau\sigma)(k)$, as desired. $\square$

Remark 2.10. The converse of Theorem 2.8 is not true!

For example, σ and σ^{-1} always commute but they are not, in general, disjoint. For instance, the permutation $\sigma \in S_5$ and its inverse are not disjoint, since they both moved 1, 2, and 4.

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & 3 & 2 & 5 \end{pmatrix}, \quad \sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 1 & 5 \end{pmatrix}$$

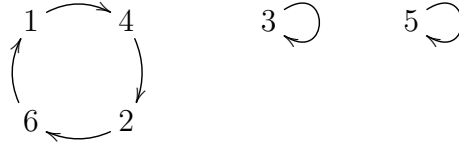
2.2 Cycles

Algebraists (or mathematicians, in general), we like separating (or breaking) complex things (algebraic structures, in our case) into their simplest component parts. This intellectual “divide and conquer” helps us to understand complicated processes and solve difficult problems.

In this section, we are going to take this approach to permutations: **we will see that every permutation can be decomposed into more simple ones called *cycles***. Let us begin with an example: consider the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 6 & 3 & 2 & 5 & 1 \end{pmatrix} \in S_6$$

and look at carefully how σ moves the elements of $X_6 = \{1, 2, 3, 4, 5, 6\}$:



In other words, σ fixes 3 and 5, and moves 1, 2, 4, 6 in a cyclic way. Due to this reason, we call σ a **cycle**, and write it in a more compact way called the **cycle notation**: $\sigma = (1 \ 4 \ 2 \ 6)$. Notice that in the cycle notation

- we have only listed the elements that are moved by sigma;
- each element is moved to its right neighbor, except the last element, which “cycles around” to the first.

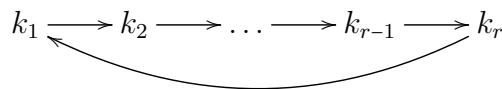
Definition 2.11. Let $k_1, k_2, \dots, k_r$ be distinct elements of X_n . A permutation $\sigma \in S_n$ fixing the remaining elements of X_n (if any) and such that

$$\sigma(k_1) = k_2, \quad \sigma(k_2) = k_3, \quad \dots, \quad \sigma(k_{r-1}) = k_r, \quad \sigma(k_r) = k_1,$$

called an **r -cycle**; r is called the **length** r of σ . We express σ as

$$\sigma = (k_1 \ k_2 \ \dots \ k_r),$$

which means that



Remark 2.12. The r -cycle $(k_1 k_2 \dots k_r)$ can be written in r different ways, depending on which k_i we choose to be the “starting point”; that is:

$$(k_1 k_2 \dots k_r) = (k_2 k_3 \dots k_r k_1) = \dots = (k_r k_1 k_2 \dots k_{r-1})$$

Notice that the **only cycle of length 1 is the identity permutation**, while a 2-cycle $(k_1 k_2)$ interchanges k_1 and k_2 , and fixes everything else; 2-cycles are called **transpositions**.

For instance, $(1 \ 2)$, $(1 \ 3)$, $(2 \ 3)$ are all the possible transpositions in S_3 . The 3-cycle $\sigma = (1 \ 4 \ 2 \ 6)$ from our previous example can also be written as

$$\sigma = (6 \ 1 \ 4 \ 2) = (2 \ 6 \ 1 \ 4) = (4 \ 2 \ 6 \ 1)$$

To reverse the action of the cycle, we simply go around the cycle in the opposite direction. More precisely:

THEOREM 2.13. *If σ is an r -cycle, then σ^{-1} is also an r -cycle. Moreover, if $\sigma = (k_1 k_2 \dots k_{r-1} k_r)$, then $\sigma^{-1} = (k_r k_{r-1} \dots k_2 k_1)$.*

PROOF. Let $\sigma = (k_1 k_2 \dots k_r)$ and $\gamma = (k_r k_{r-1} \dots k_2 k_1)$ be in S_n . The result will follow by proving that $\sigma\gamma = \varepsilon$ and $\gamma\sigma = \varepsilon$. We check here that $\sigma\gamma = \varepsilon$, please do the calculations for the remaining case. For $k \in X_n$ we need to check that $(\sigma\gamma)(k) = k$. To do so, we distinguish two cases:

1. $k \notin \{k_1, \dots, k_r\}$

This means that both σ and γ fixes k , and so

$$(\sigma\gamma)(k) = \sigma(\gamma(k)) = \sigma(k) = k.$$

2. $k \in \{k_1, \dots, k_r\}$

- If $k = k_i$ for some $i \in \{2, \dots, r\}$, then

$$(\sigma\gamma)(k_i) = \sigma(\gamma(k_i)) = \sigma(k_{i-1}) = k_i.$$

- If $k = k_1$, then

$$(\sigma\gamma)(k_1) = \sigma(\gamma(k_1)) = \sigma(k_r) = k_1.$$

□

Examples 2.14. Write the following cycles into a cycle notation and indicate their length. Use Theorem 2.13 to find their inverses.

1. $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 1 & 5 & 2 \end{pmatrix}$

Taking into account that $\sigma(1) = 4$, $\sigma(4) = 5$, $\sigma(5) = 2$, $\sigma(2) = 3$ and $\sigma(3) = 1$, we can conclude that σ is a 5-cycle, which can be written as $\sigma = (1 \ 4 \ 5 \ 2 \ 3)$. Theorem 2.13 tells us that $\sigma^{-1} = (3 \ 2 \ 5 \ 4 \ 1)$. In fact:

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 2 & 1 & 4 \end{pmatrix} = (1 \ 3 \ 2 \ 5 \ 4) = (3 \ 2 \ 5 \ 4 \ 1).$$

2. $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 4 & 7 & 1 & 6 & 5 & 2 & 3 \end{pmatrix}$

Observe that τ fixes 5, since $\tau(5) = 5$. Moreover from $\tau(1) = 4$, $\tau(4) = 6$, $\tau(6) = 2$, $\tau(2) = 7$, $\tau(7) = 3$, $\tau(3) = 1$, we deduce that τ is a 6-cycle, which can be written as $\tau = (1 \ 4 \ 6 \ 2 \ 7 \ 3)$. From Theorem 2.13 we obtain that $\tau^{-1} = (3 \ 7 \ 2 \ 6 \ 4 \ 1)$.

3. $\mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$

Since μ fixes 1 and 2, and $\mu(1) = 2$, $\mu(2) = 3$, $\mu(3) = 1$, we have that μ is a 3-cycle and so $\mu = (1 \ 2 \ 3)$. Theorem 2.13 yields that $\mu^{-1} = (3 \ 2 \ 1)$.

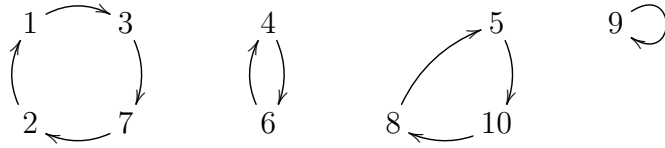
★ **BE CAREFUL!** Notice that μ is an element of S_5 that fixes 4 and 5. It is NOT A IDENTITY PERMUTATION IN S_3 . Hence, **it is important to specify the symmetric group we are dealing with in each case.**

★ **QUESTION: Is every permutation a cycle?** Let us consider the following permutation in S_{10} .

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 3 & 1 & 7 & 6 & 10 & 4 & 2 & 5 & 9 & 8 \end{pmatrix}$$

If we represent the action of σ geometrically (like in the figures below) we will see that σ partitions X_{10} into three disjoint subsets, so that, in each subset,

the elements are permuted cyclically.



In other words, σ is the composition (or product) of the following pairwise disjoint cycles:

$$(1 \ 3 \ 7 \ 2), \quad (4 \ 6), \quad (5 \ 10 \ 8)$$

Notice that σ fixes 9. On the other hand, from Theorem 2.8 we obtain that the cycles above commute. Hence, we can say that

$$\sigma = (1 \ 3 \ 7 \ 2)(4 \ 6)(5 \ 10 \ 8).$$

We will prove below, in Theorem 2.17, that **every permutation can be expressed as a product of disjoint cycles**. Before we do so, we will show with a concrete example how this decomposition algorithm works:

HOW TO FACTOR A PERMUTATION INTO A PRODUCT OF CYCLES:

Consider the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 6 & 4 & 7 & 2 & 5 & 1 & 8 & 9 & 3 \end{pmatrix} \in S_9$$

and follow these steps:

1. Write “(1”.
2. Look at $\sigma(1)$: Since $\sigma : 1 \mapsto 6$ we write “(1 6”, and we close the parentheses now since $6 \xrightarrow{\sigma} 1$. Thus σ begins “(1 6)”.
3. The first number in X_9 which has not appeared yet is 2, so write “(1 6)(2”.
4. Now $2 \xrightarrow{\sigma} 4$ and $4 \xrightarrow{\sigma} 2$, so write “(1 6)(2 4)”.
5. We have dealt with 1, 2, 4 and 6 in X_9 , so 3 is the smallest number in X_9 which has not appeared yet. We proceed as follows, and look at $\sigma(3), \sigma(\sigma(3)), \dots$

$$3 \xrightarrow{\sigma} 7 \xrightarrow{\sigma} 8 \xrightarrow{\sigma} 9 \xrightarrow{\sigma} 3$$

This gives the 4-cycle (3 7 8 9).

6. The only element left from X_9 is 5 and $\sigma(5) = 5$.

Putting altogether, we have that

$$\sigma = (1 \ 6)(2 \ 4)(3 \ 7 \ 8 \ 9)$$

Example 2.15. Factor

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \\ 5 & 12 & 2 & 1 & 9 & 11 & 4 & 3 & 7 & 10 & 13 & 8 & 6 \end{pmatrix} \in S_{13}$$

as a product of (pairwise) disjoint cycles.

Solution: Let us proceed and apply the algorithm, so we start with 1:

$$1 \xrightarrow{\sigma} 5 \xrightarrow{\sigma} 9 \xrightarrow{\sigma} 7 \xrightarrow{\sigma} 4 \xrightarrow{\sigma} 1.$$

Thus, the first cycle is $(1 \ 5 \ 9 \ 7 \ 4)$. Next, we continue with the smallest number of X_{13} , which has not been considered yet, namely 2.

$$2 \xrightarrow{\sigma} 12 \xrightarrow{\sigma} 8 \xrightarrow{\sigma} 3 \xrightarrow{\sigma} 2,$$

and so we get the 4-cycle $(2 \ 12 \ 8 \ 3)$. We now continue by examining 6:

$$6 \xrightarrow{\sigma} 11 \xrightarrow{\sigma} 13 \xrightarrow{\sigma} 6,$$

which provides us the 3-cycle $(6 \ 11 \ 13)$. The remaining member of X_{13} is 10, which is fixed by σ . Therefore:

$$\sigma = (1 \ 5 \ 9 \ 7 \ 4)(2 \ 12 \ 8 \ 3)(6 \ 11 \ 13)$$

Remark 2.16. It is clear that the cycles $\alpha = (1 \ 3 \ 5)$ and $\beta = (2 \ 7 \ 6)$ are disjoint, since the cycle notation tells us that α only moves 1, 3 and 5, while β only moves 2, 7, and 6. In a more general way:

$$\sigma \text{ and } \tau \text{ are disjoint cycles} \Leftrightarrow \{k_1, k_2, \dots, k_r\} \cap \{\ell_1, \ell_2, \dots, \ell_s\} = \emptyset,$$

where $\sigma = (k_1 \ k_2 \ \dots \ k_r)$ and $\tau = (\ell_1 \ \ell_2 \ \dots \ \ell_s)$. This follows from the fact that any $k \in \{k_1, k_2, \dots, k_r\} \cap \{\ell_1, \ell_2, \dots, \ell_s\}$ will be moved by both σ and τ .

We are now ready to prove the existence of the cycle factorisation:

THEOREM 2.17. CYCLE DECOMPOSITION THEOREM.

Every permutation $\sigma \neq \varepsilon$ (in S_n) can be factorised as a product of (one or more) disjoint cycles of length at least 2. Such factorisation is unique up to the order of the factors.

PROOF. PART 1: EXISTENCE

Let $\sigma \in S_n$ with $\sigma \neq \varepsilon$. We proceed by induction on $n \geq 2$. If $n = 2$, then $S_2 = \{\varepsilon, (1\ 2)\}$ and $\sigma = (1\ 2)$ is already a cycle (it is a transposition). Assume that the result is true for permutations in S_{n-1} and let $\sigma \in S_n$. If $\sigma(n) = n$, we can view σ as an element of S_{n-1} and by the induction hypothesis, we are done. We can then assume that $\sigma(n) \neq n$, and let $k = \sigma^{-1}(n)$. Then $\sigma(k) = \sigma(\sigma^{-1}(n)) = n$, and so $k \neq n$, because we are assuming that $\sigma(n) \neq n$. So, we can consider the transposition $\gamma = (k\ n)$, and the permutation $\tau = \sigma\gamma$. From $\gamma^2 = \varepsilon$ we obtain that $\sigma = \tau\gamma$. Moreover, $\tau(n) = \sigma\gamma(n) = \sigma(k) = n$, and so $\tau \in S_{n-1}$. Applying the induction hypothesis we obtain that τ is a product of disjoint cycles. At this point, we distinguish two cases:

- Case 1: $\tau(k) = k$.

Then τ and γ are disjoint permutations since $\tau(n) = n$, and $\gamma = (k\ n)$. We are done in this case because $\sigma = \tau(k\ n)$ and τ is a product of disjoint cycles.

- Case 2: $\tau(k) \neq k$.

Then k is moved by (exactly one) cycle factor of τ . Thus, we can write $\tau = \mu(k\ k_1\ k_2\ \dots\ k_r)$, where μ is a product of disjoint cycles fixing $k, k_1, \dots, k_r$, and n since $\tau(n) = n$. Therefore:

$$\sigma = \tau\gamma = \mu(k\ k_1\ k_2\ \dots\ k_r)(k\ n) = \mu(k\ n\ k_1\ k_2\ \dots\ k_r)$$

is a product of disjoint cycles, as desired.

PART 2: UNIQUENESS

Suppose that $\sigma = \gamma_a \dots \gamma_2 \gamma_1 = \delta_b \dots \delta_2 \delta_1$ are two factorizations of σ into disjoint cycles. We need to show that $a = b$ and $\gamma_i = \delta_i$ for all $i = 1 \dots a$ (after possible reordering).

We proceed by induction on $t := \max(a, b)$. If $t = 1$ there is nothing to prove since $\sigma = \gamma_1 = \delta_1$. Otherwise, assume that the result is true for $t - 1$ and let σ moves m . Then m occurs in exactly one γ_i and exactly one δ_j . By reordering the factors (if necessary) we can assume that m occurs in γ_1 and

in δ_1 , so we can write:

$$\gamma_1 = (m \ k_2 \ \dots \ k_r), \quad \delta_1 = (m \ \ell_2 \ \dots \ \ell_s),$$

Assume, for example, that $r \leq s$. Then:

$$\begin{aligned} k_2 &= \sigma(m) = \ell_2, \\ k_3 &= \sigma(k_2) = \sigma(\ell_2) = \ell_3, \\ &\vdots \\ k_r &= \sigma(k_{r-1}) = \sigma(\ell_{r-1}) = \ell_r. \end{aligned}$$

Now, if $r < s$ then $m = k_1 = \sigma(k_r) = \sigma(\ell_r) = \ell_{r+1}$, which is impossible since δ_1 is a cycle. Therefore $r = s$ and $\gamma_1 = \delta_1$. Write $\mu = \gamma_1 = \delta_1$. Then

$$\sigma\mu^{-1} = \gamma_a \dots \gamma_2 = \delta_b \dots \delta_2,$$

is a product of $a - 1$ ($b - 1$) disjoint cycles, and so the hypothesis induction yields that $a = b$, and (after possible reordering) $\gamma_i = \delta_i$ for $i = 2, 3, \dots, a$, which finishes the proof. $\square$

We close this section with an another interesting question:

“How should we compute the product of two cycles? Shall we use the cycle notation? Or is it better to write them using the matrix notation and compute their product as before?”

You can proceed either way, there is no right or wrong. If you feel more comfortable using the matrix notation, please do so. For example, let us compute $\alpha\beta$, for $\alpha = (2 \ 4 \ 5)$ and $\beta = (1 \ 2 \ 4)$ in S_5 . In matrix notation, we have $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 3 & 5 & 2 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 1 & 5 \end{pmatrix}$, and so

$$\begin{aligned} \beta &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 4 & 3 & 1 & 5 \end{pmatrix} \\ \alpha &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 4 & 3 & 5 & 2 \end{pmatrix} \end{aligned}$$

We have obtained that

$$\alpha\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 3 & 1 & 2 \end{pmatrix} = (1 \ 4)(2 \ 5)$$

If we choose to keep the cycle notation, to compute the product, recall that the permutation on the right goes first, and next the permutation on the left. We proceed as follows:

- β carries 1 to 2, and α carries 2 to 4, hence $\alpha\beta$ carries 1 to 4;
- β carries 2 to 4, and α carries 4 to 5, hence $\alpha\beta$ carries 2 to 5;
- β fixes 3 and so does α , hence $\alpha\beta$ fixes 3;
- β carries 4 to 1, and α fixes 1, so $\alpha\beta$ carries 4 to 1;
- β fixes 5, and α carries 5 to 2, hence $\alpha\beta$ carries 5 to 2.

We obtaine that $\alpha\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 3 & 1 & 2 \end{pmatrix}$, as expected.

2.3 The Alternating Group

In this last section, we focus our attention on transpositions. Recall that a transposition δ is a 2-cycle, so it has the form $\delta = (m \ n)$ (where $m \neq n$) and satisfies $\delta^2 = \varepsilon$, $\delta^{-1} = \delta$.

Transpositions are important since **every permutation is a product of transpositions**. This follows from the Cycle Decomposition Theorem together with the following result, which proof reduces to compute the product of certain transpositions:

THEOREM 2.18. *Every cycle of length $r \geq 1$ is a product of $r-1$ transpositions:*

$$(k_1 \ k_2 \ \dots \ k_r) = (k_1 \ k_2)(k_2 \ k_3) \dots (k_{r-2} \ k_{r-1})(k_{r-1} \ k_r)$$

Remark 2.19. In contrast to the factorisation into cycles, **factorisations into transpositions are not unique**. For example:

$$(1 \ 2)(2 \ 4)(4 \ 5) = (1 \ 2 \ 4 \ 5) = (1 \ 5)(1 \ 4)(1 \ 2).$$

Although the factorisation into transpositions of a permutation σ is far from being unique, it has a very nice property: **the numbers of transpositions into any two factorisations of σ into transpositions have the same parity**. To be more precise, if σ admits a factorisation having an even (respectively, odd) number of transpositions, then any other factorisation of σ into transpositions must have an even (respectively, odd) number of factors. This is known as the *Parity Theorem*.

THEOREM 2.20. PARITY THEOREM.

If a permutation σ has two factorisations

$$\sigma = \gamma_r \dots \gamma_2 \gamma_1 = \mu_s \dots \mu_2 \mu_1,$$

where each γ_i and μ_j is a transposition, then both r and s are even or both are odd.

The Parity Theorem allows us to introduce the following notions:

Definitions 2.21. A permutation σ is called **even** or **odd** if it can be written as a product of an even or odd number of transpositions. The set of all even permutations in S_n is denoted A_n , and is called the **alternating group of degree n** .

Before proving the *Parity Theorem*, let us see a few examples to learn how such a factorisation can be found. We will consider two cases:

- **σ is a cycle:**

In this case, Theorem 2.18 tells us that the parity of the permutation σ coincides with the parity of its length -1 . For instance, any 4-cycle can be written as a product of 3 transpositions, so any 4-cycle is odd. In fact, Theorem 2.18 yields that $\sigma = (1 \ 2 \ 4 \ 5) = (1 \ 2)(2 \ 4)(4 \ 5)$.

- **σ is not a cycle:**

We use now the *Cycle Decomposition Theorem* and Theorem 2.18. To find the parity it is enough to decompose σ into disjoint cycles, and look the parities of the corresponding cycles (see the case above to find the parity of a cycle). To determine a factorisation into transpositions, we follow these two steps:

Step 1. Factor σ into disjoint cycles;

Step 2. Factor each of the cycles obtained in Step 1 into transpositions using Theorem 2.18.

For example, consider the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 4 & 6 & 1 & 7 & 8 & 2 & 9 & 3 \end{pmatrix}$$

To determine the parity of σ we factorise into a product of disjoint cycles:

$$\sigma = (1 \ 5 \ 7 \ 2 \ 4)(3 \ 6 \ 8 \ 9).$$

Notice that σ is the product of a 5-cycle (which is even) and a 4-cycle (which is odd), and so σ is odd (even + odd = odd). To find a factorisation of σ into transpositions we apply Theorem 2.18 to the 5-cycle and the 4-cycle above:

$$\begin{aligned} (1 \ 5 \ 7 \ 2 \ 4) &= (1 \ 5)(5 \ 7)(7 \ 2)(2 \ 4), \\ (3 \ 6 \ 8 \ 9) &= (3 \ 6)(6 \ 8)(8 \ 9) \end{aligned}$$

From here, we obtain that:

$$\sigma = (1 \ 5)(5 \ 7)(7 \ 2)(2 \ 4)(3 \ 6)(6 \ 8)(8 \ 9).$$

To prove the *Parity Theorem* we need a preliminary result:

LEMMA 2.22. *The identity permutation ε can not be written as the product of an odd number of transpositions.*

PROOF. Suppose that the identity permutation ε is a product of transpositions $\varepsilon = \gamma_1 \dots \gamma_r$, and let us show that r is even. Notice that $r \geq 2$ (that is, we can not have only one transposition) because a transposition can not be the identity permutation. We proceed by (strong) induction on r :

If $r = 2$, then there is nothing to prove. Suppose then that $r > 2$ and the result true for all factorisations of the identity permutation ε onto a product of transpositions with $< r$ factors. Let $\varepsilon = \gamma_1 \gamma_2 \dots \gamma_{r-1} \gamma_r$, where the γ_j 's are all transpositions.

We first look at γ_{r-1} and γ_r . If $\gamma_r = (a \ b)$ (where $a \neq b$), then we have four possibilities for γ_{r-1} :

- $\gamma_{r-1} = (b\ a) = (a\ b)$

In this case, $\gamma_{r-1}\gamma_r = \varepsilon$, so we have that

$$\varepsilon = \gamma_1 \dots \gamma_{r-2}.$$

By the inductive hypothesis, we obtain that $r - 2$ is even, which implies that r is even.

- $\gamma_{r-1} = (a\ c)$, for $c \neq a$ and $c \neq b$

Then $\gamma_{r-1}\gamma_r = (a\ c)(a\ b) = (a\ b)(b\ c)$, and so

$$\varepsilon = \gamma_1 \dots \gamma_{r-2}(a\ b)(b\ c).$$

- $\gamma_{r-1} = (b\ c)$, for $c \neq a$ and $c \neq b$

Then $\gamma_{r-1}\gamma_r = (b\ c)(a\ b) = (a\ c)(b\ c)$, and so

$$\varepsilon = \gamma_1 \dots \gamma_{r-2}(a\ c)(b\ c).$$

- $\gamma_{r-1} = (c\ d)$, for $c \neq d$, and both c and d different from a, b

In this case, γ_{r-1} and γ_r commute, since they are disjoint transpositions.

We obtained that

$$\varepsilon = \gamma_1 \dots \gamma_{r-2}(a\ b)(c\ d).$$

Notice that in any of the last three cases, we have replaced the product $\gamma_{r-1}\gamma_r$ by the product of other two transpositions α_{r-1} and α_r such that a appears in α_{r-1} but not in α_r . If we repeat this process, at some point we will have to be in the first case and the result will follow by the inductive hypothesis. Otherwise, we would obtain a product of r transpositions $\beta_1\beta_2\dots\beta_r$ such that $\beta_1\beta_2\dots\beta_r = \varepsilon$, and a appears only in β_1 . But this is impossible because such product does not fix a , while the identity permutations does. $\square$

We are now in a position to prove the *Parity Theorem*.

PROOF OF THEOREM 2.20. Suppose that a permutation σ admits two factorizations into transpositions: $\sigma = \gamma_n \dots \gamma_2 \gamma_1 = \mu_m \dots \mu_2 \mu_1$.

We have to show that both n and m are either even or odd. Keeping in mind that $\mu_j^{-1} = \mu_j$ for all j , we obtain that

$$\varepsilon = \mu_1 \mu_2 \dots \mu_m \gamma_n \dots \gamma_2 \gamma_1.$$

An application of Lemma 2.22 tells us that $m + n$ is even, which implies that n and m are even (respectively, odd), as desired. $\square$

We finish this chapter with a few important properties of A_n ; we will talk about them a bit later when we introduce the notions of group and subgroup, in the next chapter.

THEOREM 2.23. *Let $n \geq 2$. Then:*

- (1) *The identity permutation ε is in A_n .*
- (2) *If σ and τ are in A_n , then σ^{-1} and $\sigma\tau$ are in A_n .*
- (3) $|A_n| = \frac{1}{2}n!$

PROOF. (1) Notice that ε is obviously even since, for example, $\varepsilon = (1\ 2)(1\ 2)$.

(2) Suppose that σ and τ are even permutations, which means that σ and τ are products of an even number of transpositions. Let

$$\sigma = \gamma_1 \gamma_2 \dots \gamma_r, \quad \tau = \delta_1 \delta_2 \dots \delta_s,$$

where r and s are even, and the γ_i and δ_j are transpositions. Then

$$\sigma\tau = \gamma_1 \gamma_2 \dots \gamma_r \delta_1 \delta_2 \dots \delta_s,$$

and since $r + s$ is even, we have that $\sigma\tau \in A_n$. We prove now that $\sigma^{-1} \in A_n$, which follows from the fact that $\sigma^{-1} = \gamma_r \dots \gamma_2 \gamma_1$; in fact, we have that

$$\begin{aligned} (\gamma_1 \gamma_2 \dots \gamma_{r-1} \gamma_r)(\gamma_r \gamma_{r-1} \dots \gamma_2 \gamma_1) &= \gamma_1 \gamma_2 \dots \gamma_{r-1} \gamma_r^2 \gamma_{r-1} \dots \gamma_2 \gamma_1 \\ &= \gamma_1 \gamma_2 \dots \gamma_{r-2} \gamma_{r-1}^2 \gamma_{r-2} \dots \gamma_2 \gamma_1 = \dots = \varepsilon, \end{aligned}$$

and $(\gamma_r \gamma_{r-1} \dots \gamma_2 \gamma_1)(\gamma_1 \gamma_2 \dots \gamma_{r-1} \gamma_r) = \varepsilon$, which finishes the proof of (2).

(3) Let O_n denote the set of odd permutations in S_n . Then $S_n = A_n \cup O_n$ and $A_n \cap O_n = \emptyset$ by the Parity Theorem. From $n! = |S_n| = |A_n| + |O_n|$, it is enough to show that $|A_n| = |O_n|$; in other words, the number of even permutations coincide with the number of odd permutations. To prove that $|A_n| = |O_n|$, we will provide a bijective map $f : A_n \rightarrow O_n$.

Let $\gamma = (1\ 2)$ and define $f(\sigma) = \gamma\sigma$, for all $\sigma \in A_n$. Observe that f is well-defined, namely, the permutation $\gamma\sigma$ is odd (γ is odd and σ is even). The fact that $\gamma^2 = \varepsilon$ implies that f is a bijection. In fact:

- f is one-to-one: $f(\sigma_1) = f(\sigma_2) \Rightarrow \gamma\sigma_1 = \gamma\sigma_2 \Rightarrow \gamma^2\sigma_1 = \gamma^2\sigma_2 \Rightarrow \sigma_1 = \sigma_2$
- f is surjective: Given $\tau \in O_n$, then $\sigma' = \gamma\tau \in A_n$ and

$$f(\sigma') = \gamma\sigma' = \gamma(\gamma\tau) = \gamma^2\tau = \tau.$$

Thus f is a bijection, and therefore $|A_n| = |O_n|$. □

Remark 2.24. A set of permutations is called a **group of permutations** if it contains the identity permutation, the product of any two of its members, and the inverse of any member. According to this definition, S_n is a group of permutations. Notice that Theorem 2.23 tells us that A_n is also a group of permutations.

Chapter 3

Groups

In this chapter, we introduce our main object under study: *groups*. The concept of a group dates back to the nineteenth century, with the works of Evariste Galois on the search for roots for equations.

As in happens quite often in Mathematics, the definition of group that we use today is the result of an evolutionary process: as mentioned, it all started with Galois's work (known nowadays as Galois's Theory) on groups of permutations 1830. Although, it was Arthur Cayley in 1854 the author who worded the concept of a group in a more abstract way, this terminology was not accepted until the 20th century.

3.1 What is a group?

A **group** $(G, \cdot)$ is a set G equipped with a binary operation ¹ $\cdot$ satisfying the following axioms:

- (G1) The **associative law** holds: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for every $a, b, c \in G$.
- (G2) There is a unity 1 for G . Then $1 \cdot a = a \cdot 1 = a$ for all $a \in G$.
- (G3) Every element of G has an inverse in G . In other words, for every $a \in G$ there is an element $b \in G$ such that $a \cdot b = b \cdot a = 1$

The group G is called **abelian** if, in addition, it satisfies

- (G4) The operation $\cdot$ is commutative: $a \cdot b = b \cdot a$ holds for every $a, b \in G$.

¹Read Section A.3 to recall some basic facts about binary operations

If G is finite, the number $|G|$ is called the **order** of G .

Although we write most groups multiplicatively, many important groups are written additively. Then the unity element is denoted 0 and is called **zero**, and the inverse of x is denoted $-x$ and is called the **negative** of x .

Examples 3.1.

1. $\{1\}$, $\{1, -1\}$, $\{1, -1, i, -i\}$ are all abelian groups of (complex) numbers under multiplication. Notice that -1 is self-inverse, while i and $-i$ are inverses of each other.
2. $\mathbb{Q} \setminus \{0\}$, $\mathbb{R} \setminus \{0\}$, $\mathbb{C} \setminus \{0\}$ are all abelian groups under multiplication. In each case the inverse of an element a is $a^{-1} = 1/a$.
3. The sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are all abelian groups under addition. In each case the identity element is 0 , and the inverse of a is its opposite $-a$.
4. Let $n \geq 2$ be an integer. Then $\mathbb{Z}_n$ is an additive abelian group with zero $\bar{0}$ and the negative of $\bar{a}$ being $-\bar{a} = \overline{-a}$.
5. Let R denote $\mathbb{Z}_m$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$. Then the set $U(M_n(R), \cdot)$ of invertible $n \times n$ matrices over R , that is, the matrices $A \in M_n(R)$ such that $\det(A)$ is a unit in R , is a group called the **general linear group of degree n over R** , denoted $GL_n(R)$. Moreover:

$$\text{If } R = \mathbb{Q}, \mathbb{R}, \text{ or } \mathbb{C} \text{ then } GL_n(R) = \{A \in M_n(R) \mid \det(A) \neq 0\}$$

$$GL_n(\mathbb{Z}) = \{A \in M_n(\mathbb{Z}) \mid \det(A) = \pm 1\}$$

$$GL_n(\mathbb{Z}_m) = \{A \in M_n(\mathbb{Z}_m) \mid \det(A) = \bar{a}, \text{ where } \gcd(a, m) = 1\}$$

When we refer to the groups $\mathbb{Z}_n$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$, we mean the additive group.

6. The set S_n of all permutations of $\{1, 2, \dots, n\}$ is a group under composition, called the **symmetric group of degree n** .

Remark 3.2. In what follows to ease the notation, we write ab to denote the product $a \cdot b$ in an arbitrary group G .

Definition 3.3. Let $G_1, G_2, \dots, G_n$ be groups. Their Cartesian product $G_1 \times G_2 \times \dots \times G_n$ is a group with the **component-wise operation** defined by

$$(g_1, g_2, \dots, g_n) \cdot (g'_1, g'_2, \dots, g'_n) = (g_1 g'_1, g_2 g'_2, \dots, g_n g'_n),$$

where $g_i g'_i$ refers to the product in the group G_i for each i , is a group called the **direct group** of the G_i . In fact, one can easily check that the unity of $G_1 \times G_2 \times \dots \times G_n$ is $(1, 1, \dots, 1)$ and the inverse of the element $(g_1, g_2, \dots, g_n)$ is $(g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$.

For instance, let us take $n = 2$, $G_1 = \mathbb{R}$ and $G_2 = \mathbb{Z}_2 = \{\bar{0}, \bar{1}\}$. The direct product $\mathbb{R} \times \mathbb{Z}_2 = \{(a, \bar{0}), (b, \bar{1}) \mid a, b \in \mathbb{R}\}$ is the additive group with

- operation given by

$$(a_1, \bar{c}_1) + (a_2, \bar{c}_2) = (a_1 + a_2, \overline{c_1 + c_2}),$$

where $a_1, a_2 \in \mathbb{R}$ and $c_1, c_2 \in \{0, 1\}$. For example, $(2, \bar{1}) + (-1, \bar{1}) = (-1, \bar{0})$.

- identity element $(0, \bar{0})$;

- the opposite of $(a, \bar{c})$ is $(-a, \overline{-c})$.

3.1.1 Cayley tables

Let $(G, \cdot)$ be a **finite group**. The operation $\cdot$ of G can be completely determined by giving a table, called the **group table** or **Cayley table** of G . Its rows and columns are labelled by the elements of G , and for a and b in G , the element $a \cdot b$ is the entry of the table in the row corresponding to a , and the column corresponding to b .

$\cdot$	$\dots$	b	$\dots$
$\vdots$			
a			
$\vdots$			

Example 3.4. Let us write the **Cayley table** of S_3 , the symmetric group of degree 3. Recall that

$$S_3 = \{\varepsilon, (1\ 2\ 3), (1\ 3\ 2), (1\ 2), (1\ 3), (2\ 3)\}.$$

To ease the notation, let $\sigma = (1\ 2\ 3)$, $\tau = (1\ 2)$. Using what we learnt in the previous chapter, we compute

$$\begin{aligned}\sigma^2 &= (1\ 3\ 2), & \sigma^3 &= \varepsilon = \tau^2, & \sigma\tau\sigma &= \tau, \\ \tau\sigma &= (2\ 3), & \tau\sigma^2 &= (1\ 3).\end{aligned}$$

The previous calculations allow us to write S_3 as

$$S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}.$$

The elements σ and τ are called **generators** for S_3 because any other element of S_3 can be obtained from them; the equations $\sigma^3 = \varepsilon = \tau^2$ and $\sigma\tau\sigma = \tau$ are called **relations** among the generators.

Using this new way of writing S_3 , we can easily compute the Cayley table of S_3 . For instance, from $\sigma\tau\sigma = \tau$ multiplying both sides on the equation by σ^2 on the right and using that $\sigma^3 = \varepsilon$ we obtain that $\sigma\tau = \tau\sigma^2$; similarly, multiplying now by σ^2 on the left we have that $\sigma^2\tau = \tau\sigma$. These are two of the entries of the Cayley table given below; let us compute a few more:

$$\begin{aligned}(\tau\sigma)\tau &\stackrel{\tau\sigma=\sigma^2\tau}{=} (\sigma^2\tau)\tau = \sigma^2\tau^2 \stackrel{\tau^2=\varepsilon}{=} \sigma^2, \\ \sigma^2(\tau\sigma) &= (\sigma^2\tau)\sigma \stackrel{\sigma^2\tau=\tau\sigma}{=} (\tau\sigma)\sigma = \tau\sigma^2, \\ (\tau\sigma^2)(\tau\sigma^2) &= \tau(\sigma^2\tau)\sigma^2 \stackrel{\sigma^2\tau=\tau\sigma}{=} \tau(\tau\sigma)\sigma^2 = \tau^2\sigma^3 = \varepsilon.\end{aligned}$$

The resulting table is:

S_3	ε	σ	σ^2	τ	$\tau\sigma$	$\tau\sigma^2$
ε	ε	σ	σ^2	τ	$\tau\sigma$	$\tau\sigma^2$
σ	σ	σ^2	ε	$\tau\sigma^2$	τ	$\tau\sigma$
σ^2	σ^2	ε	σ	$\tau\sigma$	$\tau\sigma^2$	τ
τ	τ	$\tau\sigma$	$\tau\sigma^2$	ε	σ	σ^2
$\tau\sigma$	$\tau\sigma$	$\tau\sigma^2$	τ	σ^2	ε	σ
$\tau\sigma^2$	$\tau\sigma^2$	τ	$\tau\sigma$	σ	σ^2	ε

Example 3.5. Consider the direct product $\mathbb{Z}_2 \times \mathbb{Z}_2$ of two copies of the additive group $\mathbb{Z}_2$. Recall that

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1}), (\bar{1}, \bar{0}), (\bar{1}, \bar{1})\}$$

Keeping in mind that $\bar{0} + \bar{0} = \bar{0} = \bar{1} + \bar{1}$ and $\bar{0} + \bar{1} = \bar{1} = \bar{1} + \bar{0}$, and that the operation in $\mathbb{Z}_2 \times \mathbb{Z}_2$ was defined component-wise, that is,

$$(\bar{a}, \bar{b}) + (\bar{c}, \bar{d}) = (\overline{a+c}, \overline{b+d}),$$

for $a, b, c, d \in \{0, 1\}$, one can easily compute the Cayley table of $\mathbb{Z}_2 \times \mathbb{Z}_2$.

$\mathbb{Z}_2 \times \mathbb{Z}_2$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$
$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$

Some features of the Cayley table of a group

The **Cayley table of a group** $(G, \cdot)$ provide us a lot of information about G . We can discover many properties of G by simply taking a quick look to its Cayley table; to more precise,

- G is **Abelian** if and only if its Cayley table is symmetric with respect its main diagonal.
- an element $e \in G$ is the **identity element** 1 of G if and only if the e th row and e th column look the same as the labels.
- the **inverse of a is b** if and only if the entry in row a and column b ($a \cdot b$) and the entry in row b and column a ($b \cdot a$) turn out to be 1.

For instance, the group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is abelian, while S_3 is not; $(\bar{0}, \bar{0})$ (respectively, ε) is the identity element for $\mathbb{Z}_2 \times \mathbb{Z}_2$ (respectively, S_3); every nonzero element in $\mathbb{Z}_2 \times \mathbb{Z}_2$ is self-inverse; in S_3 besides σ and σ^2 which are inverse of each other, the rest of elements are self-inverses.

3.1.2 Inverses. Exponent Laws. Cancellation Laws

Let G be a group and $g \in G$. If $n \geq 0$ denotes an integer, we can inductively define the **n th power** g^n of g as follows:

$$g^0 = 1, \quad g^n = g \cdot g^{n-1}$$

The **definition of n th powers could be extended to include negative powers** by considering the inverse of g . More precisely, we define

$$g^{-n} = (g^{-1})^n = (g^n)^{-1}$$

We can now define the order of g in the group G ; to do so, the only thing one has to do is to compute the powers $g, g^2, g^3, \dots$ (or multiples $g, g + g, g + g + g, \dots$ of g in the additive case) until we reach the identity element of G . Let us say that $g^{10} = 1$ (respectively, $10g = 0$), then the order of g is 10. A natural question arises: what would happen if we never reach the identity? Well, in that case, we will say that the order of g is infinite. More formally:

Definition 3.6. Let G be a group and $g \in G$. The **order of the element g** is defined as the **smallest positive integer n** such that $g^n = 1$. If $g^k \neq 1$ for all $k \geq 1$, then we say that g has **infinite order**. We write $\text{o}(g)$ (or sometimes $|g|$) to denote the order of g .

If $\text{o}(g) = n$ and $g^\ell = 1$, for some $\ell \geq 1$, one can prove that $n|\ell$. Any idea?



Examples 3.7. Notice that in additive groups, the order of g is the smallest positive integer n such that $ng = 0$.

1. Elements of order 2 are self-inverses; for instance, $\tau \in S_3$ and $(\bar{1}, \bar{0}) \in \mathbb{Z}_2 \times \mathbb{Z}_2$ satisfy that $\tau^2 = \varepsilon$ and $2(\bar{1}, \bar{0}) = (\bar{0}, \bar{0})$.
2. Consider the additive group $\mathbb{Z}_4 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$. The order of $\bar{2}$ is 2 (and therefore, $\bar{2}$ is self-inverse) since

$$\bar{2} + \bar{2} = \overline{2+2} = \bar{4} = \bar{0};$$

while the order of $\bar{3}$ in $\mathbb{Z}_4$ is 4:

$$\begin{aligned}\bar{3} + \bar{3} &= \overline{3+3} = \bar{6} = \bar{2}, & \bar{3} + \bar{3} + \bar{3} &= \bar{2} + \bar{3} = \overline{2+3} = \bar{5} = \bar{1}, \\ \bar{3} + \bar{3} + \bar{3} + \bar{3} &= \bar{1} + \bar{3} = \overline{1+3} = \bar{4} = \bar{0}.\end{aligned}$$

The following result collect very important properties of computing inverses in groups:

THEOREM 3.8. *Let $g, h, g_1, \dots, g_n$ denote elements of a group G .*

- (1) $1^{-1} = 1$.
- (2) $(g^{-1})^{-1} = g$.
- (3) $(gh)^{-1} = h^{-1}g^{-1}$.
- (4) $(g_1g_2 \dots g_{n-1}g_n)^{-1} = g_n^{-1}g_{n-1}^{-1} \dots g_2^{-1}g_1^{-1}$ for all $n \geq 1$.

PROOF. (1) follows from (G3), while (2) follows from (G4) (applied to g).

(3) We need to check $(gh)(h^{-1}g^{-1}) \stackrel{?}{=} (h^{-1}g^{-1})(gh) \stackrel{?}{=} 1$. Applying (G4) to g and h we obtain that $gg^{-1} = g^{-1}g = 1 = hh^{-1} = h^{-1}h$. From here we have:

$$\begin{aligned}(gh)(h^{-1}g^{-1}) &= g(hh^{-1})g^{-1} = gg^{-1} = 1, \\ (h^{-1}g^{-1})(gh) &= h^{-1}(g^{-1}g)h = h^{-1}h = 1.\end{aligned}$$

(4) We proceed by induction on n . For $n = 1$ the result trivially follows. Suppose that $n > 1$ and the result true for $n - 1$, that is,

$$(g_1g_2 \dots g_{n-1})^{-1} = g_{n-1}^{-1} \dots g_2^{-1}g_1^{-1},$$

and we prove it for n :

$$\begin{aligned}(g_1g_2 \dots g_{n-1}g_n)^{-1} &= ((g_1g_2 \dots g_{n-1})g_n)^{-1} \stackrel{(3)}{=} g_n^{-1}(g_1g_2 \dots g_{n-1})^{-1} \\ &\stackrel{(1.H)}{=} g_n^{-1}g_{n-1}^{-1} \dots g_2^{-1}g_1^{-1}.\end{aligned}$$

And (4) follows by the Principle of Mathematical Induction. $\square$

The following result, known as the *Exponent Laws*, is very important and we will use it **all the time!** Its proof is quite technical since it involves several proofs that can not be done directly by induction. Recall that the *Principal Mathematical of Induction* is a property of the natural numbers, not the integers.

THEOREM 3.9. Exponent Laws. *Let G be a group and $g, h \in G$.*

- (1) *If $gh = hg$, then $g^n h = hg^n$ for all $n \in \mathbb{Z}$.*
- (2) *If $gh = hg$, then $(gh)^n = g^n h^n$ for all $n \in \mathbb{Z}$.*
- (3) *$g^n g^m = g^{n+m}$ for all $n, m \geq 0$.*
- (4) *$g^n g^m = g^{n+m}$ for all $n, m \in \mathbb{Z}$.*
- (5) *$(g^n)^m = g^{nm}$ for all $n, m \in \mathbb{Z}$.*

PROOF. (1) Suppose that $gh = hg$. For $n = 0$, we have that $g^0 h = h = hg^0$, since $g^0 = 1$ by definition. We prove (1) for $n > 0$ using induction. It trivially holds for $n = 1$, since $g^1 h = gh = hg = hg^1$. Take $n > 1$ and suppose that $g^{n-1} h = hg^{n-1}$. Using the definition of powers and the inductive hypothesis we have that

$$\begin{aligned} g^n h &= (gg^{n-1})h = g(g^{n-1}h) \stackrel{(\text{I.H.})}{=} g(hg^{n-1}) = (gh)g^{n-1} = (hg)g^{n-1} \\ &= h(gg^{n-1}) = hg^n. \end{aligned}$$

By the Principle of Mathematical Induction the equality (1) follows for all $n > 0$. Take now $n < 0$. Then $-n > 0$ and $g^{-n} = (g^{-1})^n$ by definition of the powers. Notice that g^{-1} and h commute:

$$gh = hg \Rightarrow h = g^{-1}hg \Rightarrow hg^{-1} = g^{-1}h.$$

We could then use (1) with h, g^{-1} and $-n$:

$$h(g^{-1})^{-n} = (g^{-1})^{-n}h \Rightarrow hg^n = g^n h,$$

which finishes the proof of (1).

(2) Suppose that g and h commute, that is, $gh = hg$. We prove first that $hg^n = g^n h$ for all $n \geq 0$. We proceed by induction on n : for $n = 0$ there is nothing to prove, since it follows from the definition of powers:

$$(gh)^0 = 1 = 1 \cdot 1 = g^0 h^0.$$

Take $n > 1$ and suppose that $(gh)^{n-1} = g^{n-1} h^{n-1}$. Using the definition of powers we have

$$\begin{aligned} (gh)^n &= (gh)(gh)^{n-1} \stackrel{(\text{I.H.})}{=} (gh)(g^{n-1} h^{n-1}) \stackrel{gh=hg}{=} (hg)(g^{n-1} h^{n-1}) \\ &= h(gg^{n-1})h^{n-1} = (hg^n)h^{n-1} \stackrel{hg^n=g^nh}{=} (g^n h)h^{n-1} = g^n(hh^{n-1}) = g^n h^n. \end{aligned}$$

Suppose now that $n < 0$; then $-n > 0$. On the other hand, notice since we are assuming that $gh = hg$, using Theorem 3.8 (3) we have that g^{-1} and h^{-1} commute; in fact:

$$g^{-1}h^{-1} = (gh)^{-1} = (gh)^{-1} = h^{-1}g^{-1}.$$

We apply Theorem 3.8 (3), equality (2) to g^{-1} , h^{-1} and $-n > 0$, and Theorem 3.8 (2) to obtain

$$\begin{aligned} ((gh)^{-1})^{-n} &= ((hg)^{-1})^{-n} = ((g^{-1}h^{-1})^{-n} = (g^{-1})^{-n}(h^{-1})^{-n} \\ &= ((g^{-1})^{-1})^n((h^{-1})^{-1})^n = g^n h^n, \end{aligned}$$

concluding the proof of (2).

(3) trivially holds if either $n = 0$ or $m = 0$. Fix $m > 0$ and let us prove the equality by induction on n . In this case, we proceed by induction on n . For $n = 1$, we have

$$g^1 g^m = g \cdot g^m = g^{m+1}.$$

Suppose now that $n > 1$ and that $g^{n-1}g^m = g^{n+m-1}$, and let us prove that $g^n g^m = g^{n+m}$. Using the definition of powers and the inductive hypothesis we obtain:

$$g^n g^m = (g \cdot g^{n-1})g^m = g(g^{n-1}g^m) \stackrel{\text{(I.H.)}}{=} g \cdot g^{n+m-1} = g^{1+(n+m-1)} = g^{n+m},$$

as desired.

To prove (4) we distinguish a few cases:

(C1) $n < 0$ and $m < 0$.

In this case, $-n > 0$ and $-m > 0$, so we can apply (3) with the element g^{-1} to obtain:

$$(g^{-1})^{-n}(g^{-1})^{-m} = (g^{-1})^{-n-m},$$

which implies $g^n g^m = g^{n+m}$, by the definition of powers.

(C2) $n > 0$ and $m < 0$. We need to consider three subcases here:

- $n = -m$.

Then $n + m = 0$ and so $g^{n+m} = g^0 = 1$. We need to show that $g^n g^m = 1$.

- $n > -m$.

Notice that $-m > 0$ and $n + m > 0$. We can then write n as $n = (n + m) + (-m)$ and apply (3) and the previous subcase to obtain:

$$g^n g^m = g^{n+m+(-m)} g^m \stackrel{(3)}{=} (g^{n+m} g^{-m}) g^m = g^{n+m} (g^{-m} g^m) = g^{n+m}.$$

- $n > -m$.

Then $n + m < 0$ and $-n < 0$. In this case, we will apply (C1) and the first subcase:

$$g^n g^m = g^n g^{(-n)+(n+m)} = g^n (g^{-n} g^{n+m}) = (g^n g^{-n}) g^{n+m} = g^{n+m}.$$

(C3) $n < 0$ and $m > 0$. The proof of this case is similar to the previous one.

(5) Fix $n \in \mathbb{Z}$, and let $h := g^n \in G$. We then need to show that $h^m = g^{nm}$ for all $m \in \mathbb{Z}$. This trivially holds for $m = 0$. We prove it for $m > 0$ using induction. For $m = 1$ this is trivial since $h = h^1 = g^n$. Suppose that $h^{m-1} = g^{n(m-1)}$ and let us prove that $h^m = g^{nm}$. Using the definition of powers and the inductive hypothesis we have

$$h^m = h h^{m-1} \stackrel{(1.H)}{=} h g^{n(m-1)} = g^n g^{n(m-1)} \stackrel{(3),(4)}{=} g^{n+n(m-1)} = g^{nm},$$

as desired. The Principle of Mathematical Induction tells us that $(g^n)^m = g^{nm}$ for all $g \in G$, $n \in \mathbb{Z}$ and $m > 0$. It remains to prove (5) for $m < 0$. But if $m < 0$, then $-m > 0$ and we can apply what we have just proved to g^{-n} to obtain that

$$(g^{-n})^{-m} = g^{(-n)(-m)} = g^{nm}$$

But from the definition of powers we have

$$(g^{-n})^{-m} = ((g^{-n})^{-1})^m = \left(((g^n)^{-1})^{-1} \right)^m = (g^n)^m.$$

Putting it together we have proved that $(g^n)^m = g^{nm}$. □

Using that every element of a group has an inverse (in the group), we can obtain the result below. The proof is very easy, since it consists on multiplying on the left or on the right by the inverse of appropriate elements.

THEOREM 3.10. *Let g , h , and f be elements of a group.*

- (1) *If $gh = gf$, then $h = f$.* **Left cancellation Law**
- (2) *If $hg = fg$, then $h = f$.* **Right cancellation Law**
- (3) *The equation $gx = h$ has a unique solution $x = g^{-1}h$ in G .*
- (4) *The equation $xg = h$ has a unique solution $x = hg^{-1}$ in G .*

As a consequence of the *Cancellation Laws*, we can prove a very important and useful property of the Cayley table:

COROLLARY 3.11. *Every row (and column) of the Cayley table of a finite group G contains every element of G exactly once.*

PROOF. For $g \in G$, the row of the Cayley table corresponding to g consists on the elements of the form gx , where $x \in G$. Now this row contains any other element $h \in G$ because the equation $gx = h$ is always solvable (in G) for each h (Theorem 3.10 (3)) and it contains h only once because the solution is unique by Theorem 3.10 (3). A similar argument works for the columns, using now Theorem 3.10 (4). □

Recap about the ADDITIVE NOTATION:

In general, when referring to groups we have adopted a multiplicative notation. If a group G is written additively, for example $(\mathbb{Z}, +)$, the **unity is denoted 0** and the **inverse of $g \in G$ is denoted $-g$** . The exponent g^n (in multiplicative notation) becomes ng . The exponents laws look like:

Multiplicative notation

$$g^{n+m} = g^n g^m$$

$$(g^n)^m = g^{nm}$$

$$(gh)^n = g^n h^n$$

if g and h commute

Additive notation

$$(n+m)g = ng + mg$$

$$m(ng) = (mn)g$$

$$n(g+h) = ng + nh$$

3.1.3 A taste of isomorphic groups

A finite group is completely determined by its Cayley table since all group axioms (associativity, existence of the unity and inverses) depend entirely on the operation of the group. Keeping this in mind, let us take a quick look to the Cayley tables of the (multiplicative) group $\mathbb{Z}^* = \{1, -1\}$ (of units of $\mathbb{Z}$) and the (additive) group $\mathbb{Z}_2 = \{\bar{0}, \bar{1}\}$:

$\mathbb{Z}^*$	1	-1
1	1	-1
-1	-1	1

$\mathbb{Z}_2$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{0}$

Notice that **these two tables are the same** in the sense that the Cayley table of $\mathbb{Z}^*$ becomes that of $\mathbb{Z}_2$ if we replace the symbols 1 and -1 by $\bar{0}$ and $\bar{1}$, respectively. Put it this way **$\mathbb{Z}^*$ and $\mathbb{Z}_2$ are the same groups except for notation**; in such a case we will say that they are **isomorphic**, or that **they are the same up to isomorphism**.

We will talk more about this whole topic later on; but for now we will call **two (finite) groups isomorphic if they have the same Cayley table except for notation**. In what follows, we will prove that, up to isomorphism, there is only one group of order 1, 2, or 3; to do so, we will use the property of the Cayley tables proved in Corollary 3.11. If G denotes a finite group of order $|G| \leq 3$, then we will show that

- If $|G| = 1$, then $G = \{1\}$.
- If $|G| = 2$, then $G = \{1, g\}$, where $g \neq 1$ and $g^2 = 1$.
- If $|G| = 3$, then $G = \{1, g, g^2\}$, where $g \neq 1$, $g^2 \neq 1$, $g \neq g^2$ and $g^3 = 1$.

In each case, we will prove that the Cayley table is completely determined by the exponent laws. In order to show this, it is enough to prove that there is only one way to fill in the Cayley table that produces a group. For **$|G| = 1$** , there is nothing to do, the group reduces to the identity element $G = \{1\}$. Next, for **$|G| = 2$** we have that $G = \{1, g\}$, where $g \neq 1$. Let us fill in the Cayley table of G :

G	1	g
1	1	g
g	g	?

We have two possibilities for g^2 , namely, $g^2 = 1$ or $g^2 = g$. But $g^2 = g$ is impossible because it implies that $g = 1$ by the cancellation laws. Thus $g^2 = 1$ and the Cayley table of $G = \{1, g\}$ is the following:

G	1	g
1	1	g
g	g	1

This shows, in particular, that any two groups of order 2 are isomorphic. To finish, let us analyse what happens when $|G| = 3$. Write $G = \{1, g, h\}$, where $g \neq h$ and both g and h are different from the identity element. By the cancellation laws we have that $gh \neq g$ and $gh \neq h$, so we must have that $gh = 1$; the same applies to $hg = 1$. So far, we have that

G	1	g	h
1	1	g	h
g	g	?	1
h	h	1	?

To find out the values of g^2 and h^2 we apply Corollary 3.11. Notice that $g^2 \neq 1$ (respectively, $g^2 \neq g$) because otherwise, if $g^2 = 1$ (respectively, $g^2 = g$) the element 1 (respectively, g) would appear twice in the column of g . Thus $g^2 = h$. Similarly $h^2 = g$. Therefore, the Cayley table of G is:

G	1	g	h
1	1	g	h
g	g	h	1
h	h	1	g

Moreover, since $g^2 = h$ we have that $G = \{1, g, g^2\}$ and

$$gh = g \cdot g^2 = g^3 = g^2 \cdot g = hg,$$

which gives the table:

G	1	g	g^2
1	1	g	g^2
g	g	g^2	1
g^2	g^2	1	g

Groups whose elements are all powers of one element g are called **cyclic**. We write C_n to denote the cyclic group $\{1, g, \dots, g^{n-1}\}$ of order n . We have proved that **every group of order 1, 2, or 3 is cyclic**. However, **this is**

not the case for groups of order 4. For instance, the direct product $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1}), (\bar{1}, \bar{0}), (\bar{1}, \bar{1})\}$ of the additive group $\mathbb{Z}_2$ is a noncyclic group, since all its elements have order 2:

$$\begin{aligned}(\bar{0}, \bar{1}) + (\bar{0}, \bar{1}) &= (\bar{0} + \bar{0}, \bar{1} + \bar{1}) = (\overline{0+0}, \overline{1+1}) = (\bar{0}, \bar{2}) = (\bar{0}, \bar{0}), \\(\bar{1}, \bar{0}) + (\bar{1}, \bar{0}) &= (\bar{1} + \bar{1}, \bar{0} + \bar{0}) = (\overline{1+1}, \overline{0+0}) = (\bar{2}, \bar{0}) = (\bar{0}, \bar{0}), \\(\bar{1}, \bar{1}) + (\bar{1}, \bar{1}) &= (\bar{1} + \bar{1}, \bar{1} + \bar{1}) = (\overline{1+1}, \overline{1+1}) = (\bar{2}, \bar{2}) = (\bar{0}, \bar{0}).\end{aligned}$$

Remark 3.12. The (additive) groups $\mathbb{Z}_2$ and $\mathbb{Z}_3$ of orders 2 and 3, respectively, are isomorphic to the (multiplicative) cyclic groups $C_2 = \{1, g\}$ and $C_3 = \{1, g, g^2\}$ (respectively), whose Cayley tables are given above. To convince yourself, write down the Cayley tables of $\mathbb{Z}_2$ and $\mathbb{Z}_3$ and compare them with the corresponding counterparts.

We will see in the workshop sessions that, up to isomorphism, **there are only two groups of order 4**, the **cyclic group C_4** and a **noncyclic group K_4** with Cayley table:

K_4	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

The group $K_4 = \{1, a, b, c\}$ is called the **Klein group**. The multiplication can be described as $a^2 = b^2 = c^2 = 1$ (all its elements have order 2) and the product of any two of a , b , and c is the third. Notice that $\mathbb{Z}_2 \times \mathbb{Z}_2$ and K_4 are isomorphic, while $\mathbb{Z}_4$ is isomorphic to C_4 .

3.2 Subgroups

In this section, we will learn **how to determine whether a subset H of a group G is itself a group**. The motivation behind is that there are many important groups which arise as subsets of other known groups.

Definition 3.13. A subset H of a group G is called a **subgroup of G** if H is itself a group using the operation of G .

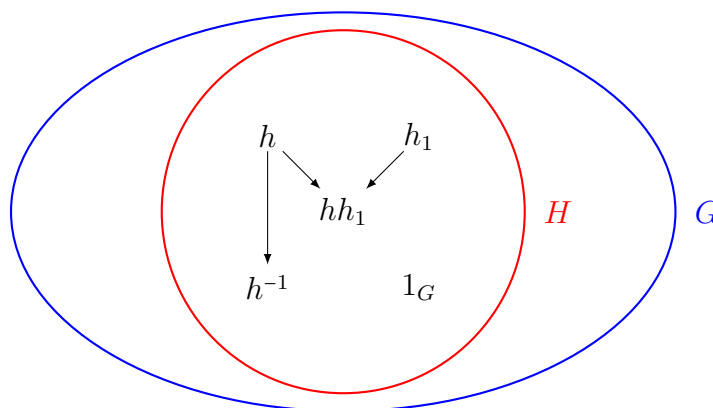
Examples 3.14.

1. Each of the **additive groups** $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$ is a subgroup of the larger ones.
2. The subset of even permutations A_n is a subgroup of S_n , called the **alternating group of degree n** .

SUBGROUP TEST. A subset H of a group G is a subgroup of G if and only if the following three conditions are satisfied.

- (1) $1_G \in H$, where 1_G is the identity element of G .
- (2) If $h \in H$ and $h_1 \in H$, then $hh_1 \in H$.
- (3) If $h \in H$, then $h^{-1} \in H$. Here h^{-1} denotes the inverse of h in G .

In this case, H has the same unity as G , and if $h \in H$, its inverse in H is the same as its inverse in G .



A few comments before we prove the *Subgroup Test*. In some texts, conditions (2) and (3) are replaced by the following condition:

- (2') If $h \in H$ and $h_1 \in H$, then $hh_1^{-1} \in H$.

One can check that (1), (2) and (3) are equivalent to (1) and (2'). So, if you want to check whether a subset H of a group G is a subgroup of G , you can either check that H satisfies

- Conditions (1), (2) and (3), or
- Conditions (1) and (2').

I personally find easier at first to prove (1), (2) and (3), but please go ahead and check (1) and (2') if this other option works better for you.

PROOF. Suppose that H satisfies (1), (2) and (3). Then H is closed under the operation of G by (2); the unity of G is the unity of H by (1), and the inverse in G of $h \in H$ coincides with the inverse of h in H by (3). As H inherits the associative law from G , we have that H is a subgroup of G .

Conversely, assume that H is a subgroup of G , and let e denote the unity of H . Using that e is the identity of H we have that $e^2 = e$, and using that e is an element of G and 1_G is the identity of G we have that $e 1_G = e$. Thus:

$$e^2 = e = e 1_G,$$

so $e = 1_G$ by the cancellation laws in G ; this proves (1).

(2) follows because H is closed under the operation of G , since we are assuming that it is a subgroup of G . Lastly, if $h \in H$, let h' denote its inverse in H , that is, $hh' = h'h = e = 1_G$, by the calculations above. If h^{-1} is the inverse in G , we need to show that $h' = h^{-1}$. This follows from the cancellation laws in G :

$$hh' = e = 1_G = hh^{-1} \Rightarrow h' = h^{-1}.$$

This finishes the proof. □

Example 3.15. Let $n \geq 0$ be an integer. Then the set of all multiples of n , namely, $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ is a **subgroup of the additive group** $(\mathbb{Z}, +)$. In fact, the identity element of $\mathbb{Z}$ is 0, and $0 = n \cdot 0 \in n\mathbb{Z}$. If a and b are in $n\mathbb{Z}$, then $a = nk$ and $b = nm$, for some $k, m \in \mathbb{Z}$, and

$$a + b = nk + nm = n(k + m) \in n\mathbb{Z},$$

$$-a = -(nk) = n(-k) \in n\mathbb{Z}.$$

From the Subgroup Test we get that $n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$. These are indeed all the subgroups of $\mathbb{Z}$! We will prove this when we study cyclic groups in the next section.

FINITE SUBGROUP TEST. If H is a **finite** nonempty subset of a group G , then H is a subgroup of G if and only if H is closed under the operation of G :

$$h, h_1 \in H \Rightarrow hh_1 \in H, \quad \forall h, h_1 \in H.$$

PROOF. If H reduces to the identity element of G , that is, $H = \{1\}$, then there is nothing to prove. Suppose now that $|H| > 1$. Assume first that H is closed, and let $h \in H$ such that $h \neq 1$. Then all the powers $h, h^2, h^3, \dots$ of h are in H because we are assuming that H is closed. Since H is finite, there exists $\ell > 2$ such that $h = h^\ell = h^{\ell-1} \cdot h$. This implies that $1 = h^{\ell-1} \in H$, which shows that H contains the identity element of G . Moreover, $1 = h^{\ell-1} = h \cdot h^{\ell-2}$ and so $h^{-1} = h^{\ell-2} \in H$. By the Subgroup Test we can conclude that H is a subgroup of G ; the converse is obvious. $\square$

Example 3.16. Using the *Finite Subgroup Test* we can easily check that $H = \{\bar{2}, \bar{4}, \bar{6}\}$ is a subgroup of the additive group $\mathbb{Z}_6$. In fact, it is enough to show that H is closed, which is clear since

$$\bar{2} + \bar{2} = \bar{4} \in H, \quad \bar{2} + \bar{4} = \bar{6} \in H, \quad \bar{4} + \bar{4} = \bar{8} = \bar{2} \in H.$$

Thus, H is a subgroup of $\mathbb{Z}_6$.

Example 3.17. Consider the cyclic group of order 4: $C_4 = \{1, a, a^2, a^3\}$, where $a^4 = 1$. We can use the *Finite Subgroup Test* to check that $\{1\}$, $\{1, a^2\}$ and C_4 are **subgroups of C_4** .

In fact, $\{1\}$ and C_4 are clearly subgroups of C_4 . Concerning to $H = \{1, a^2\}$, this follows because $1 \cdot a^2 = a^2 \cdot 1 = a^2 \in H$, and $a^2 \cdot a^2 = a^4 = 1$. Moreover, these three are the only subgroups of C_4 . In fact, suppose that K is a subgroup of C_4 distinct from $\{1\}$ and H , and let us show that $K = C_4$. Notice that either $a \in K$ or $a^3 \in K$.

- If $a \in K$, then a^2 and $a^3 \in K$ because K is closed. Hence $K = C_4$.
- If $a^3 \in K$, then

$$(a^3)^2 = a^6 = a^4 \cdot a^2 = 1 \cdot a^2 = a^2 \in K, \quad (a^3)^3 = a^9 = (a^4)^2 \cdot a = 1 \cdot a = a \in K,$$

which implies that $K = C_4$.

Example 3.18. Let us now find all the subgroups of the Klein group:

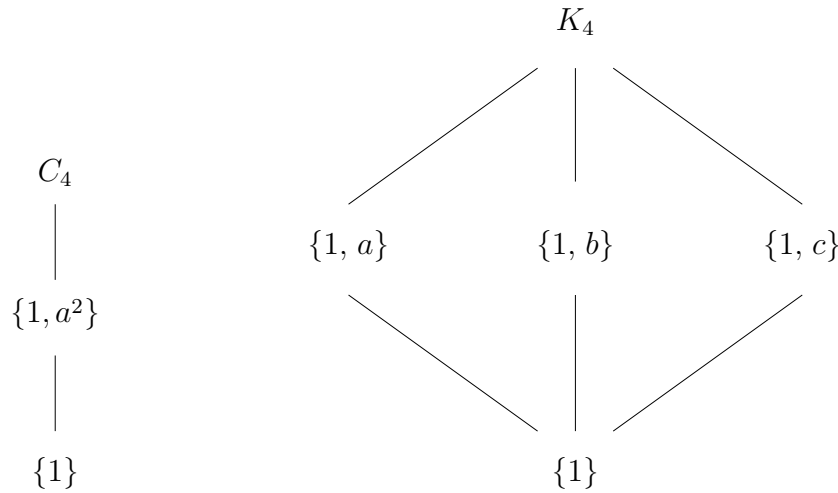
$$K_4 = \{1, a, b, c\}, \text{ where } a^2 = b^2 = c^2 = 1,$$

and the product of two of a , b , and c is the third.

An application of the *Finite Subgroup Test* tells us that $\{1, a\}$, $\{1, b\}$ and $\{1, c\}$ are subgroups of K_4 . On the other hand, if H is a subgroup of K_4 with $|H| \geq 3$ then H must contain 1 and two of a , b and c , and so H must contain the other one (their product); it shows that $H = K_4$ and we can conclude that the subgroups of K_4 are $\{1\}$, $\{1, a\}$, $\{1, b\}$, $\{1, c\}$ and K_4 .

We will be able to provide a different proof when we talk about *Lagrange's Theorem* close to the end of this course.

It is quite useful to picture the subgroups drawing, what is called, the **lattice diagram of the subgroups of a group G** . To do so, given K and H subgroups of G , we draw a line from K up to H whenever $K \subseteq H$. Below you could find the lattice diagrams of C_4 and K_4 .



The following is perhaps the most important subgroup of any group:

Definition 3.19. The **centre** $Z(G)$ of a group G is the set of elements of G that commutes with every element of G ; that is:

$$Z(G) = \{z \in G \mid zg = gz \text{ for all } g \in G\}.$$

Notice that **G is abelian if and only if $Z(G) = G$.**

PROPOSITION 3.20. *The centre $Z(G)$ of a group G is a subgroup.*

PROOF. We will use the *Subgroup Test*. It is clear that $1 \in Z(G)$. For $z, z' \in Z(G)$ we need to check that $zz' \in Z(G)$ and $z^{-1} \in Z(G)$. To do so, we need to check that zz' and z^{-1} commute with any element g of G ; in fact, for $g \in G$ we have that

$$g(zz') = (gz)z' \stackrel{z \in Z(G)}{=} (zg)z' = z(gz') \stackrel{z' \in Z(G)}{=} z(z'g) = (zz')g,$$

which shows that $zz' \in Z(G)$. Now, since $z \in Z(G)$ we have that

$$zg = gz \Rightarrow z^{-1}(zg) = z^{-1}(gz) \Rightarrow g = (z^{-1}g)z \Rightarrow gz^{-1} = z^{-1}g,$$

proving that $z^{-1} \in Z(G)$. □

Examples 3.21.

1. $Z(\mathbb{Z}_4) = \mathbb{Z}_4$ and $Z(K_4) = K_4$ because both groups are abelian.

2. $Z(S_3) = \{\varepsilon\}$

Recall that $S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}$, where $\sigma = (1\ 2\ 3)$ and $\tau = (1\ 2)$. From the Cayley table of S_3 computed in Example 3.4 we can see that σ^i (for $i = 1, 2$) and τ do not commute, since

$$\begin{aligned}\sigma\tau &= \tau\sigma^2 \neq \tau\sigma, \\ \sigma^2\tau &= \tau\sigma \neq \tau\sigma^2.\end{aligned}$$

This implies that $\sigma^i, \tau \notin Z(S_3)$. Now, $\tau\sigma$ and $\tau\sigma^2$ do not commute, so $\tau\sigma, \tau\sigma^2 \notin Z(S_3)$. More precisely:

$$\begin{aligned}\tau\sigma &\neq \tau\sigma^2 = \sigma\tau, \\ \tau\sigma^2 &\neq \tau\sigma = \sigma^2\tau, \\ (\tau\sigma)(\tau\sigma^2) &= \sigma \neq \sigma^2 = (\tau\sigma^2)(\tau\sigma).\end{aligned}$$

This shows that $Z(S_3)$ reduces to the identity element of S_3 , that is, $Z(S_3) = \{\varepsilon\}$.

3.3 Cyclic groups

Recall that **cyclic groups** are groups whose elements are all powers (or multiples) of a fixed element, called the **generator of the group**. More precisely, given $n \geq 1$, the **cyclic group of order n** is the group C_n of order n consisting of powers of a : $C_n = \{1, a, a^2, \dots, a^{n-1}\}$, where $a^n = 1$. We write $C_n = \langle a \rangle$ and call a a **generator of C_n** . By $|C_n| = n$, we mean that the elements $1, a, a^2, \dots, a^{n-1}$ are **all distinct elements** of C_n .

The **Cayley table of C_n** is determined completely by the **exponent laws** and the condition $a^n = 1$.

C_n	1	a	a^2	$\dots$	a^{n-2}	a^{n-1}
1	1	a	a^2	$\dots$	a^{n-2}	a^{n-1}
a	a	a^2	a^3	$\dots$	a^{n-1}	1
a^2	a^2	a^3	a^4	$\dots$	1	a
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
a^{n-2}	a^{n-2}	a^{n-1}	1	$\dots$	a^{n-4}	a^{n-3}
a^{n-1}	a^{n-1}	1	a	$\dots$	a^{n-3}	a^{n-2}

When filling in the Cayley table of C_n , we will deal with elements of the form a^k , where $k > n$. What do we do? We need to find $i \in \{0, 1, \dots, n-1\}$ such that $a^k = a^i$; in other words, we need to reduce the exponent k modulo n . To do so, we will apply the *Division Algorithm*, the exponent laws and the condition $a^n = 1$. More precisely, using the *Division Algorithm* we can write $k = qn + r$, where $0 \leq r \leq n-1$. Then an application of the exponent laws gives

$$a^k = a^{qn+r} = (a^n)^q \cdot a^r \stackrel{a^n=1}{=} 1^q \cdot a^r = a^r.$$

This section is devoted to the study of cyclic groups and their subgroups.

Definition 3.22. Let g be an element of a group G , the subset of G consisting of all the powers of g , namely, $\langle g \rangle = \{g^k \mid k \in \mathbb{Z}\}$ is a **subgroup of G** (this follows from the exponent laws, please fill in the details), called the **cyclic subgroup of G generated by g** . Moreover, $\langle g \rangle$ is the **smallest subgroup of G containing the element g** .

We will say that G is a **cyclic group** if there exists $g \in G$ such that $G = \langle g \rangle$. Notice that if g has order n , then G is a **cyclic group of order n** ; if g has order infinity, we say that G is a **cyclic group of order infinity**.

Remark 3.23. If the operation in the group G is **written additively**, the **cyclic subgroup generated by g** consists on the multiples of g :

$$\langle g \rangle = \{kg \mid k \in \mathbb{Z}\} = \mathbb{Z}g.$$

Remember that in additive notation “powers” are replaced by “multiples”.

Examples 3.24.

- 1.** The cyclic subgroup generated by $\sigma \in S_3$ is $\langle \sigma \rangle = \{\varepsilon, \sigma, \sigma^2\}$.
- 2.** The cyclic subgroup generated by $2 \in \mathbb{Z}$ consists on the even numbers.
- 3.** The group $(\mathbb{Z}, +)$ is cyclic and 1 and -1 are the only generators.

If $k \in \mathbb{Z}$ we have $k = k \cdot 1 \in \langle 1 \rangle$ and $k = (-k) \cdot (-1) \in \langle -1 \rangle$. Thus $\mathbb{Z} = \langle 1 \rangle = \langle -1 \rangle$. Moreover, if n is an integer different from 1 and -1 then $n\mathbb{Z} \neq \mathbb{Z}$ since, for example, $n + 1 \notin n\mathbb{Z}$.

- 4.** The group $(\mathbb{Z}_n, +)$ is cyclic with generator $\bar{1}$ and order n .

Recall that $\mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$. Given $\bar{k} \in \mathbb{Z}_n$ we have that $\bar{k} = k\bar{1}$ and so $\bar{k} \in \langle \bar{1} \rangle$. It follows that $\mathbb{Z}_n = \langle \bar{1} \rangle$, as desired.

If we compare the additive cyclic group $\mathbb{Z}_n$ with the multiplicative cyclic group $C_n = \langle a \rangle = \{1 = a^0, a, a^2, \dots, a^{n-1}\}$, we will notice that they are quite similar! In fact, there is a bijective correspondence:

$$\begin{array}{ccccccc} C_n = & \{a^0, & a^1, & a^2, & \dots, & a^{n-1}\} \\ & \updownarrow & \updownarrow & \updownarrow & & \updownarrow \\ \mathbb{Z}_n = & \{\bar{0}, & \bar{1}, & \bar{2}, & \dots, & \overline{n-1}\} \end{array}$$

Their Cayley tables are the same except for notation, so these two groups are isomorphic. See Theorem 3.42 below for a more formal proof.

- 5.** $\mathbb{Z}$ is a cyclic group of order infinity.
- 6.** The (multiplicative) group $\mathbb{Z}_9^* = \{\bar{1}, \bar{2}, \bar{4}, \bar{5}, \bar{7}, \bar{8}\}$ consisting on the units of $\mathbb{Z}_9$ is a cyclic group. In fact, $\mathbb{Z}_9^* = \langle \bar{2} \rangle$ because

$$\bar{2}^0 = \bar{1}, \bar{2}^1 = \bar{2}, \bar{2}^2 = \bar{4}, \bar{2}^3 = \bar{8}, \bar{2}^4 = \bar{16} = \bar{7}, \bar{2}^5 = \bar{14} = \bar{5}, \bar{2}^6 = \bar{10} = \bar{9},$$

which shows that all the elements of $\mathbb{Z}_8^*$ are powers of $\bar{2}$.

7. The (multiplicative) **group** $\mathbb{Z}_8^* = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$ **is not cyclic**. Recall that $\mathbb{Z}_8^*$ consists on the units of $\mathbb{Z}_8$, that is, elements of $\mathbb{Z}_8 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7}\}$ having a multiplicative inverse.

To prove that $\mathbb{Z}_8^*$ is not cyclic, we will show that $\langle a \rangle \neq \mathbb{Z}_8^*$ for every $a \in \mathbb{Z}_8^*$. It is clear that $\langle \bar{1} \rangle = \{\bar{1}\}$. Moreover:

$$\begin{aligned}\langle \bar{3} \rangle &= \{\bar{1}, \bar{3}\}, & \text{since } \bar{3}^2 = \bar{9} = \bar{1} \text{ in } \mathbb{Z}_8; \\ \langle \bar{5} \rangle &= \{\bar{1}, \bar{5}\}, & \text{since } \bar{5}^2 = \bar{25} = \bar{1} \text{ in } \mathbb{Z}_8; \\ \langle \bar{7} \rangle &= \{\bar{1}, \bar{7}\}, & \text{since } \bar{7}^2 = \bar{49} = \bar{1} \text{ in } \mathbb{Z}_8.\end{aligned}$$

We have proved that every element of $\mathbb{Z}_8^*$ has order 2, since $\mathbb{Z}_8^*$ has four elements, we can conclude that $\mathbb{Z}_8^*$ is isomorphic to the Klein group.

In what follows, we study the cyclic subgroup $\langle g \rangle$ generated by an element g more closely.

THEOREM 3.25. *Let g be an element of a group G .*

(i) **If the cyclic subgroup $\langle g \rangle$ is finite**, then there exists a smallest positive integer n such that $g^n = 1$. In such a case:

- (a) $g^k = 1$ if and only if $n|k$.
- (b) $g^k = g^m$ if and only if $k \equiv m \pmod{n}$.
- (c) $\langle g \rangle = \{1, g, g^2, \dots, g^{n-1}\}$ and the elements $1, g, g^2, \dots, g^{n-1}$ are all distinct.

(ii) **If the cyclic subgroup generated $\langle g \rangle$ is infinite**, then:

- (d) $g^k = 1$ if and only if $k = 0$.
- (e) $g^k = g^m$ if and only if $k = m$.
- (f) $\langle g \rangle = \{\dots, g^{-2}, g^{-1}, 1, g, g^2, \dots\}$ and all these powers of g are distinct.

PROOF. (i) Suppose that $\langle g \rangle$ is finite. Then the powers $g, g^2, g^3, \dots$ can not be all distinct, so we can find two distinct positive integers k and m such that $g^k = g^m$. Assume, for example, that $k < m$. Then $g^{m-k} = 1$, where $m - k > 0$, so by the *Well-Ordering Axiom* there exists a smallest positive integer n such that $g^n = 1$.

(a) Assume first that $n|k$. Then we can write $k = qn$ for some $q \in \mathbb{Z}$. Thus:

$$g^k = g^{qn} = (g^n)^q = 1^q = 1.$$

Conversely, if $g^k = 1$, applying the *Division Algorithm* we can write $k = qn + r$ with $0 \leq r < n$. Then

$$g^r = g^{k-qn} = g^k (g^n)^{-q} = 1,$$

which implies $r = 0$ since $r < n$ and n is the smallest integer with that property. Thus $k = qn$ and $n|k$, concluding the proof of (a).

(b) $g^k = g^m \Leftrightarrow g^{k-m} = 1 \Leftrightarrow n|(k-m) \Leftrightarrow k \equiv m \pmod{n}$

(c) Clearly, $\{1, g, \dots, g^{n-1}\} \subseteq \langle g \rangle$. Let us prove that $\langle g \rangle \subseteq \{1, g, \dots, g^{n-1}\}$. To this end, take $g^k \in \langle g \rangle$. Proceed like in the proof of (a) and write $k = nq + r$ with $0 \leq r < n$. Then:

$$g^k = g^{nq+r} = (g^n)^q g^r = g^r \in \{1, g, \dots, g^{n-1}\}.$$

To finish assume that two of $1, g, g^2, \dots, g^{n-1}$ are equal, say $g^k = g^m$, where $0 \leq k < m < n$. Then $g^{m-k} = 1$ and since $0 < m - k < n$ the minimality of n implies that $m - k = 0$. Thus the elements $1, g, g^2, \dots, g^{n-1}$ are distinct.

(ii) Assume that $\langle g \rangle$ is infinite.

(d) It is clear that $g^0 = 1$. If $g^k = 1$ with $k \neq 0$ then also $g^{-k} = (g^k)^{-1} = 1$. But it would imply that $\langle g \rangle$ is finite, which would contradict our hypothesis. Thus $g^k = 1$ implies $k = 0$.

(e) $g^k = g^m \Leftrightarrow g^{k-m} = 1 \Leftrightarrow k - m = 0$ by (d).

(f) By definition $\langle g \rangle = \{\dots, g^{-2}, g^{-1}, 1, g, g^2, \dots\}$. We only need to point out that all the powers of g are distinct by (e). $\square$

A direct consequence follows:

COROLLARY 3.26. *The order $\text{o}(g)$ of an element g coincides with the cardinal of the cyclic subgroup $\langle g \rangle$ generated by g .*

Example 3.27. We proved in Example 3.24 (7) that the order of each element in $\mathbb{Z}_8^* = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$ is 2. Corollary 3.26 yields $|\langle \bar{3} \rangle| = |\langle \bar{5} \rangle| = |\langle \bar{7} \rangle| = 2$.

3.3.1 Some important properties of Cyclic Groups

Notice that **a cyclic group can have more than one generator**. For instance, consider the additive cyclic group $\mathbb{Z}_{12} = \{\bar{0}, \bar{1}, \bar{2}, \dots, \bar{11}\}$, which we know is generated by $\bar{1}$. We can easily prove that $\bar{5}$ also generates $\mathbb{Z}_{12}$, that is to say that $\mathbb{Z}_{12} = \langle \bar{5} \rangle = \{k\bar{5} \mid k \in \mathbb{Z}\}$.

To prove that $\bar{5}$ is a generator of $\mathbb{Z}_{12}$, we need to show that any element $\bar{x} \in \mathbb{Z}_{12}$ can be written as $\bar{x} = \bar{5} + \bar{5} + \dots^{(k)} + \bar{5} = k\bar{5}$ for some $k \in \mathbb{Z}$. In fact:

$$\begin{aligned} \bar{0} &= 0 \cdot \bar{5}, \\ \bar{1} &= 5 \cdot \bar{5}, & \text{since } \overline{25} = \bar{1}, & \text{ because } 25 \equiv 1 \pmod{12}, \\ \bar{2} &= 10 \cdot \bar{5}, & \text{since } \overline{50} = \bar{2}, & \text{ because } 50 \equiv 2 \pmod{12}, \\ \bar{3} &= 3 \cdot \bar{5}, & \text{since } \overline{15} = \bar{3}, & \text{ because } 15 \equiv 3 \pmod{12}, \\ \bar{4} &= 8 \cdot \bar{5}, & \text{since } \overline{40} = \bar{4}, & \text{ because } 40 \equiv 4 \pmod{12}, \\ &\vdots \end{aligned}$$

In the next result, we provide, among other properties of cyclic groups, a necessary and sufficient condition for an element of a (finite) cyclic group to be a generator.

THEOREM 3.28. *Let $G = \langle g \rangle$ be a cyclic group with generator g . Then:*

- (1) G is abelian.
- (2) Every subgroup of G is cyclic.
- (3) Suppose that $\text{o}(g) = n$. Then $G = \langle g^k \rangle$ if and only if $\gcd(k, n) = 1$.

PROOF. (1) If $x, y \in G$, then $x = g^k$, $y = g^m$, for $k, m \in \mathbb{Z}$. Then the exponent laws give

$$xy = g^k g^m = g^{k+m} = g^{m+k} = g^m g^k = yx,$$

which proves that G is abelian.

(2) Suppose that H is a subgroup of G . If $H = \{1\}$, then we are done because $H = \langle 1 \rangle$ is cyclic. Otherwise, let $g^k \in H$, $k \neq 0$. We can assume (without loss of generality) that $k > 0$, since $g^{-k} = (g^k)^{-1} \in H$ because H is a subgroup of G . By the *Well-Ordering Principle* we can choose m to be the smallest positive integer such that $g^m \in H$. We claim that $\langle g^m \rangle = H$. In fact, it is clear that $\langle g^m \rangle \subseteq H$, since $g^m \in H$ and H is a subgroup of G . For the other containment, take $g^s \in H$ and apply the *Division Algorithm* to write $s = qm + r$, with $0 \leq r < m$. Our goal is to show that $r = 0$; if $r \neq 0$, then $g^r = (g^m)^{-q}g^s \in H$, by the exponent laws. But it contradicts the minimality of m . Therefore $r = 0$ and $g^s = (g^m)^q \in \langle g^m \rangle$, as desired.

(3) Suppose first that g^k is a generator, that is, $G = \langle g^k \rangle$. Then, in particular, $g \in \langle g^k \rangle$. Thus $g = (g^k)^m = g^{km}$, for some $m \in \mathbb{Z}$. Theorem 4.5 implies that $n \mid (1 - km)$, and so $1 - km = qn$ for $q \in \mathbb{Z}$; that is, $1 = km + qn$, which implies that $\gcd(k, n) = 1$. Conversely, if $\gcd(k, n) = 1$, then $1 = xk + yn$ for some integers x and y . Hence:

$$g = g^1 = g^{xk+yn} = (g^k)^x (g^n)^y = (g^k)^x (1)^y = (g^k)^x \in \langle g^k \rangle,$$

which implies that $G = \langle g^k \rangle$. □

Remarks 3.29. We have just proved in Theorem 3.28 (1) that **every cyclic group is abelian**. The **converse does NOT hold**, i.e., there are abelian groups which are not cyclic. For instance, $\mathbb{Z}_8^*$.

Coming back to $\mathbb{Z}_{12}$, Theorem 3.28 (3) tells us that all its generators are the elements $\bar{k}$ such that $\gcd(k, 12) = 1$, namely: $\bar{1}$, $\bar{5}$, $\bar{7}$, and $\bar{11}$.

You might be wondering what can be said about the generators of an infinite cyclic group. Well ... I am so sorry to say this, but we are not going to solve this mystery here. We will do it in the tutorial sessions.

3.3.2 Fundamental Theorem of Finite Cycle Groups

The *Fundamental Theorem of Finite Cycle Groups*, gives a **precise description of all subgroups of a finite cyclic group** G . It tells us not only how many subgroups G has but also how to obtain them. To be more precise: it says that G has a unique cyclic subgroup of order k for every divisor k of n and those are the only subgroups of G .

THEOREM 3.30. FUNDAMENTAL THEOREM OF FINITE CYCLIC GROUPS.

Let $G = \langle g \rangle$ be a cyclic group of order n .

- (1) If H is a subgroup of G , then $H = \langle g^d \rangle$ for some $d|n$.
- (2) If H is any subgroup of G with order $|H| = k$, then $k|n$.
- (3) If $k|n$, then $\langle g^{n/k} \rangle$ is the unique subgroup of G of order k .

Another interpretation of the *Fundamental Theorem of Finite Cyclic Groups*:

the subgroups of G are in one-to-one correspondence with the positive divisors of n

In particular, the order of any subgroup of G divides the order of G . This result, known as *Lagrange's Theorem*, is true for any finite group, cyclic or not. To prove the *Fundamental Theorem of Finite Cyclic Groups*, we first need to a preliminary result:

LEMMA 3.31. Let g be an element of a group G of order n . If $d|n$ then the order of the element g^d is n/d .

PROOF. Let $k = \frac{n}{d}$. Then $(g^d)^k = g^{dk} = g^n = 1$. It remains to show that k is the smallest positive integer with that property. Suppose that $(g^d)^r = 1$ for some $r \geq 1$. Then $g^{dr} = 1$, so $n|dr$ by Theorem 4.5 (a). We then can write that $dr = qn$, for some $q \geq 1$. But then

$$dr = qn = q(dk) = d(qk),$$

and so $r = qk$, which implies that $r \geq k$, as desired. □

PROOF OF THEOREM 3.30.

(1) The result is clear for $|H| = 1$. Otherwise, from Theorem 3.28 (2) we get that $H = \langle g^m \rangle$ for some $m > 0$. Let $d = \gcd(m, n)$. Then $d|n$ and we only need to show that $H = \langle g^d \rangle$.

From $d|m$, we have that $m = qd$ for some $q \in \mathbb{Z}$. Thus $g^m = (g^d)^q \in \langle g^d \rangle$ and therefore $H = \langle g^m \rangle \subseteq \langle g^d \rangle$. On the other hand, by Bézout's Lemma we have that $d = xm + yn$, where $x, y \in \mathbb{Z}$, and so

$$g^d = g^{xm+yn} = (g^m)^x (g^n)^y = (g^m)^x (1)^y = (g^m)^x \in \langle g^m \rangle = H \Rightarrow \langle g^d \rangle \subseteq H,$$

finishing the proof of (1).

(2) Let H be a subgroup of G of order k . From (1) we obtain that $H = \langle g^d \rangle$, for some $d|n$. If $k = |H|$, then by Lemma 3.31 we have that $k = |H| = o(g^d) = n/d$, which shows that $k|n$.

(3) Suppose that K is any subgroup of G of order k . By (1) we have that $K = \langle g^m \rangle$ where $m|n$. Then from Lemma 3.31 we obtain that

$$k = |K| = o(g^m) = \frac{n}{m} \Rightarrow m = \frac{n}{k} \Rightarrow K = \langle g^{\frac{n}{k}} \rangle,$$

which finishes the proof. $\square$

COROLLARY 3.32. Let $G = \langle g \rangle$ be a cyclic group of order n . Then for any positive integer m we have that $\langle g^m \rangle = \langle g^d \rangle$, where $d = \gcd(m, n)$. Moreover, the order of g^m is n/d .

PROOF. Take m an arbitrary positive integer and let $d = \gcd(m, n)$. Proceeding like in the proof of (1) in the *Fundamental Theorem of Finite Cyclic Groups*, we have that $\langle g^m \rangle = \langle g^d \rangle$. On the other hand, we know that

$$o(g^m) = |\langle g^m \rangle| = |\langle g^d \rangle| = o(g^d) = \frac{n}{d},$$

by Lemma 3.31. $\square$

Example 3.33. In the (additive) cyclic group $\mathbb{Z}_{30}$, find the order of $\overline{18}$.

Notice that $\mathbb{Z}_{30} = \langle \overline{1} \rangle$ and $\overline{18} = 18 \cdot \overline{1}$. We will use Corollary 3.32. In this case, $n = 30$ and $m = 18$. Since $\gcd(30, 18) = 6$, we have that $\langle \overline{18} \rangle = \langle \overline{6} \rangle$ and the order of $\overline{18}$ in $\mathbb{Z}_{30}$ is $\frac{30}{6} = 5$. In other words, the subgroup generated by $\overline{18}$ in $\mathbb{Z}_{30}$ has 5 elements: $\langle \overline{18} \rangle = \langle \overline{6} \rangle = \{\overline{0}, \overline{6}, \overline{12}, \overline{18}, \overline{24}\}$, because in $\mathbb{Z}_{30}$ we have that $0 \cdot \overline{6} = \overline{0}$, $1 \cdot \overline{6} = \overline{6}$, $2 \cdot \overline{6} = \overline{12}$, $3 \cdot \overline{6} = \overline{18}$, $4 \cdot \overline{6} = \overline{24}$, $5 \cdot \overline{6} = \overline{30} = \overline{0}$.

The *Fundamental Theorem of Finite Cyclic Groups* provides us an easy way to find all the subgroups of a finite cyclic group. Suppose that $G = \langle g \rangle$ is a cyclic group of order n . To draw the corresponding lattice diagram of G notice that

$$\langle g^m \rangle \subseteq \langle g^k \rangle \Leftrightarrow k|m$$

In fact, suppose first that $\langle g^m \rangle \subseteq \langle g^k \rangle$ and let us show that $k|m$. From $\langle g^m \rangle \subseteq \langle g^k \rangle$ we have, in particular, that $g^m \in \langle g^k \rangle$, so there exists an integer t such that $g^m = (g^k)^t = g^{kt}$. Thus $m \equiv kt \pmod{n}$ by Theorem 4.5 (i) (b). But then $n|(m - kt)$, which yields that $m - kt = nu$ for some integer u . On the other hand, since we are assuming that $k|n$ we obtain that $n = kv$ for some integer v . Putting it altogether we have

$$m = kt + nu \stackrel{n=kv}{=} kt + (kv)u = k(t + vu) \Rightarrow k|m$$

Suppose now that $k|m$ and let us show that $\langle g^m \rangle \subseteq \langle g^k \rangle$. To do so, it is enough to prove that $g^m \in \langle g^k \rangle$. From $k|m$ we derive that $m = ku$ for some integer u , and using the Exponent Laws we have

$$g^m = g^{ku} = (g^k)^u \in \langle g^k \rangle.$$

Example 3.34. Find all subgroups of the multiplicative cyclic group C_{12} and draw its lattice diagram.

Write $C_{12} = \langle g \rangle$, where $\text{o}(g) = 12$. Since the positive divisors of 12 are 1, 2, 3, 4, 6, 12, the *Fundamental Theorem of Finite Cyclic Groups* allows us to conclude that C_{12} has exactly 6 subgroups. To determine the unique subgroup of each of these orders, we use Lemma 3.31. Clearly, $\{1\} = \langle 1 \rangle = \langle g^{12} \rangle$.

$$\text{Order 2: } 2 = \text{o}(g^d) = \frac{12}{d} \Rightarrow d = 6; \quad \text{the subgroup of order 2 is } \langle g^6 \rangle;$$

$$\text{Order 3: } 3 = \text{o}(g^d) = \frac{12}{d} \Rightarrow d = 4; \quad \text{the subgroup of order 3 is } \langle g^4 \rangle;$$

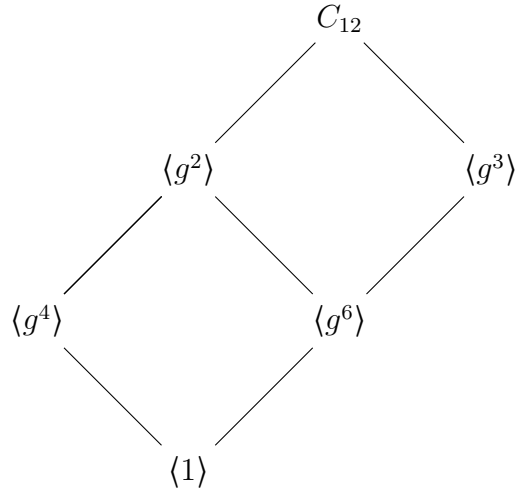
$$\text{Order 4: } 4 = \text{o}(g^d) = \frac{12}{d} \Rightarrow d = 3; \quad \text{the subgroup of order 4 is } \langle g^3 \rangle;$$

$$\text{Order 6: } 6 = \text{o}(g^d) = \frac{12}{d} \Rightarrow d = 2; \quad \text{the subgroup of order 6 is } \langle g^2 \rangle;$$

To picture the diagram, notice that

$$\langle g^6 \rangle \subseteq \langle g^2 \rangle \quad \text{and also} \quad \langle g^6 \rangle \subseteq \langle g^3 \rangle$$

$$\langle g^4 \rangle \subseteq \langle g^2 \rangle$$



Example 3.35. Find all subgroups of the additive cyclic group $\mathbb{Z}_{100}$ and draw the lattice diagram.

Since the positive divisors of 100 are 1, 2, 4, 5, 10, 20, 25, 50 and 100, by the *Fundamental Theorem of Finite Cyclic Groups* we can assert that $\mathbb{Z}_{100}$ has exactly 9 subgroups of orders the mentioned divisors of 100. Clearly, $\{\bar{0}\} = \langle \bar{0} \rangle$. To determine the remaining subgroups, we use Lemma 3.31 in its additive version:

$$2 = o(d\bar{1}) = \frac{100}{d} \Rightarrow d = 50; \text{ the subgroup of order 2 is } \langle \bar{50} \rangle = 50\mathbb{Z}_{100};$$

$$4 = o(d\bar{1}) = \frac{100}{d} \Rightarrow d = 25; \text{ the subgroup of order 4 is } \langle \bar{25} \rangle = 25\mathbb{Z}_{100};$$

$$5 = o(d\bar{1}) = \frac{100}{d} \Rightarrow d = 20; \text{ the subgroup of order 5 is } \langle \bar{20} \rangle = 20\mathbb{Z}_{100};$$

$$10 = o(d\bar{1}) = \frac{100}{d} \Rightarrow d = 10; \text{ the subgroup of order 10 is } \langle \bar{10} \rangle = 10\mathbb{Z}_{100};$$

$$20 = o(d\bar{1}) = \frac{100}{d} \Rightarrow d = 5; \text{ the subgroup of order 20 is } \langle \bar{5} \rangle = 5\mathbb{Z}_{100};$$

$$25 = o(d\bar{1}) = \frac{100}{d} \Rightarrow d = 4; \text{ the subgroup of order 25 is } \langle \bar{4} \rangle = 4\mathbb{Z}_{100};$$

$$50 = o(d\bar{1}) = \frac{100}{d} \Rightarrow d = 2; \text{ the subgroup of order 50 is } \langle \bar{2} \rangle = 2\mathbb{Z}_{100};$$

To picture the diagram we use that

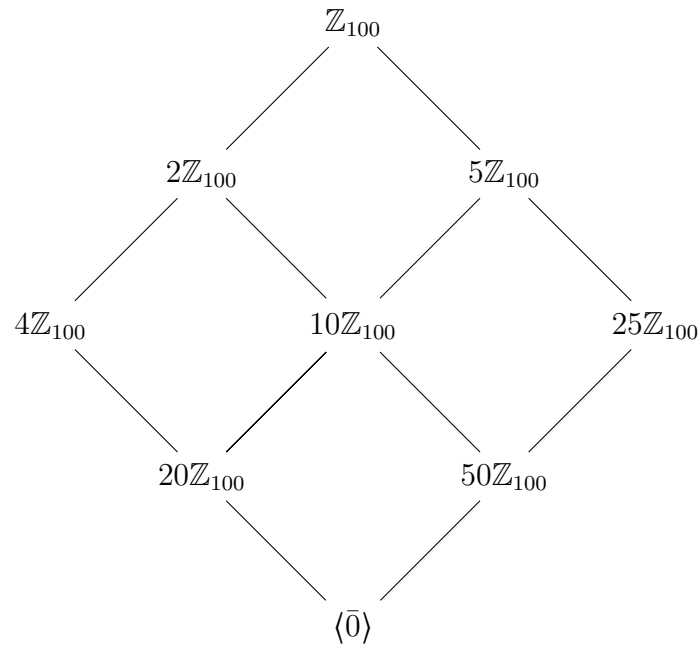
$$m\mathbb{Z} \subseteq k\mathbb{Z} \Leftrightarrow k|m$$

In our case, we have the following containments:

$$20\mathbb{Z}_{100} \subseteq 4\mathbb{Z}_{100} \subseteq 2\mathbb{Z}_{100} \quad \text{and} \quad 20\mathbb{Z}_{100} \subseteq 10\mathbb{Z}_{100} \subseteq 5\mathbb{Z}_{100}$$

$$50\mathbb{Z}_{100} \subseteq 10\mathbb{Z}_{100} \subseteq 2\mathbb{Z}_{100} \quad \text{and} \quad 50\mathbb{Z}_{100} \subseteq 25\mathbb{Z}_{100} \subseteq 5\mathbb{Z}_{100}$$

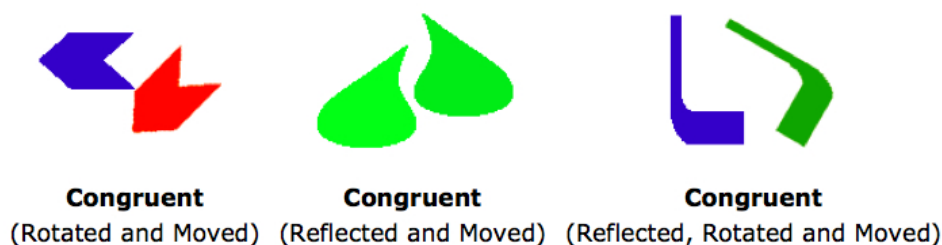
Using the considerations above, we obtain the diagram below. Please, don't worry if you do not come up with a nice diagram at first. It takes a bit of practice. The important thing here is that you draw all the lines, since this is what it tells us the containment relations between the subgroups.



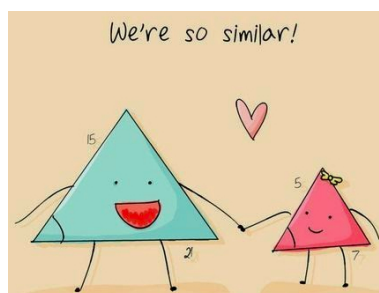
3.4 Homomorphisms and isomorphisms

Two objects are “isomorphic” if they have the same shape or structure. In other words, two objects might appear to be different might they are identical in form. This idea of being isomorphic (i.e. having the same structure) appears in all the branches of Mathematics.

In Geometry there are several kinds of isomorphism, for instance, congruence and similarity. Two geometry figures are congruent if there exists a plane motion (translation, rotation or reflection) which makes one figure coincide with the other; in other words, the two figures have the same shape and size. For example:



Regarding to similarity, two geometric figures are similar if one can be obtained from the other one by scaling (enlarging or shrinking) in a uniform way; with additional translation, rotation and reflection might be needed. Similar figures have the same shape but not necessarily the same size.



Here, we will study maps from one group onto another group that preserve the group operation. Roughly speaking, an isomorphism between two groups is a way to transform one group into the other one via a bijective map satisfying a certain property. In such a case, we say that such two groups are isomorphic. More precisely: let G_1 and G_2 be two groups. An **isomorphism** from G_1

to G_2 is a bijective correspondence φ from G_1 to G_2 satisfying the following property: for every pair of elements a and b in G_1 ,

$$\text{if } \varphi(a) = a' \text{ and } \varphi(b) = b', \text{ then } \varphi(ab) = a'b' = \varphi(a)\varphi(b) \quad (*)$$

In §3.1.1 we called two finite groups isomorphic if they have the same Cayley table except for notation. Observe that this terminology is consistent with the one that we have just introduced. If $\varphi : G_1 \rightarrow G_2$ is an isomorphism between two finite groups, then using $(*)$ we can see that φ transform the Cayley table of G_1 into the Cayley table of G_2 ; in fact:

G_1	b	$\xrightarrow{\text{For every } x \text{ replace } x \text{ by } \varphi(x)}$	G_2	b'
a	$\dots \quad \vdots \quad ab$		a'	$\dots \quad \vdots \quad a'b'$

The idea behind two isomorphic groups is the following: If G_1 and G_2 are isomorphic, then we can say that they are indeed the same, except that their elements have different labels. The group G_1 becomes G_2 after relabelling its elements; the map which does the relabelling is precisely an isomorphism from G_1 to G_2 . More formal definitions follow.

Definitions 3.36. Let $(G_1, \cdot_1)$ and $(G_2, \cdot_2)$ be two groups (not necessarily of finite order). A mapping $\varphi : (G_1, \cdot_1) \rightarrow (G_2, \cdot_2)$ is called a **group homomorphism** if

$$\varphi(a \cdot_1 b) = \varphi(a) \cdot_2 \varphi(b),$$

for all $a, b \in G_1$. A **monomorphism** is an injective homomorphism, while a surjective homomorphism is called an **epimorphism**. Homomorphisms which are bijections (one-to-one and onto) are said to be **isomorphisms**. When an isomorphism exists from G_1 to G_2 we say that G_1 is **isomorphic to** G_2 , and we write $G_1 \cong G_2$. Isomorphisms from a group to itself are called **automorphisms**.

At this point a question arises: **How does one recognize if two groups are isomorphic?** Unfortunately, this is not an easy task! There is no way of spontaneously recognize whether two given groups G_1 and G_2 are isomorphic.

To answer it, one needs to find explicitly an isomorphism between them. I would suggest to go through the following steps:

1. Use your intuition to come up with a map $\varphi : G_1 \rightarrow G_2$, which looks like it could be an isomorphism. The more exercises you do, the easier this first step will become.
2. Check that φ is bijective.
3. Take two arbitrary elements a and b in G_1 , compute $\varphi(a)$, $\varphi(b)$ and $\varphi(ab)$ and check that the equality $\varphi(ab) = \varphi(a)\varphi(b)$ holds.

Let us see an example: **the additive group $\mathbb{R}$ is isomorphic to the multiplicative group $\mathbb{R}^+$** . To prove this, we go through the previous steps:

1. The exponential function $f : \mathbb{R} \rightarrow \mathbb{R}^+$, $f(x) = e^x$ might work.
2. f is one-to-one: if $f(a) = f(b)$, that is, $e^a = e^b$, then taking the natural logarithm $\ln$ on both sides, we obtain $a = b$.

 f is surjective: if y is any positive real number, then $y = e^{\ln y} = f(\ln y)$.
3. We all know that $e^{a+b} = e^a \cdot e^b$, for all $a, b \in \mathbb{R}$. It says that

$$f(a+b) = f(a) \cdot f(b)$$

Observe that the operation of $\mathbb{R}$ is $+$, whereas the operation of $\mathbb{R}^+$ is $\cdot$. It justifies why we have used $+$ on the left hand side of the preceding equation, and $\cdot$ on the right hand side of the equation.

Examples 3.37.

1. The map $\varphi : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ given by $\varphi(a) = 3a$ is a **monomorphism** because φ is one-to-one

$$\varphi(a) = \varphi(b) \Rightarrow 3a = 3b \Rightarrow 3(a-b) = 0 \Rightarrow a-b = 0 \Rightarrow a = b,$$

and it satisfies that

$$\varphi(a+b) = 3(a+b) = 3a + 3b = \varphi(a) + \varphi(b),$$

for all $a, b \in \mathbb{Z}$. Observe that φ is not an isomorphism since it is not surjective: there is no integer a such that $\varphi(a) = 4$ because the equation $3x = 4$ has no solutions in $\mathbb{Z}$.

2. The **absolute value map** $\psi : \mathbb{C}^* \rightarrow \mathbb{R}^+$ given by $\psi(z) = |z|$ for all $z \in \mathbb{C}^*$ is a homomorphism (in fact, epimorphism) since

$$\psi(zw) = |zw| = |z||w| = \psi(z)\psi(w), \quad \text{for all } z, w \in \mathbb{C}^*.$$

3. The **determinant map** $GL_n(\mathbb{R}) \rightarrow \mathbb{R}^*$ given by $A \mapsto \det(A)$ is a homomorphism because $\det(AB) = \det(A)\det(B)$, for all matrices A and B , and $\det(AB) \neq 0$. Moreover, the determinant map is an epimorphism. **Why?** Is it a monomorphism?

In what follows, we collect some basic properties of automorphisms and isomorphisms.

THEOREM 3.38. *Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism. Then:*

- (1) φ preserves the identity element: $\varphi(1_{G_1}) = 1_{G_2}$
- (2) φ preserves inverses: $\varphi(g^{-1}) = \varphi(g)^{-1}$ for all $g \in G_1$.
- (3) φ preserves powers: $\varphi(g^k) = \varphi(g)^k$ for all $g \in G_1$ and $k \in \mathbb{Z}$.

PROOF. (1) We have that $\varphi(1_{G_1}) = \varphi(1_{G_1} \cdot 1_{G_1}) = \varphi(1_{G_1}) \cdot \varphi(1_{G_1})$, and by the cancellation law in G_2 we obtain that $\varphi(1_{G_1}) = 1_{G_2}$.

(2) For $g \in G_1$ we have that

$$1_{G_2} \stackrel{(1)}{=} \varphi(1_{G_1}) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}),$$

which implies that $\varphi(g^{-1}) = \varphi(g)^{-1}$.

(3) For $k = 0$, the result follows from (1). We proceed by induction on $k > 0$. For $k = 1$, we have that

$$\varphi(g^1) = \varphi(g) = \varphi(g)^1.$$

Assume that $\varphi(g^k) = \varphi(g)^k$ for some $k > 1$, and consider $\varphi(g^{k+1})$.

$$\varphi(g^{k+1}) = \varphi(g \cdot g^k) = \varphi(g) \cdot \varphi(g^k) = \varphi(g) \cdot \varphi(g)^k = \varphi(g)^{k+1}.$$

Thus (3) holds for $k > 0$ by the Principle Mathematical of Induction. If $k < 0$, write $k = -m$ for $m > 0$. Then by (2) and the previous calculations we have

$$\varphi(g^k) = \varphi(g^{-m}) = \varphi((g^m)^{-1}) = \varphi(g^m)^{-1} = (\varphi(g)^m)^{-1} = \varphi(g)^{-m} = \varphi(g)^k.$$

Thus $\varphi(g^k) = \varphi(g)^k$ for all $k \in \mathbb{Z}$. $\square$

The proof of the following two results is left as an exercise.

THEOREM 3.39. *Let G , G_1 and G_2 denote groups.*

- (1) *The identity map is an isomorphism for every group.*
- (2) *If $\varphi : G \rightarrow G_1$ is an isomorphism, then the inverse $\varphi^{-1} : G_1 \rightarrow G$ is also an isomorphism.*
- (3) *If $\varphi : G \rightarrow G_1$ and $\psi : G_1 \rightarrow G_2$ are isomorphisms, then $\psi \circ \varphi : G \rightarrow G_2$ is also an isomorphism.*

COROLLARY 3.40. *The isomorphic relation $\cong$ is an equivalence relation.*

COROLLARY 3.41. *Then the set $\text{aut}(G)$ consisting of all automorphisms of a group G forms a group under composition. It is called the **automorphism group of G** .*

And what's about non-isomorphic groups? **How can we show that two groups are not isomorphic?** You might not believe that it is actually easier to show that two groups are not isomorphic than to show that they are. Keeping in mind that isomorphic groups are essentially the same (just except for notation, i.e., different name for their elements and/or operations) it is natural to think that they will share the same properties. More precisely:

If a group G_1 has a property and G_2 does not have it, or vice versa, we can conclude that G_1 and G_2 are not isomorphic.

Here are some examples of properties to look out for are:

1. G_1 is commutative, and G_2 is not.
 2. G_1 is cyclic, and G_2 is not.
 3. G_1 is generated by two elements and G_2 is not generated by any choice of two of its elements.
-

4. Every element of G_1 is the square of an element of G_1 , whereas G_2 does not have this property.

For example, the additive groups $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_4$ are not isomorphic, since $\mathbb{Z}_4$ is cyclic and $\mathbb{Z}_2 \times \mathbb{Z}_2$ isn't (recall that all the elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$ have order 2). Let us see another example: **the additive group $\mathbb{R}$ is not isomorphic to the multiplicative group $\mathbb{R}^*$** . Notice that $\mathbb{R}^*$ have an element distinct from 1 such that its square is 1, namely, $(-1)^2 = (-1)(-1) = 1$. Observe that 1 is the identity element of $\mathbb{R}^*$ and we have just seen that isomorphisms preserve the identity element. So this property interpreted in the additive group $\mathbb{R}$ would tell us that $\mathbb{R}$ has $a \in \mathbb{R}$, $a \neq 0$, such that $a + a = 0$, which clearly does not happen. Therefore, we can conclude that these two groups are not isomorphic.

We close the section by proving that there are essentially two cyclic groups:

THEOREM 3.42. ISOMORPHISM OF CYCLIC GROUPS.

- (1) *For every positive integer n , every cyclic group of order n is isomorphic to $\mathbb{Z}_n$. Thus, any two cyclic groups of order n are isomorphic.*
- (2) *Every cyclic group of order infinity is isomorphic to $\mathbb{Z}$, and therefore any two cyclic groups of order infinity are isomorphic.*

PROOF. (1). Let $C_n = \langle a \rangle$ be a (multiplicative) cyclic group of order n . Then

$$\langle a \rangle = \{1, a, a^2, \dots, a^{n-1}\}.$$

Comparing this group with the additive cyclic group $\mathbb{Z}_n = \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}\}$,

$$\begin{array}{ccccccc} \langle a \rangle = & \{a^0, & a^1, & a^2, & \dots, & a^{n-1}\} \\ & \updownarrow & \updownarrow & \updownarrow & & \updownarrow \\ \mathbb{Z}_n = & \{\bar{0}, & \bar{1}, & \bar{2}, & \dots, & \overline{n-1}\} \end{array}$$

one realises that the map $\varphi : \mathbb{Z}_n \rightarrow \langle a \rangle$ given by $\varphi(i) = a^i$ is a bijective map. But this map also satisfies that

$$\varphi(i+j) = a^{i+j} = a^i a^j = \varphi(i)\varphi(j),$$

which shows that φ is an isomorphism from $\mathbb{Z}_n$ to $\langle a \rangle$.

(2). Suppose that $\langle a \rangle$ is a cyclic group of order infinity. Then

$$\langle a \rangle = \{\dots, a^{-2}, a^{-1}, 1, a, a^2, \dots\}.$$

Proceeding like in (1) we compare this group with $\mathbb{Z}$:

$$\begin{array}{ccccccccc} \langle a \rangle = & \{\dots, & a^{-2}, & a^{-1}, & 1, & a^1 & a^2, & \dots\} \\ & & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \\ \mathbb{Z} = & \{\dots, & -2, & -1, & 0, & 1, & 2, & \dots\} \end{array}$$

Then it is clear that the map $\varphi : \mathbb{Z} \rightarrow \langle a \rangle$ given by $\varphi(i) = a^i$ is the desired isomorphism. $\square$

Chapter 4

Lagrange's Theorem

In this short chapter we will prove Lagrange's Theorem, one of the most important results in (finite) group theory. To do so, we first need to introduce the required machinery: cosets.

4.1 Cosets

The notion of a coset was introduced by E. Galois in 1930 but was G. A. Miller the one who name them cosets.

Definition 4.1. Let H be a subgroup of a group G and $a \in G$. The set $Ha = \{ha \mid h \in H\}$ is called the **right coset** of H in G **generated** by a , while $aH = \{ah \mid h \in H\}$ is called the **left coset** of H in G **generated** by a .

Observe that in **additive notation the left and right cosets** are:

$$H + a = \{h + a \mid h \in H\}, \quad a + H = \{a + h \mid h \in H\}$$

It is important to notice that **cosets in a group G are subsets of G .**

Cosets are not elements in G !

Example 4.2. Find all the left and right cosets of $H = \{\varepsilon, (1\ 3)\}$ in S_3 . Recall that $S_3 = \{\varepsilon, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$. Then we have

$$\begin{aligned}\varepsilon H &= H = (1\ 3)H, \\ (1\ 2)H &= \{(1\ 2)\varepsilon, (1\ 2)(1\ 3)\} = \{(1\ 2), (1\ 3\ 2)\} \\ (2\ 3)H &= \{(2\ 3)\varepsilon, (2\ 3)(1\ 3)\} = \{(2\ 3), (1\ 2\ 3)\}\end{aligned}$$

We have that $(1\ 2)H = (1\ 3\ 2)H$ and $(2\ 3)H = (1\ 2\ 3)H$. You will understand why I can make this affirmation without performing any calculations after we study the coset properties. In the meantime, do the calculations yourself to practice a bit. These are all the left cosets of H . Its right cosets are computed in a similar way:

$$\begin{aligned} H\varepsilon &= H = H(1\ 3), \\ H(1\ 2) &= \{(1\ 2), (1\ 2\ 3)\} = H(1\ 2\ 3), \\ H(2\ 3) &= \{(2\ 3), (1\ 3\ 2)\} = H(1\ 3\ 2). \end{aligned}$$

Notice that, for instance, $H(1\ 2) \neq (1\ 2)H$.

Example 4.3. Find all the cosets of the subgroup $4\mathbb{Z}$ in the additive group $\mathbb{Z}$. In this case, since $\mathbb{Z}$ is an abelian group, it is enough to compute the right (or the left) cosets.

Given $a \in \mathbb{Z}$ the right coset generated by a is $4\mathbb{Z} + a$. But

$$4\mathbb{Z} + a = a + 4\mathbb{Z} = \bar{a} \quad \text{in } \mathbb{Z}_4.$$

We only have four right (or left) cosets, which are the four residue classes modulo 4:

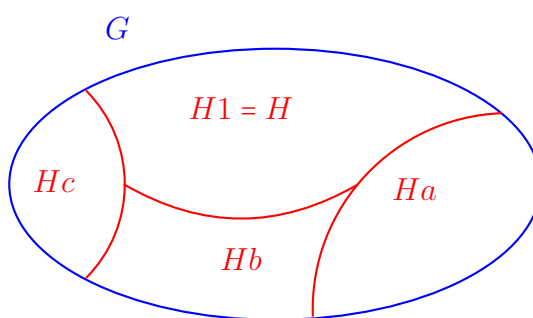
$$\begin{aligned} 4\mathbb{Z} + 0 &= \{\dots, -12, -8, -4, 0, 4, 8, 12, \dots\} \\ 4\mathbb{Z} + 1 &= \{\dots, -11, -7, -3, 1, 5, 9, 13, \dots\} \\ 4\mathbb{Z} + 2 &= \{\dots, -10, -6, -2, 2, 6, 10, 14, \dots\} \\ 4\mathbb{Z} + 3 &= \{\dots, -9, -5, -1, 3, 7, 11, 15, \dots\} \end{aligned}$$

4.2 Cosets properties

In this section we will prove several properties of the right and left cosets. Please pay special attention to these since we will be using them several times in the next few chapters, and you will need them next year if you decide to continue this algebra journey. It is therefore very important that you understand them and learn how to apply them.

Let $a \in G$ and H a subgroup of G . Then:

1. $H1 = H = 1H$, that is, H is a right and left coset of itself.
2. It is obvious that $a \in Ha$ and $a \in aH$ since $a = 1 \cdot a = a \cdot 1$ for $1 \in H$.
3. If G is **abelian**, then $Ha = aH$ for all $a \in G$ and all subgroups H of G .
4. **3** might not be true if G is not abelian.
5. If $a \in Hb$, then $Ha = Hb$.
6. $Ha = Hb$ if and only if $ab^{-1} \in H$.
7. $Ha = H$ if and only if $a \in H$.
8. There exists a bijective map from H to Ha . In particular, if H is finite, then any coset Ha has the same number of elements as H and therefore all the cosets have the same number of elements.
9. The family of all the cosets of Ha , as a ranges over G , is a partition of G . It means that $G = \cup_{a \in G} Ha$, and $Ha \neq Hb$ implies $Ha \cap Hb = \emptyset$.



Notice that, in additive notation, properties **6** and **7** become

$$H + a = H + b \Leftrightarrow a - b \in H; \quad H + a = H \Leftrightarrow a \in H$$

For instance, $6\mathbb{Z} + 7 = 6\mathbb{Z} + 1$ since $7 - 1 = 6 \in 6\mathbb{Z}$.

PROOF. Properties **1**, **2** and **3** trivially hold.

4. For $G = S_3$ and $H = \{\varepsilon, (1\ 3)\}$, we have

$$(1\ 2)H = \{(1\ 2), (1\ 3\ 2)\} \neq \{(1\ 2), (1\ 2\ 3)\} = H(1\ 2)$$

5. To show that the cosets Ha and Hb are equal, we must show that they are equal sets (remember that cosets are, by definition, subsets of G). Suppose that $a \in Hb$. Then there exists $h \in H$ such that $a = hb$. If $x \in Ha$, then $x = h_1a$ for some $h_1 \in H$. From $a = hb$ we get that $x = h_1a = h_1(hb) = (hh_1)b \in Hb$. This proves that $Ha \subseteq Hb$. The proof of the containment $Hb \subseteq Ha$ is similar, using now that $b = h^{-1}a$.

6. If $ab^{-1} \in H$, then $ab^{-1} = h$ for some element h in H . From here we obtain that $a = hb \in Hb$. An application of **5** gives that $Ha = Hb$. If $Ha = Hb$, then $a \in Hb$ so $a = hb$ for some $h \in H$. This gives that $ab^{-1} = h \in H$, as desired.

7 follows from **6** by taking $b = 1$.

8. Define $\psi : H \rightarrow Ha$ by $\psi(h) = ha$. Clearly, ψ is surjective because every element of Ha is of the form ha for some $h \in H$ and so $ha = \psi(h)$. Lastly, if $\psi(h_1) = \psi(h_2)$, then $h_1a = h_2a$ and applying the cancellation law in G we get that $h_1 = h_2$, which proves that ψ is injective.

9. We first need to show that any two cosets, say Ha and Hb , are either disjoint or equal. If they are disjoint, then we are done. Otherwise, suppose that $Ha \cap Hb \neq \emptyset$, and let $x \in Ha \cap Hb$. Then there exists $x \in G$ such that $x \in Ha$ and $x \in Hb$, which yield $x = h_1a = h_2b$ for some $h_1, h_2 \in H$. Solving the equation $h_1a = h_2b$ for a , we obtain $a = (h_1^{-1}h_2)b \in Hb$. Hence, property **5** implies that $Ha = Hb$. To finish, we need to check that G is the union of all the cosets. In order to do so, we must show that every element $g \in G$ is in one of the cosets of H . But this clearly holds, since $g = 1 \cdot g \in Hg$.

The considerations above show that the family of all cosets of H is a partition of G , as desired. $\square$

4.3 Lagrange's Theorem

Using cosets properties, proving Lagrange's Theorem becomes an easy task.

THEOREM 4.4. LAGRANGE'S THEOREM. *Let H be a subgroup of a finite group G . Then the order of G is a multiple of the order of H .*

PROOF. Suppose that $Ha_1, Ha_2, \dots, Ha_k$ are the distinct cosets of H in G . Then $G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_k$, which is a disjoint union by Property 9. Next, using Property 8 we have that $|Ha_i| = |H|$ for each i , so

$$|G| = |Ha_1| + \dots + |Ha_k| = |H| + \dots + |H| = k|H|,$$

which shows that the order of G is a multiple of the order of H , as desired. $\square$

A few applications of Lagrange's Theorem follow.

COROLLARY 4.5. *The order of any element of a finite group divides the order of the group.*

PROOF. Let G be a finite group and $g \in G$. By definition the order of g coincides with the order of the cyclic subgroup $\langle g \rangle$ of G generated by g . The result now follows by applying Lagrange's Theorem to $\langle g \rangle$ and G . $\square$

COROLLARY 4.6. *Every group of prime order is cyclic.*

PROOF. Let p be a prime and G a group of order p . Take $g \in G$ with $g \neq 1$ and consider the subgroup $\langle g \rangle$ of G , which is non-trivial because $g \neq 1$. By Lagrange's Theorem we have that $|\langle g \rangle| \mid p$, and so $|\langle g \rangle| = p$, that is, $G = \langle g \rangle$. $\square$

We have seen that every group of order $4 = 2^2$ is either cyclic or isomorphic to the Klein group. We can now use Lagrange's Theorem to extend this result to any prime p .

COROLLARY 4.7. *If $|G| = p^2$ where p is a prime, then either G is cyclic or $g^p = 1$ for every element g in G .*

PROOF. Assume that G is not cyclic and take $g \in G$. Write $H = \langle g \rangle$ for the cyclic subgroup of G generated by g . Then by Corollary 4.5 the order of H ,

namely, $\text{o}(g) = |H|$ divides the order of G , i.e., p^2 . From here we derive that $\text{o}(g) = 1, p$, or p^2 . But $\text{o}(g) \neq p^2$ because G is not cyclic, so $\text{o}(g)$ is either 1 or p . In any case, $g^p = 1$, concluding the proof. $\square$

Definition 4.8. Let G be a group and H a subgroup of G . The **index of H in G** is the number of cosets of H in G . We denote it by $|G : H|$.

Remark 4.9. If G is finite, from the proof of Lagrange's Theorem, we get that the number of elements in G is equal to the number of elements in H , multiplied by the number of cosets of H in G . In other words:

$$|G : H| = \frac{\text{order of } G}{\text{order of } H} = \frac{|G|}{|H|}$$

A subgroup H can have finite index in a group G even if both H and G are infinite. For example, $H = 4\mathbb{Z}$ has index 4 in $G = \mathbb{Z}$ and both $4\mathbb{Z}$ and $\mathbb{Z}$ are infinite.

Chapter 5

Factor groups

In this last chapter, we will learn about a special types of subgroups, called *normal subgroups*, and how to use them to construct a new kind of groups, called *factor groups*, whose elements are cosets.

5.1 Normal subgroups

We have already seen that, in general, left and right cosets do not coincide. When we have the equality for all of them, we say that such a subgroup is normal, although this might not look very normal ...

Definition 5.1. We say that H is a **normal subgroup** of G , or that H is normal in G , if $gH = Hg$ for all $g \in G$. In such a case, we write $H \triangleleft G$.

Examples 5.2.

1. Let G be any group. Then $\{1\} \triangleleft G$ and $G \triangleleft G$ since $g\{1\} = \{g\} = \{1\}g$, $gG = G = Gg$, for all $g \in G$. The centre $Z(G)$ of G is always a **normal subgroup** of G .
2. Every subgroup of an abelian group is normal.
3. Let $G = S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}$, where $\sigma^3 = \tau^2 = \varepsilon$ and $\sigma\tau\sigma = \tau$. Write $H = \{\varepsilon, \tau\}$ and $K = \{\varepsilon, \sigma, \sigma^2\}$. Then one can check that $K \triangleleft S_3$ but H is not normal in S_3 . In fact:

$$H\sigma = \{h\sigma \mid h \in H\} = \{\varepsilon\sigma, \tau\sigma\} = \{\sigma, \tau\sigma\},$$

$$\sigma H = \{\sigma h \mid h \in H\} = \{\sigma\varepsilon, \sigma\tau\} = \{\sigma, \tau\sigma^2\},$$

since $\sigma\tau = \sigma^2\tau\sigma = \sigma^3\tau\sigma^2 = \tau\sigma^2$. Thus $H\sigma \neq \sigma H$, and H is not normal in S_3 . On the other hand, $K \triangleleft S_3$ because

$$\begin{aligned}\alpha K &= K = K\alpha, \text{ for all } \alpha \in K, \\ \tau K &= \{\tau, \tau\sigma, \tau\sigma^2\} = K\tau, \\ \tau\sigma K &= \{\tau\sigma, \tau\sigma^2, \tau\} = K\tau\sigma, \\ \tau\sigma^2 K &= \{\tau\sigma^2, \tau, \tau\sigma\} = K\tau\sigma^2.\end{aligned}$$

A good exercise would be to perform some of these calculations yourself.

Take a look to the Cayley table of S_3 computed in Example 3.4.

The definition of a normal subgroup does not provide us a practical way to check whether a given subgroup is normal, since it requires the computation of all left and right cosets of such subgroup. In what follows, we will learn a more convenient method to test whether a subgroup is normal.

THEOREM 5.3. NORMALITY TEST. *The following conditions are equivalent for a subgroup H of a group G .*

- (1) H is normal in G .
- (2) $gHg^{-1} \subseteq H$ for all $g \in G$.

PROOF. (1) $\Rightarrow$ (2). Suppose that H is a normal subgroup of G . This means that $Ha = aH$ for all $a \in G$. Take $g \in G$ and let us prove that $gHg^{-1} \subseteq H$. Take $x \in gHg^{-1}$, then there exists $h \in H$ such that $x = ghg^{-1}$. Consider the element gh and notice that $gh \in gH \stackrel{(1)}{=} Hg$. Thus $gh = h_1g$ for some $h_1 \in H$. From here we obtain that

$$x = ghg^{-1} = (gh)g^{-1} = (h_1g)g^{-1} = h_1(gg^{-1}) = h_1 \in H,$$

which proves (2).

(2) $\Rightarrow$ (1). Suppose now that $gHg^{-1} \subseteq H$ for all $g \in G$ and let us show that H is normal in G . To do so, we need to check that $gH = Hg$ for all $g \in G$. Take $g \in G$ and apply (2) with the element g^{-1} to get that $g^{-1}H(g^{-1})^{-1} = g^{-1}Hg \subseteq H$. From here, using that $1 = gg^{-1} = g^{-1}g$ we obtain

$$H = (gg^{-1})H(gg^{-1}) = g(g^{-1}Hg)g^{-1} \subseteq gHg^{-1} \stackrel{(2)}{\subseteq} H,$$

which yields that $H = gHg^{-1}$ and so $Hg = gH$, as required. $\square$

PROPOSITION 5.4. *Let $\varphi : G \rightarrow G'$ be a group homomorphism. Then the kernel of φ , defined as,*

$$\ker \varphi = \{k \in G \mid \varphi(k) = 1_{G'}\}$$

is a normal subgroup of G . Moreover, φ is a monomorphism if and only if $\ker \varphi = \{1_G\}$.

PROOF. We first need to check that $\ker \varphi$ is a subgroup of G . To do so, we use the Subgroup Test: take $k, k' \in \ker \varphi$, then using that φ is a group homomorphism we have that

$$\begin{aligned}\varphi(kk') &= \varphi(k)\varphi(k') \stackrel{k, k' \in \ker \varphi}{=} 1_{G'}1_{G'} = 1_{G'} \\ \varphi(k^{-1}) &= \varphi(k)^{-1} \stackrel{k \in \ker \varphi}{=} 1_{G'}\end{aligned}$$

which imply that $kk', k^{-1} \in \ker \varphi$, and so $\ker \varphi$ is a subgroup of G . It remains to show that $\ker \varphi$ is normal in G . To do so, we check that condition (2) in the *Normality Test* is satisfied. Take $g \in G$ and $k \in \ker \varphi$ and let us show that $gkg^{-1} \in \ker \varphi$, that is, $\varphi(gkg^{-1}) = 1_{G'}$. In fact, using that φ is a group homomorphism and that $k \in \ker \varphi$ we have:

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) = \varphi(g)\varphi(g^{-1})\varphi(g)\varphi(g)^{-1},$$

which shows that $gkg^{-1} \in \ker \varphi$, as required.

Suppose first that φ is a monomorphism, which means that φ is an injective group homomorphism. Since φ is a group homomorphism, we have that $\varphi(1_G) = 1_{G'}$. For $k \in \ker \varphi$ we have that

$$\varphi(k) = 1_{G'} = \varphi(1_G),$$

which implies $k = 1_G$ since φ is injective. This shows that $\ker(\varphi) = \{1_G\}$.

Conversely, assume that $\ker(\varphi) = \{1_G\}$. Since we are assuming that φ is a group homomorphism, we only need to check and let us show that φ is injective. To do so, suppose that g_1 and g_2 are in G such that $\varphi(g_1) = \varphi(g_2)$, and let us show that $g_1 = g_2$. From $\varphi(g_1) = \varphi(g_2)$ multiplying by the inverse of $\varphi(g_2)$, which is, $\varphi(g_2)^{-1} = \varphi(g_2^{-1})$,

$$1_{G'} = \varphi(g_1)\varphi(g_2)^{-1} = \varphi(g_1)\varphi(g_2^{-1}) = \varphi(g_1g_2^{-1}) \Rightarrow g_1g_2^{-1} \in \ker(\varphi) = \{1_G\},$$

which implies that $g_1 g_2^{-1} = 1_G$ and so $g_1 = g_2$. This shows that φ is injective, finishing the proof. $\square$

Example 5.5. Consider the general linear group

$$GL(2, \mathbb{R}) = \{A \in M_2(\mathbb{R}) : \det(A) \neq 0\}.$$

Then the subset $SL(2, \mathbb{R}) = \{A \in M_2(\mathbb{R}) : \det(A) = 1\}$ of G consisting on the matrices with determinant 1, called the *special linear group* $\hat{A}$ is a normal subgroup of $GL(2, \mathbb{R})$.

To ease the notation let $H = SL(2, \mathbb{R})$. Using the properties of the determinant, it is straightforward to check that H is a subgroup of G (please, complete the details and let me know whether you have any questions). We prove here that H satisfies (2) in the *Normality Test*. Given $A \in GL(2, \mathbb{R})$, we need to show that $AHA^{-1} \subseteq H$. If $B \in H$, then

$$\det(ABA^{-1}) = \det(A) \det(B) \frac{1}{\det(A)} = \det(B) = 1 \Rightarrow ABA^{-1} \in H,$$

as desired.

5.2 Factor groups

Let G be a group and H a subgroup of G . We can introduce an intuitive operation in the set of all the cosets of H in G , called the *coset operation*, consisting on multiplying the generators of the cosets:

the coset of a operated with the coset of b is defined to be the coset of ab .

In symbols,

$$Ha \cdot Hb = H(ab)$$

This definition seems to be quite innocent but it is not that simple! The difficulty arises from the fact that the result of operate two cosets Ha and Hb might not be uniquely defined. Recall that Ha may be the same coset as Hc (it happens if $c \in Ha$), and similarly, Hb may be the same coset as Hd . Therefore, $Ha \cdot Hb$ should be the same as $Hc \cdot Hd$. Unfortunately, it may happen that $H(ab)$ is not the same coset as $H(cd)$. We can express this

situation using symbols as

$$\begin{array}{ccc} Ha \cdot Hb & = & H(ab) \\ \parallel & & \parallel \\ Hc \cdot Hd & = & H(cd) \end{array}$$

For example, if $G = S_3$ and $H = \{\varepsilon, (1\ 2)\}$, then

$$H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\} = H(1\ 3\ 2)$$

$$H(2\ 3) = \{(2\ 3), (1\ 2\ 3)\} = H(1\ 2\ 3)$$

and yet $H(1\ 3)(2\ 3) = H(1\ 3\ 2) \neq H = H\varepsilon = H(1\ 3\ 2)(1\ 2\ 3)$.

Therefore, coset multiplication does not work as an operation in the cosets of $H = \{\varepsilon, (1\ 2)\}$ in S_3 . Why not? **What did it go wrong?** Well ...

Although H is a subgroup, H is not a normal subgroup of S_3 . So then the next questions are “what if H were a normal subgroup?” “Would coset operation work?” Good news! It does! ☺☺ Check out the following theorem!

THEOREM 5.6. *Let H be a normal subgroup of a group G . If $Ha = Hc$ and $Hb = Hd$ then $H(ab) = H(cd)$.*

PROOF. Assume that $Ha = Hc$ and $Hb = Hd$. Then, in particular, $a \in Hc$ and $b \in Hd$, hence $a = h_1c$ and $b = h_2d$ for some $h_1, h_2 \in H$. Thus:

$$ab = (h_1c)(h_2d) = h_1(ch_2)d.$$

But $ch_2 \in cH = Hc$ because H is normal, and so $ch_2 = h_3c$ for some $h_3 \in H$. Returning to ab we get that

$$ab = h_1(ch_2)d = h_1(h_3c)d = (h_1h_3)(cd) \in H(cd)$$

An application of the Cosets Property **5** gives that $H(ab) = H(cd)$. □

Let G be a group and H a normal subgroup of G . Let G/H be the **set consisting of all the cosets of H in G** . Put it this way, if $Ha, Hb, Hc, \dots$ are the cosets of H in G , then $G/H = \{Ha, Hb, Hc, \dots\}$ or, more generally,

$$G/H = \{Ha, Hb, Hc, \dots\} = \{Hg \mid g \in G\}.$$

We have just proved that the coset operation is indeed an operation on G/H . In our next result, we prove that G/H is a group under coset operation.

THEOREM 5.7. *Let G be a group and H a normal subgroup of G . Then G/H is a group under coset operation: $Ha \cdot Hb = H(ab)$*

PROOF. 1. Coset operation is associative:

$$\begin{aligned} Ha \cdot (Hb \cdot Hc) &= Ha \cdot H(bc) = H(a(bc)) = H((ab)c) = H(ab) \cdot Hc = \\ &= (Ha \cdot Hb) \cdot Hc \end{aligned}$$

2. The unity for G/H is H :

$$Ha \cdot H = Ha \cdot H1 = H(a1) = H(1a) = H1 \cdot Ha = H \cdot Ha$$

3. The inverse of Ha is Ha^{-1} :

$$Ha \cdot Ha^{-1} = H(aa^{-1}) = H1 = H, \quad Ha^{-1} \cdot Ha = H(a^{-1}a) = H1 = H.$$

□

Recall that $|G : H|$ denotes the number of distinct cosets of H in G , and it is called the **index of H in G** . The group G/H is said to be the **factor group**, or **quotient group** of G by H . Note that $|G/H| = |G : H|$.

An important map is the **coset map** (also called **canonical projection**) $\pi : G \rightarrow G/H$ that assigns each element of G to its coset in H , that is, $\pi(a) = Ha$. It is straightforward to check that the coset map is a **group epimorphism** with kernel $\ker \pi = H$.

Example 5.8. Let $\mathbb{Z}$ be the additive cyclic group of the integers, and let $\langle 6 \rangle = 6\mathbb{Z}$ be the cyclic subgroup of $\mathbb{Z}$ consisting of all the multiples of 6. Notice that $6\mathbb{Z}$ is a normal subgroup of $\mathbb{Z}$ because $\mathbb{Z}$ is abelian (so all its subgroups are normal). Therefore, we can consider the factor group $\mathbb{Z}/6\mathbb{Z}$. The elements of this group are the cosets of $6\mathbb{Z}$ in $\mathbb{Z}$, namely:

$$\begin{aligned} 6\mathbb{Z} + 0 &= \{\dots, -18, -12, -6, 0, 6, 12, 18, \dots\} \\ 6\mathbb{Z} + 1 &= \{\dots, -17, -11, -5, 1, 7, 13, 19, \dots\} \\ 6\mathbb{Z} + 2 &= \{\dots, -16, -10, -4, 2, 8, 14, 20, \dots\} \\ 6\mathbb{Z} + 3 &= \{\dots, -15, -9, -3, 3, 9, 15, 21, \dots\} \\ 6\mathbb{Z} + 4 &= \{\dots, -14, -8, -2, 4, 10, 16, 22, \dots\} \end{aligned}$$

$$6\mathbb{Z} + 5 = \{\dots, -13, -7, -1, 5, 11, 17, 23, \dots\}$$

These are all the different cosets of $6\mathbb{Z}$. Observe that $6\mathbb{Z} + 6 = 6\mathbb{Z} + 0$ (since $6 \in 6\mathbb{Z} + 0$), $6\mathbb{Z} + 7 = 6\mathbb{Z} + 1$ (since $7 \in 6\mathbb{Z} + 1$), $6\mathbb{Z} + 8 = 6\mathbb{Z} + 2$, etc.

Since the operation in $\mathbb{Z}$ is $+$, we use $+$ for the coset operation, as well. But nothing has changed except for the name: $(6\mathbb{Z} + a) + (6\mathbb{Z} + b) = 6\mathbb{Z} + (a + b)$. To ease our notation, we write the coset in the following shorter form:

$$\bar{0} = 6\mathbb{Z} + 0 \quad \bar{1} = 6\mathbb{Z} + 1 \quad \bar{2} = 6\mathbb{Z} + 2 \quad \bar{3} = 6\mathbb{Z} + 3 \quad \bar{4} = 6\mathbb{Z} + 4 \quad \bar{5} = 6\mathbb{Z} + 5$$

Then $\mathbb{Z}/6\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$, and its Cayley table is:

$+$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$
$\bar{5}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$

Notice that the factor group $\mathbb{Z}/6\mathbb{Z}$ is isomorphic to the additive cyclic group $\mathbb{Z}_6$. In general, we have that $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Some group properties are preserved by the factor group construction. We present two of them in our next result:

THEOREM 5.9. *Let H be a normal subgroup of a group G .*

- (1) *If G is abelian, then G/H is abelian.*
- (2) *If $G = \langle g \rangle$ is cyclic, then G/H is also cyclic: $G/H = \langle Hg \rangle$.*

PROOF. (1) For $a, b \in G$ we need to check that $Ha \cdot Hb = Hb \cdot Ha$, which follows by the definition of the coset operation and the fact that we are assuming that G is abelian. In fact:

$$Ha \cdot Hb = H(ab) = H(ba) = Hb \cdot Ha.$$

(2) Suppose that G is a cyclic group and $g \in G$ is a generator of G , that is, the elements of G are all powers of the element g . We claim that Hg is a

generator for the factor group G/H . To show this, we need to prove that all the elements of G/H are powers of Hg . Take Ha an arbitrary element of Hg , where $a \in G$. Since g is a generator of G and $a \in G$ we can find some integer k such that $a = g^k$. From here using the coset operation we obtain that $(Hg)^k = Hg^k = Ha$, finishing the proof. $\square$

Remark 5.10. In Example 5.8 we proved, by computing the Cayley table of $\mathbb{Z}/6\mathbb{Z}$, that this factor group is isomorphic to the cyclic group $\mathbb{Z}_6$. An alternative proof is the following:

Since $\mathbb{Z}$ is cyclic, Theorem 5.9 (2) tells us that $\mathbb{Z}/6\mathbb{Z}$ is a cyclic group. Now keeping in mind that $\mathbb{Z}/6\mathbb{Z}$ has six elements, an application of Theorem 3.42 (1) reveals that $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}_6$ are isomorphic.

5.3 The First Isomorphism Theorem

Let G and G' be two groups. We say that G' is a **homomorphic image** of G if there exists an epimorphism (i.e. a surjective group homomorphism) from G onto G' . Using this terminology, the canonical projection (or coset mapping) tells us that

any factor group G/H is a homomorphic image of G .

This means that when we construct factor groups of G , we are, in fact, constructing homomorphic images of G . In other words, the factor group construction gives us a way of producing homomorphic images of a group G . In this section, we will show that this is indeed a way of manufacturing **all** the homomorphic images of G . To be more precise, we will prove that

every homomorphic image of G is isomorphic to a factor group

of G . As a consequence, we will obtain that a group G' is a homomorphic image of G if and only if G' is (or, is isomorphic to) a factor group of G . We begin by proving a simpler result.

THEOREM 5.11. *Let $\varphi : G \rightarrow G'$ be a group homomorphism with kernel K . Then $\varphi(a) = \varphi(b)$ if and only if $Ka = Kb$.*

PROOF. If a and b are two elements in G , then

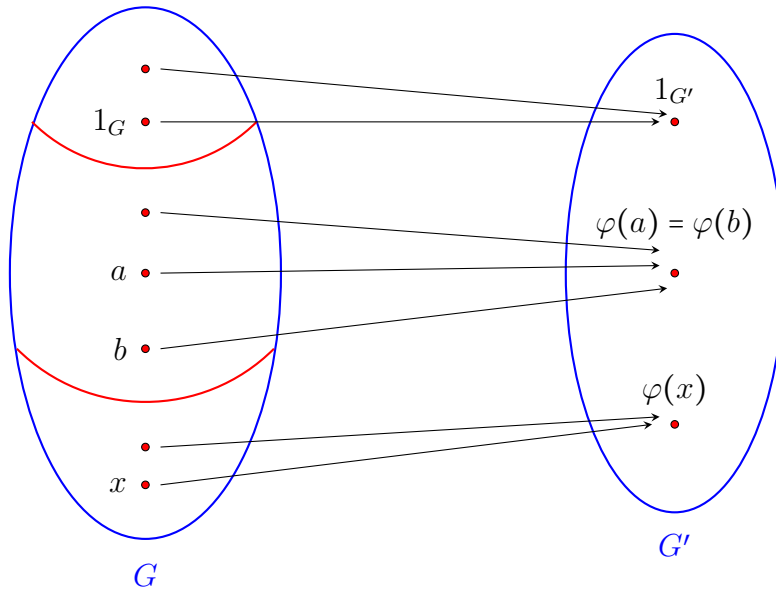
$$\varphi(a) = \varphi(b) \Leftrightarrow 1_{G'} = \varphi(a)\varphi(b)^{-1} = \varphi(a)\varphi(b^{-1}) = \varphi(ab^{-1}) \Leftrightarrow ab^{-1} \in K$$

Applying Cosets Property **6** we get

$$\varphi(a) = \varphi(b) \Leftrightarrow ab^{-1} \in K \Leftrightarrow Ka = Kb,$$

as desired. $\square$

Suppose that $\varphi : G \rightarrow G'$ is a group epimorphism with kernel K . Theorem 5.11 tells us that all the elements in the coset Kx have the same image under φ , and, conversely, two elements a and b in G have the same image under φ if and only if they are in the same coset of K .



We have then built a bijective map that matches cosets of K , namely, elements of the factor group G/K , with elements in G' . To be more precise, the coset Kx is matched with the element $\varphi(x) \in G'$. It remains to check that this correspondence is indeed an isomorphism.

THEOREM 5.12. FIRST ISOMORPHISM THEOREM. *Let $\varphi : G \rightarrow G'$ be a group epimorphism with kernel $K = \ker \varphi$. Then $G/K \cong G'$.*

PROOF. To show that G/K is isomorphic to G' , we need find a group isomorphism from G/K to G' . Let us try with the map $\tilde{\varphi} : G/K \rightarrow G'$ defined by

$\tilde{\varphi}(Kx) = \varphi(x)$. We first need to make sure that $\tilde{\varphi}(Kx)$ is uniquely defined. To do so, we must make sure that if Ka is the same coset as Kb , then $\varphi(a) = \varphi(b)$, but this follows from Theorem 5.11. We now prove that $\tilde{\varphi}$ is bijective:

- $\tilde{\varphi}$ is injective: If $\tilde{\varphi}(Ka) = \tilde{\varphi}(Kb)$, then $\varphi(a) = \varphi(b)$ and by Theorem 5.11, we get that $Ka = Kb$.
- $\tilde{\varphi}$ is surjective: since we are assuming that φ is surjective we have that every element y of G' is of the form $y = \varphi(x)$. But, by definition, $\tilde{\varphi}(Kx) = \varphi(x) = y$, which shows that $\tilde{\varphi}$ is surjective.

Finally, $\tilde{\varphi}(Ka \cdot Kb) = \tilde{\varphi}(K(ab)) = \varphi(ab) = \varphi(a)\varphi(b) = \tilde{\varphi}(Ka)\tilde{\varphi}(Kb)$, which shows $\tilde{\varphi}$ is a group homomorphism. We can then conclude that $\tilde{\varphi}$ is a group isomorphism, and so, $G/K \cong G'$. $\square$

The *First Isomorphism Theorem* asserts that **every homomorphic image of G is isomorphic to a factor group of G** . If we are given an homomorphism $\varphi : G \rightarrow G'$, not necessarily surjective, we can consider the restriction $\varphi : G \rightarrow \varphi(G)$ to the image, which is now an epimorphism, and apply the *First Isomorphism Theorem* to it. More precisely:

COROLLARY 5.13. *Let $\varphi : G \rightarrow G'$ be a group homomorphism with kernel K . Then $G/K \cong \text{Im}\varphi$.*

Example 5.14. Let $G = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$. Show that $Z(G) \cong \mathbb{R}$ and $G/Z(G) \cong \mathbb{R} \times \mathbb{R}$.

We first prove that G is a group under matrix multiplication. In fact:

- G is closed under matrix multiplication:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a+a' & b'+ac'+b \\ 0 & 1 & c+c' \\ 0 & 0 & 1 \end{pmatrix}$$

- The identity matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is in G , and

- G is closed under inverses: $\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}$.

Since matrix multiplication is an associative operation, we can conclude that G is a group under matrix multiplication. Next, we need to compute the center $Z(G)$ of G , that is, the elements of G that commutes with any other element of G . If $\begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \in Z(G)$, then

$$\begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix}, \quad \forall a, b, c \in \mathbb{R}.$$

It implies that $b' + ac' + b = b + a'c + b'$, for all $a, b, c \in \mathbb{R}$. In particular, taking $a = 1$ and $c = 0$ gives $c' = 0$. Next, if we take $c = 1$ we obtain that $a' = 0$. Thus:

$$Z(G) = \left\{ \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : b \in \mathbb{R} \right\}$$

The map $\varphi : \mathbb{R} \rightarrow Z(G)$ given by

$$b \mapsto \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

is the desired isomorphism. It is clearly bijective (complete the details and let me know whether you have any questions) and it is a group homomorphism:

$$\varphi(b_1 b_2) = \begin{pmatrix} 1 & 0 & b_1 b_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & b_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & b_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \varphi(b_1) \varphi(b_2), \quad \forall b_1, b_2 \in \mathbb{R}.$$

We have proved that $\mathbb{R} \stackrel{\varphi}{\cong} Z(G)$. It remains to show that $G/Z(G) \cong \mathbb{R} \times \mathbb{R}$. To do this, we apply the *First Isomorphism Theorem*. We need to find a group epimorphism $G \rightarrow \mathbb{R} \times \mathbb{R}$ with kernel $Z(G)$. Taking into account the previous calculations, we can try with the map $\psi : G \rightarrow \mathbb{R} \times \mathbb{R}$ defined by

$$\psi \left(\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \right) = (a, c).$$

I am thinking about this map because I have some practice and I can see that its kernel will be $Z(G)$. Notice that $\mathbb{R} \times \mathbb{R}$ is an additive group, so its identity

element is $(0, 0)$. Keeping this in mind we have

$$\begin{aligned} \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \in \ker \psi &\Leftrightarrow \psi \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} = (0, 0) \Leftrightarrow (a, c) = (0, 0) \Leftrightarrow a = c = 0 \\ &\Leftrightarrow \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in Z(G) \end{aligned}$$

Clearly, ψ is surjective: for $(x, y) \in \mathbb{R} \times \mathbb{R}$, we have $(x, y) = \psi \begin{pmatrix} 1 & x & 0 \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \in G$.

Lastly, ψ is a group homomorphism:

$$\begin{aligned} \psi \left[\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \right] &= \psi \begin{pmatrix} 1 & a+a' & b'+ac'+b \\ 0 & 1 & c+c' \\ 0 & 0 & 1 \end{pmatrix} = \\ &= (a+a', c+c') = (a, c) + (a', c') = \psi \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} + \psi \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

An application of the *First Isomorphism Theorem* gives that $G/\ker \psi \cong \mathbb{R} \times \mathbb{R}$, that is, $G/Z(G) \cong \mathbb{R} \times \mathbb{R}$.

Example 5.15. Let G be a group of order 5 such that there exists a group homomorphism $\varphi : \mathbb{Z}_{30} \rightarrow G$. We can use the *First Isomorphism Theorem* and *Lagrange's Theorem* to determine $\ker \varphi$.

Thanks to the *Fundamental Theorem of Finite Cyclic Groups* we know how all the subgroups of the (additive) cyclic group $\mathbb{Z}_{30}$ look like. So, in particular, we have a good understanding of $\ker \varphi$. An application of the *First Isomorphism Theorem* tells us that $\mathbb{Z}_{30}/\ker \varphi \cong \text{im } \varphi$. From here, we obtain:

$$|\mathbb{Z}_{30}/\ker \varphi| = |\text{im } \varphi|.$$

Next we use *Lagrange's Theorem* to obtain some information about the order of $\text{im } \varphi$, which is a subgroup of G . Since G has order 5 (a prime), we have that either $|\mathbb{Z}_{30}/\ker \varphi| = |\text{im } \varphi| = 1$ or 5. We consider both cases:

- If $|\mathbb{Z}_{30}/\ker \varphi| = 1$, then $\ker \varphi = \mathbb{Z}_{30}$ (and φ is the trivial homomorphism).
- If $|\mathbb{Z}_{30}/\ker \varphi| = 5$, then $\ker \varphi$ is a subgroup of $\mathbb{Z}_{30}$ of order 6, namely, $\ker \varphi = 5\mathbb{Z}_{30}$.

Appendix A

Some useful stuffs

This appendix provides a summary of some algebra concepts needed for this course. If you learnt them last year and you do not need to revisit them, please feel free to skip these pages. I just want to make sure we are all in the same page, since it will set yourself up for success!

Here, we will talk about **sets**, **maps** and **binary operations**.

A.1 Sets

A **set** X in Mathematics is nothing more than a collection of objects, called **elements**. If a is an element of the set X , we will say that a **lies in** X , or a **belongs to** X , or X **contains** a , and we write $a \in X$. Otherwise, if a is not an element of X , we write $a \notin X$.

For example, the number 1 belongs to the **set of natural numbers** $\mathbb{N}$, while the number -1 does not. So, $1 \in \mathbb{N}$ and $-1 \notin \mathbb{N}$.

An interesting set is the set that contains no elements, which is called the **empty set** and denoted by $\emptyset$. For example, consider the set consisting of the rational numbers whose square equal 2, that is,

$$\{q \in \mathbb{Q} \mid q^2 = 2\}.$$

Such a set has no elements, since we know that $\sqrt{2}$ is not a rational number, that is, it is the empty set: $\{q \in \mathbb{Q} \mid q^2 = 2\} = \emptyset$.

Definition A.1. Let X and Y be two sets. If every element of X belongs to Y , then we say that X is a **subset** of Y , or X is **contained in** Y , or Y **contains** X ; we write $X \subseteq Y$, or $Y \supseteq X$.

We say that

- X and Y are **equal** and we write $X = Y$ if $X \subseteq Y$ and $Y \subseteq X$;
- X is a **proper subset** of Y if $X \subseteq Y$ and $X \neq Y$; we write $X \subsetneq Y$ or $X \subset Y$.

Example A.2. $\mathbb{Z}$ is a proper subset of $\mathbb{Q}$.

Definition A.3. We say that a set X is **finite** if either $X = \emptyset$ or there exists a natural number n such that X has exactly n elements. We call n the **cardinality** of X and write $|X| = n$. We define $|\emptyset| = 0$. Lastly, a set that is not finite is called **infinite**.

Example A.4. $|\{x \in \mathbb{Z} \mid x^2 = 4\}| = 2$ and $\{x \in \mathbb{Z} \mid x^2 = 4\} = \{-2, 2\}$.

Definition A.5. The **Cartesian product** of two (nonempty) sets X and Y , denoted $X \times Y$, is the set consisting on all the (ordered) pairs (a, b) , where $a \in X$ and $b \in Y$. In general:

$$X_1 \times \dots \times X_n = \{(a_1, \dots, a_n) \mid a_i \in X_i \text{ for all } i\}.$$

A very important particular case is the **Cartesian product of X with itself**: $X \times X = \{(a, b) \mid a, b \in X\}$; this is denoted by X^2 .

Example A.6. $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is the Cartesian plane.

Definition A.7. Let X and Y be two sets.

- The **intersection** $\hat{\cap}$ of X and Y is the set $X \cap Y$ of all elements that belong to both X and Y . If $X \cap Y = \emptyset$, then we say that X and Y are **disjoint**.
- The **union** $\hat{\cup}$ of X and Y is the set $X \cup Y$ of all elements that belong to either X or Y (not necessarily to both).
- The **difference** $\hat{-}$ of X and Y is the set $X \setminus Y$ (or $X - Y$) of all elements that belong X but not to Y .

Example A.8. For $X = \{-1, 2, 3, 7\}$ and $Y = \{0, -5, 7\}$ we have that

$$X \cap Y = \{7\}, \quad X \cup Y = \{-5, -1, 0, 1, 2, 3, 7\}, \quad X \setminus Y = \{-1, 2, 3\}.$$

A.2 Maps

Let X and Y be two sets. Informally, we can say that a **map** f from X to Y , denoted $f : X \rightarrow Y$ is a rule that assigns an element $a \in X$ to **exactly one** element $b \in Y$. We write that $f(a) = b$, or $a \mapsto f(a) = b$. We call X and Y the **domain** and **codomain** of f , respectively. For instance, the map assigning any real number to its square is an example of map from $\mathbb{R}$ to $\mathbb{R}$. That is, $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$ for all $x \in \mathbb{R}$.

Suppose that U is a subset of X . The map $f|_U : U \rightarrow Y$ given by

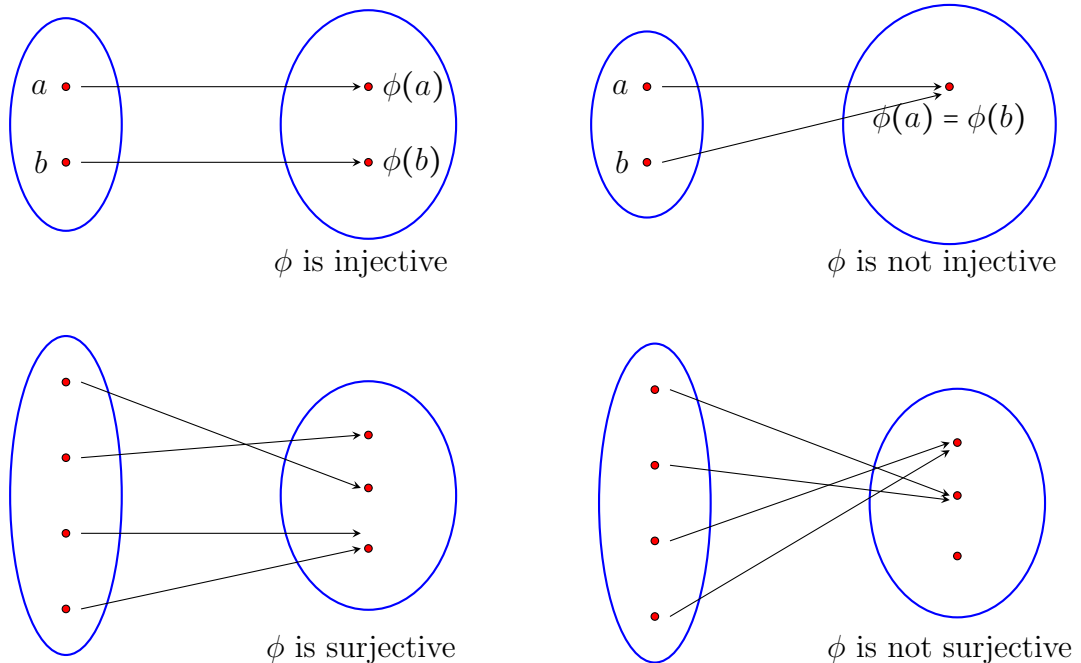
$$f|_U(a) = f(a), \text{ for all } a \in U,$$

is called the **restriction** of f to U . Roughly speaking, $f|_U$ is the “same” map but with a smaller domain.

Definition A.9. Let X and Y be two sets. We say that a map $f : X \rightarrow Y$ is

- **injective** if $f(a_1) = f(a_2)$ implies $a_1 = a_2$.
- **surjective** or **onto** if for every $b \in Y$ there exists $a \in X$ such that $f(a) = b$.
- **bijective** if f is both injective and surjective.

Venn’s diagrams provide a cool (and easy) visualisation of these concepts:



Example A.10. The map $f : \mathbb{R} \rightarrow \{x \in \mathbb{R} \mid x \geq 0\}$ given by $f(x) = x^2$ is surjective but not injective, since $f(-2) = 4 = f(2)$; the map $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = e^x$ is injective but not surjective (the negative numbers do not have a pre-image); the map $h : \mathbb{R} \rightarrow \mathbb{R}$ defined by $h(x) = x^3$ is injective and surjective, and so is bijective.

As you move along in your algebra journey, you will realise that we (algebraist) pay special attention to bijective maps from a set (or other algebraic structures) X to itself. A very simple example, is the **identity map** Id_X , defined by $\text{Id}_X(x) = x$ for all $x \in X$.

Remark A.11. If X is a finite set, then a map $f : X \rightarrow X$ is injective if and only if f is surjective (and hence, bijective).

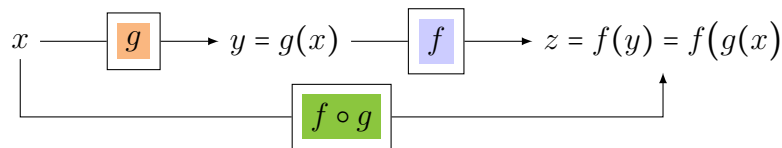
Definition A.12. Let X, Y, Z be sets, and $g : X \rightarrow Y$, $f : Y \rightarrow Z$ maps. We can construct a new map $f \circ g : X \rightarrow Z$ (or fg), called the **composition** (or **composite**) of f and g , as follows:

$$(f \circ g)(x) = f(g(x)), \text{ for all } x \in X.$$

This is really important! You might find yourself coming back to this comment later on, when we talk about permutations in Chapter 2. **When computing $(fg)(x)$ we do the following:**

1. **we first calculate** $g(x)$, let us call it y , that is, $y = g(x)$, and then
2. **we calculate** f **of** y : $z = f(y) = f(g(x))$.

In some texts, g is called the **inner map** and f is called the **outer map** of the composition. In a visual way, this is how composition of maps work:



Example A.13. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$ and $g(x) = x + 1$. Then:

$$(f \circ g)(x) = f(g(x)) = f(x + 1) = (x + 1)^2,$$

$$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 1;$$

Notice that $f \circ g \neq g \circ f$, since $(f \circ g)(2) = (2 + 1)^2 = 9$ and $(g \circ f)(2) = 2^2 + 1 = 5$.

A.3 Binary operations

In this section, we will generalise the addition and multiplication of numbers. Notice that the addition process for numbers assigns to any pair (a, b) of numbers a new number, their sum, usually denoted $a + b$. Similarly, multiplication assigns the product ab to the pair (a, b) . Thus, in order to come up with a more general notion, we will need a set X (which will play the role of the numbers) and “algorithm” with input a pair of elements (x, y) in our set, and output the result of operating x and y . We will represent such an algorithm by a map $X \times X \rightarrow X$. More formally:

Definitions A.14. A **binary operation** on a nonempty set X is a map $\ast : X \times X \rightarrow X$. We usually write $x \ast y$ to denote $\ast(x, y)$.

Examples A.15.

1. Addition and multiplication are binary operations on $\mathbb{N}$ and on $\mathbb{Z}$.
2. Subtraction is a binary operation on $\mathbb{Z}$.
3. Division is not a binary operation on $\mathbb{Q}$, because we can not divide by 0. It is a binary operation on $\mathbb{Q}^\times = \mathbb{Q} - \{0\}$.
4. Let X denote a nonempty set and $\text{Map}(X)$ the set of all maps from X onto X . The map composition $(f, g) \mapsto f \circ g$ is an operation on $\text{Map}(X)$.

Definition A.16. Let X be a nonempty set and $\ast$ a binary operation on X . We say that

- $\ast$ is **commutative** if $x \ast y = y \ast x$ for all $x, y \in X$.
- $\ast$ is **associative** if $x \ast (y \ast z) = (x \ast y) \ast z$ for all $x, y, z \in X$.
- An element $e \in X$ is called an **identity element** for $\ast$ if $e \ast x = x = x \ast e$ for every $x \in X$.

THEOREM A.17. *If a binary operation $\ast$ has an identity element, then it is unique.*

PROOF. Assume that e and e' are identities elements for $\ast$. Then $e' = e \ast e'$, since e is an identity element, and $e \ast e' = e$ because e' is an identity element. Thus $e = e'$. □

Examples A.18.

1. Addition and multiplication on $\mathbb{N}$ are both commutative and associative. Moreover, 0 is the identity element for $+$ and 1 for $\times$.
2. Multiplication on $M_n(\mathbb{R})$ is not commutative. It is associative and the $n \times n$ identity matrix I_n is the identity element.
3. Composition on $\text{Map}(X)$ is associative, but not commutative if $|X| > 1$. In fact, take a and b are two distinct elements of X , and consider the maps $f, g : X \rightarrow X$ given by $f(x) = a$ and $g(x) = b$, for all $x \in X$, respectively. Then:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(b) = a, \\ (g \circ f)(x) &= g(f(x)) = g(a) = b,\end{aligned}$$

and since we are assuming that $a \neq b$, we have that $f \circ g \neq g \circ f$.

Definition A.19. Let $*$ be a binary operation on a (nonempty) set X and $e \in X$ the identity element for $*$. We say that $y \in X$ is an **inverse** for $x \in X$ if $x * y = e = y * x$. Elements with inverses are called **units**. We will denote by $U(X, *)$, or simply $U(X)$, the **set consisting of all the units** of X .

THEOREM A.20. Let $*$ be an associative binary operation on a (nonempty) set X and $e \in X$ the identity element for $*$. If $x \in X$ has an inverse in X , then such an inverse is unique.

PROOF. Suppose that y and y' are inverses of x , then $x * y = e = y * x$ and $x * y' = e = y' * x$. From here we get that

$$y' = y' * e = y' * (x * y) = (y' * x) * y = e * y = y,$$

and we are done. □

Examples A.21.

1. Consider $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$. In all cases, the identity element is 0 and the usual negative $-x$ of an element x is the (additive) inverse. Then every element is a unit and hence

$$U(\mathbb{Z}, +) = \mathbb{Z}, \quad U(\mathbb{Q}, +) = \mathbb{Q}, \quad U(\mathbb{R}, +) = \mathbb{R}, \quad U(\mathbb{C}, +) = \mathbb{C}.$$

2. Notice that $U(\mathbb{Z}, \cdot) = \{1, -1\}$.
3. Multiplication on $\mathbb{Q}$ has the identity $1 \in \mathbb{Q}$. Every nonzero $x \in \mathbb{Q}$ has an inverse x^{-1} with respect to multiplication, but $0 \in \mathbb{Q}$ has no inverse. Thus: $U(\mathbb{Q}, \cdot) = \mathbb{Q} - \{0\}$. The same applies to $(\mathbb{R}, \cdot)$ and $(\mathbb{C}, \cdot)$.
4. We have that $U(M_n(\mathbb{R}), +) = M_n(\mathbb{R})$. But regarding to matrix multiplication, $A \in M_n(\mathbb{R})$ has an inverse if and only if $\det(A) \neq 0$. Thus:

$$U(M_n(\mathbb{R}), \cdot) = \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}.$$

5. In $\text{Map}(X)$, a map $f : X \rightarrow X$ has an inverse if and only if it is bijective.

Algebra (or Mathematics in general) will be getting more and more abstract as you continue diving into it. Binary operations are denoted by many different symbols in mathematics. For example, we use $+$ and $\cdot$ to denote the addition and multiplication of numbers, but these symbols are also used for the addition and multiplication of matrices. Similarly, $\cap$ and $\cup$ are well-established notations in set theory. When discussing binary operations in general, we have been using $*$ for the binary operation. But algebraists, we do not do this! We usually adopt one of the following two formats:

- **Multiplicative Notation:** $a * b$ is written ab (or $a \cdot b$) and called the **product of a and b** . The **multiplicative unity** is denoted 1 .
- **Additive Notation:** $a * b$ is written $a + b$ and is called the **sum of a and b** . The **additive unity** is denoted 0 .

Multiplicative notation is the most popular format among algebraists. Here we will use multiplicative notation, and denote the unity by 1 . Keep in mind that ab can mean many different things, depending the set under discussion. Neither a nor b need to be numbers, so ab might be nothing to do with the idea of multiplication that we have. Please, don't stress about it. You will get used to it.

A.4 Equivalence relations

Let us begin by talking about binary relations. A **binary relation** R on a set X is a subset of the Cartesian product $X \times X$. We write xRy if $(x, y) \in R$. For instance, take $X = \mathbb{R}$ and $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x \leq y\}$. Then R is a binary relation on $\mathbb{R}$; we have that $1R2$ and $-4R0$ since $1 \leq 2$ and $-4 \leq 0$. Notice that this is just the usual $\leq$ relation, written in a more fancy way.

The equality $=$ relation on $\mathbb{R}$ is nothing more than the subset of $\mathbb{R} \times \mathbb{R}$ consisting on all the pairs, having the two components equal:

$$\{(x, x) \mid x \in \mathbb{R}\}.$$

Definition A.22. An **equivalence relation** R on a set X is a binary relation on X satisfying the three following properties:

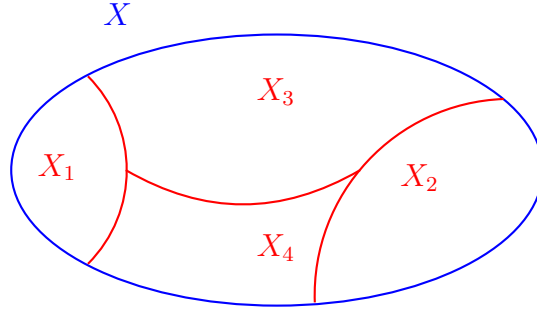
1. **Reflexive property:** aRa for all $a \in X$.
2. **Symmetric property:** aRb implies bRa .
3. **Transitive property:** aRb and bRc imply aRc .

A more practical notation for an equivalence relation is the following: for $a, b \in X$ we write $a \sim b$ if aRb (that is, $(a, b) \in R$).

Examples A.23.

1. The binary relation $\leq$ on $\mathbb{R}$ is reflexive and transitive but it is not symmetric, so $\leq$ is not an equivalence relation on $\mathbb{R}$. The equality relation $=$ is clearly an equivalence relation on $\mathbb{R}$.
2. Let $\mathcal{F} = \mathbb{R}[x]$ denote the set of all polynomials with coefficients in $\mathbb{R}$. On $\mathcal{F}$ we define a binary relation $f \sim g$ if $f' = g'$. You can easily check that $\sim$ is an equivalence relation on $\mathcal{F}$.

Let X be a nonempty set. Recall that a **partition** $\hat{A}$ of X is a set of (nonempty) pairwise disjoint subsets of X such that the union of all of them is X . For instance, $\{X_1, X_2, X_3, X_4\}$ is a partition of the set X drawn below.



A more concrete example: $\{\{1\}, \{a, b\}, \{-1, 2, 3\}\}$ and $\{\{1, a, b, -1\}, \{2, 3\}\}$ are partitions of the set $X = \{a, b, -1, 1, 2, 3\}$.

Any equivalence relation $\sim$ on a (nonempty) set X produces a partition of X ; in fact, for $x \in X$, we define the **equivalence class of x** as the set of elements of X related to x , that is:

$$[x] = \{y \in X \mid x \sim y\}$$

Notice that $[x] = [x']$ if and only if $x \sim x'$. Moreover, $[x] \neq \emptyset$, since $x \sim x$ because $\sim$ is reflexive. We claim that $\{[x] \mid x \in X\}$ is a partition of X . In fact, if $a \in [x] \cap [y]$, then we need to show that $[x] = [y]$, or equivalently that $x \sim y$. But this is true because $\sim$ is symmetric and transitive:

$$\begin{aligned} a \in [x] &\Rightarrow x \sim a, \\ a \in [y] &\Rightarrow y \sim a \Rightarrow a \sim y, \end{aligned}$$

imply that $x \sim y$, as desired. It remains to show that $X = \cup_{a \in X} [a]$, but this is trivial since $x \in [x]$, for each $x \in X$.

Example A.24. Consider the equivalence relation $\sim$ on $\mathcal{F} = \mathbb{R}[x]$ defined on Example A.23 (2). The equivalence class of a polynomial f consists on all the polynomial that are related to f via $\sim$, that is, all the polynomials that have the same derivative of f . Since the constant polynomials have zero derivative, we have that $[f] = \{f + c \mid c \in \mathbb{R}\}$.